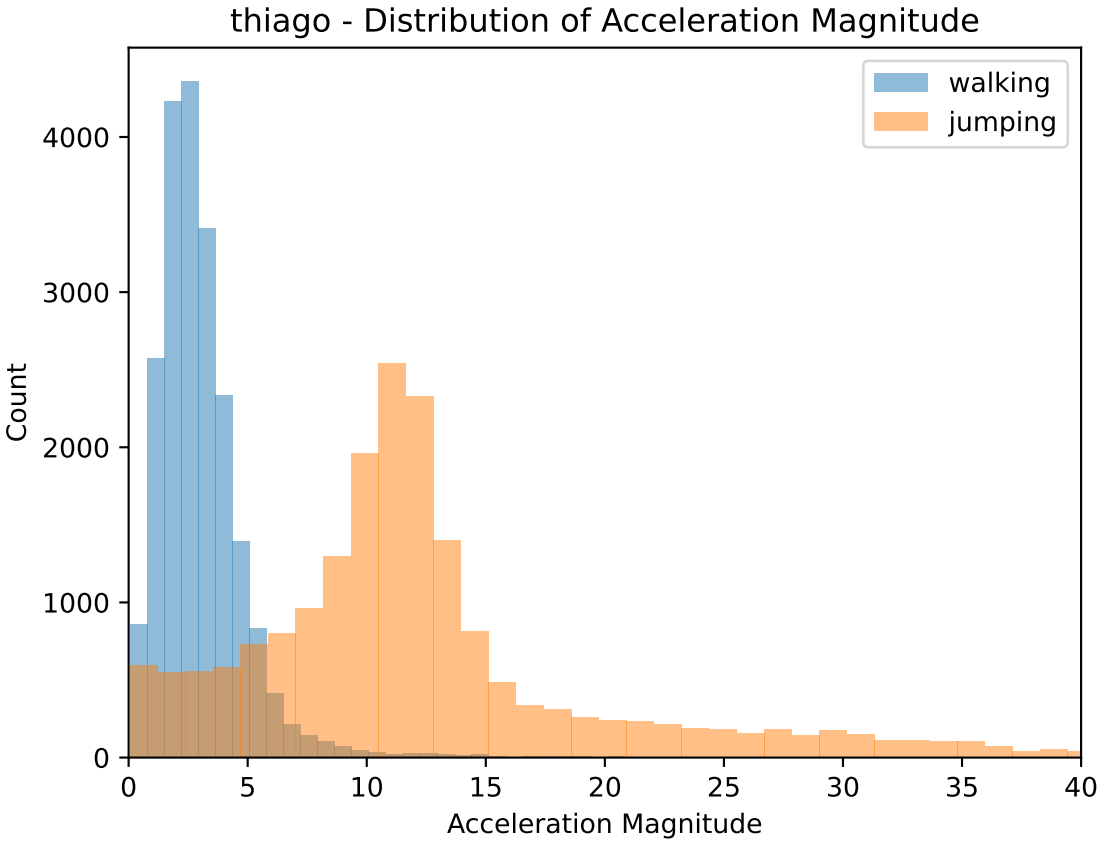
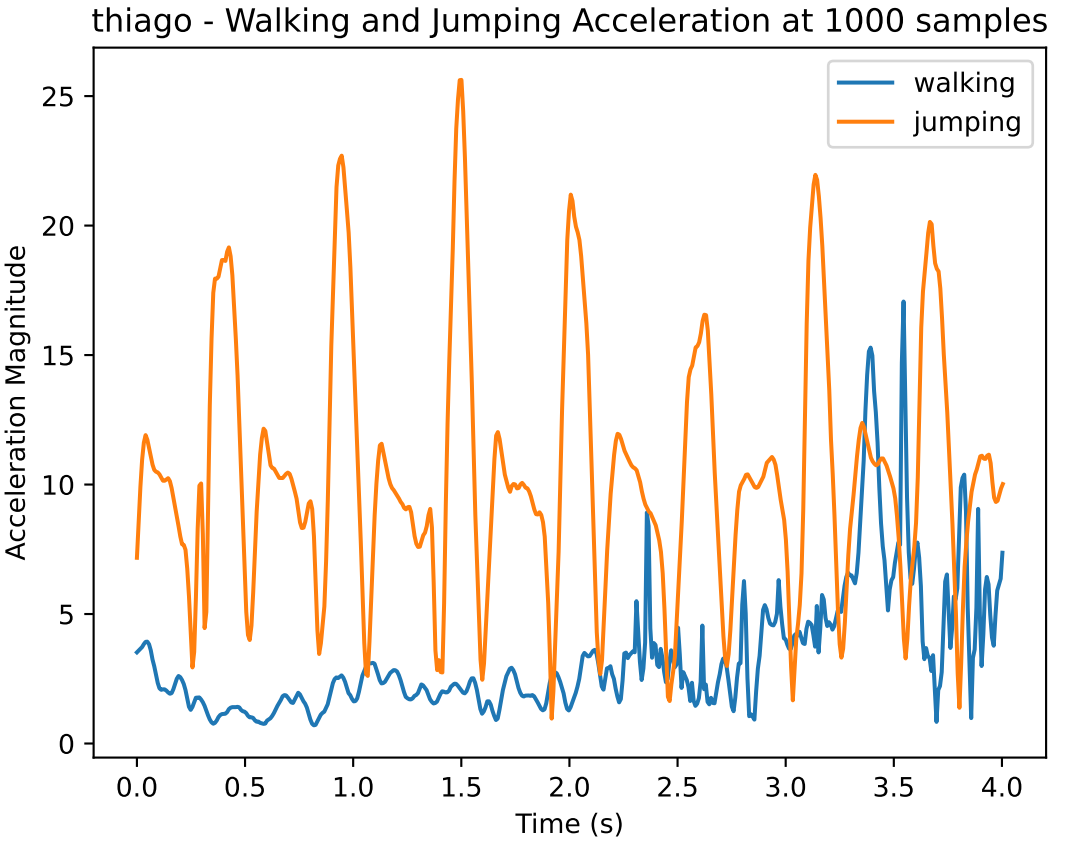
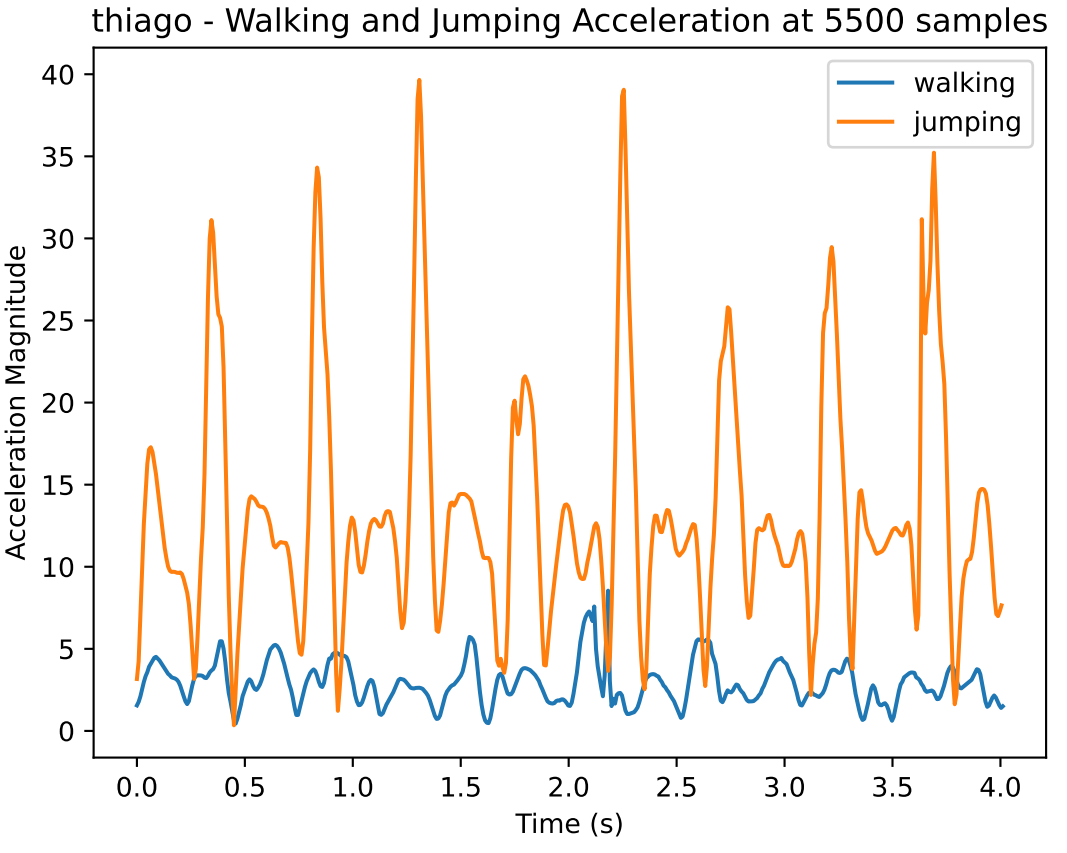
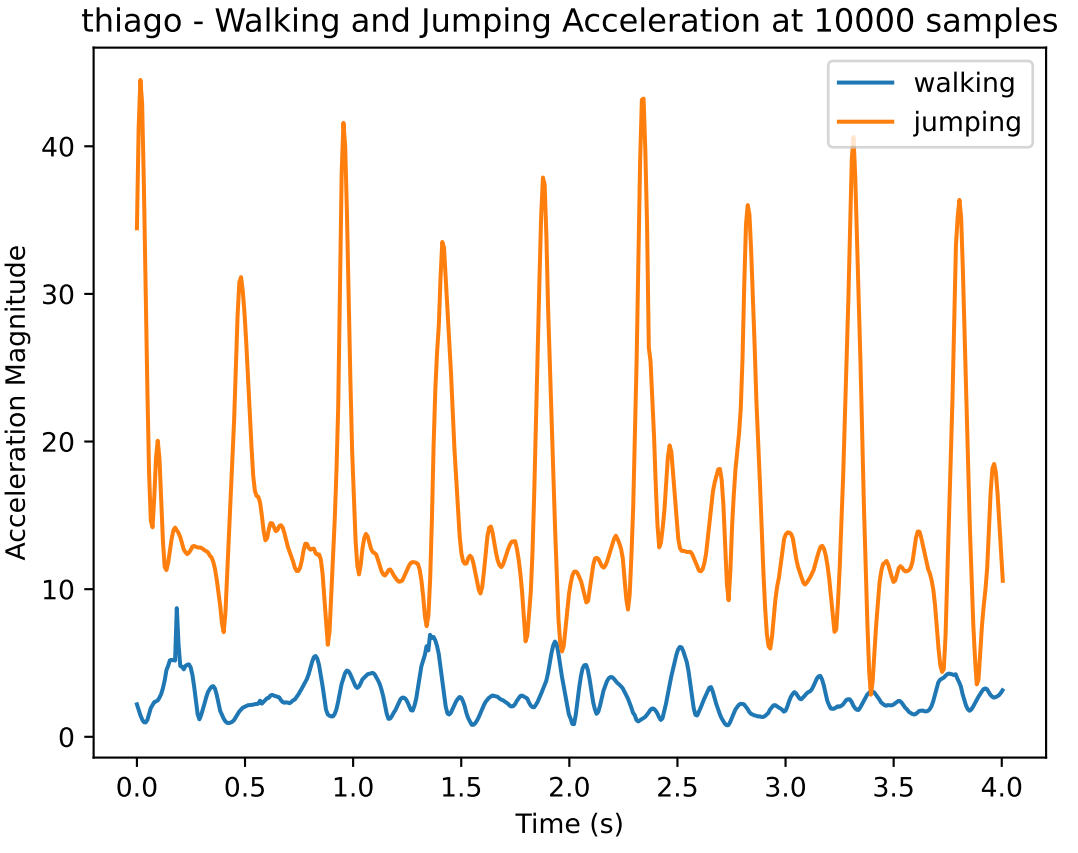
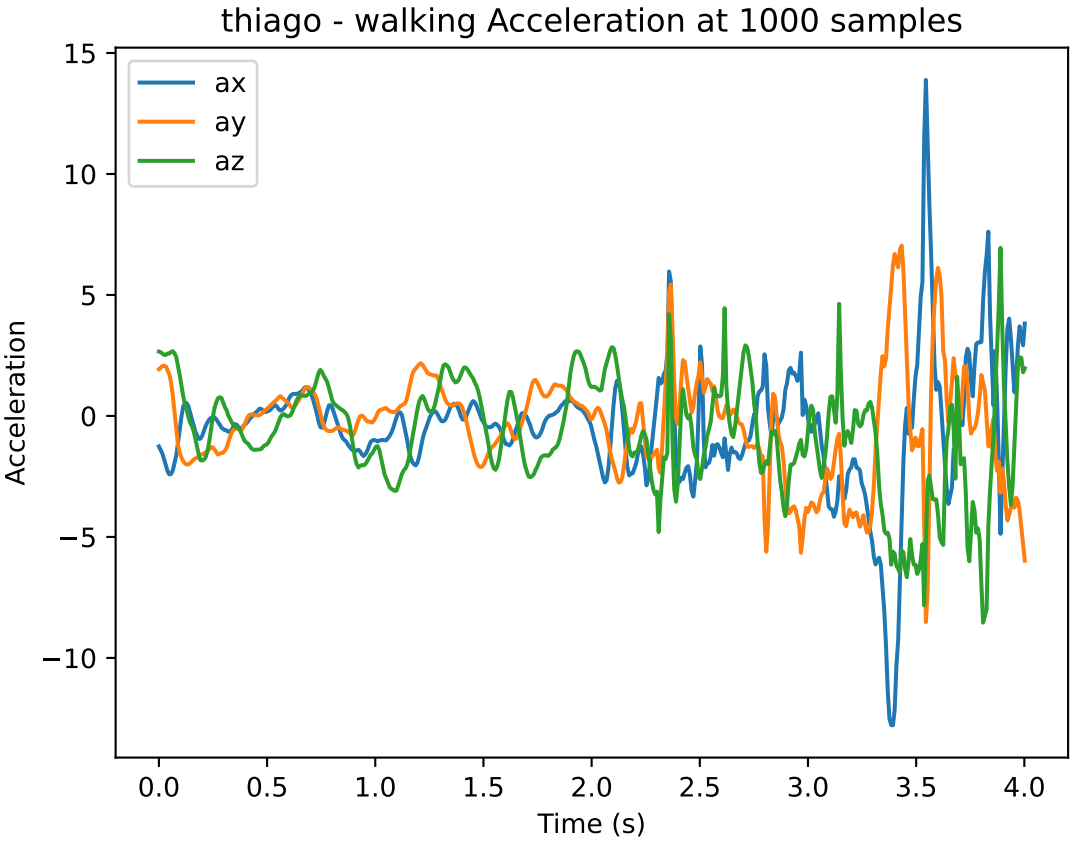
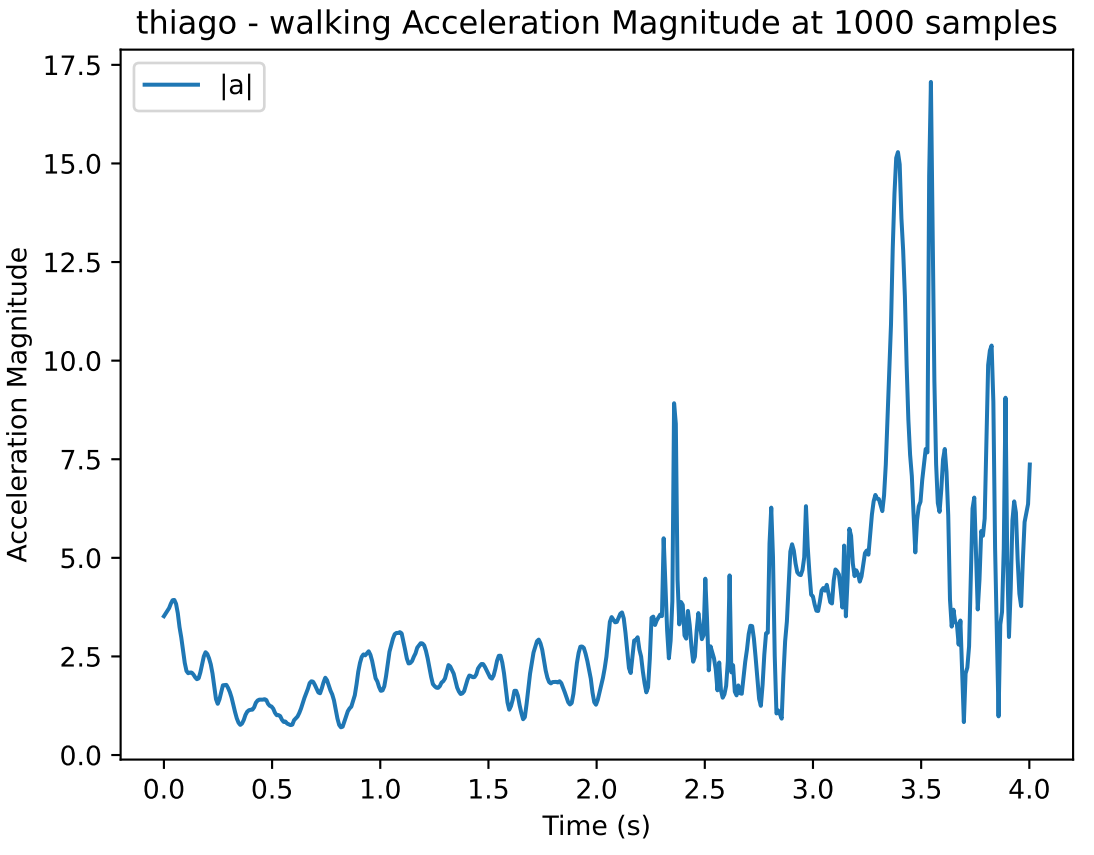

thiago - Distribution of Acceleration Magnitude
walking
jumping
Count
0
1000
2000
3000
4000
0
5
10
15
20
25
30
35
40
Acceleration Magnitude

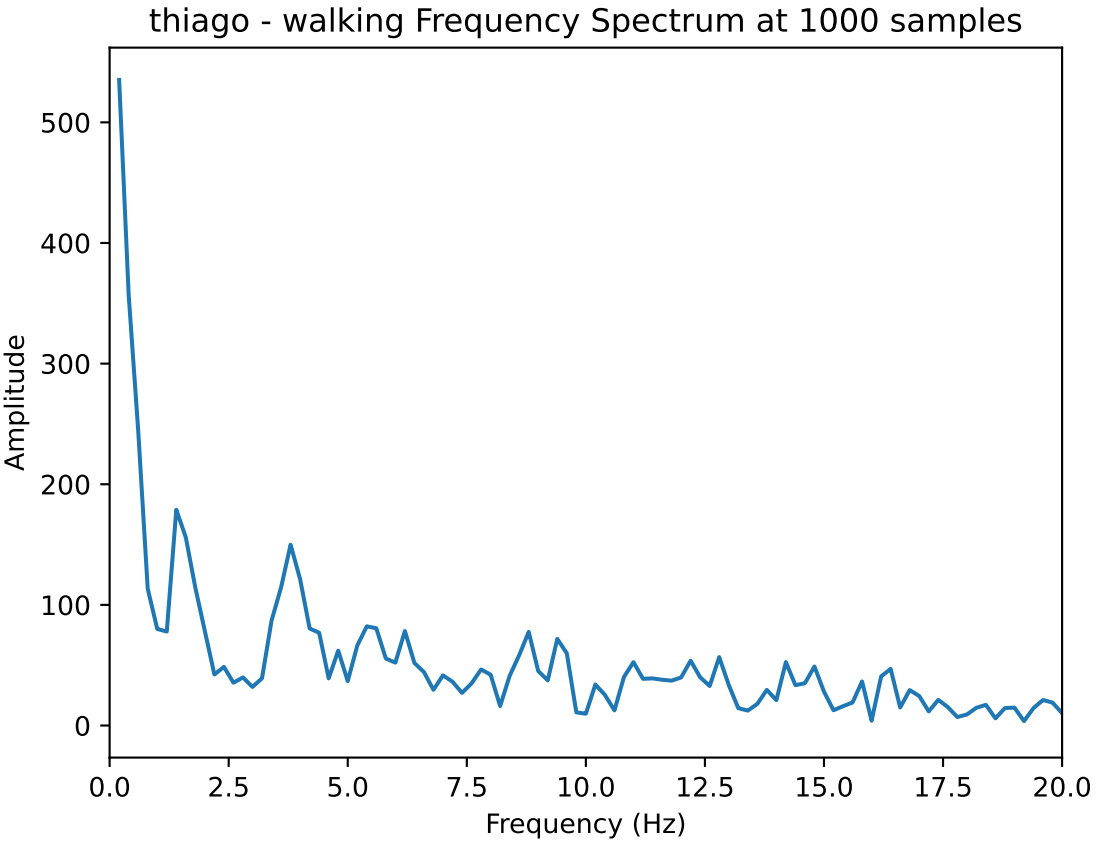

thiago - Walking and Jumping Acceleration at 1000 samples
walking
jumping
Acceleration Magnitude
Time (s)
0
5
10
15
20
25
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

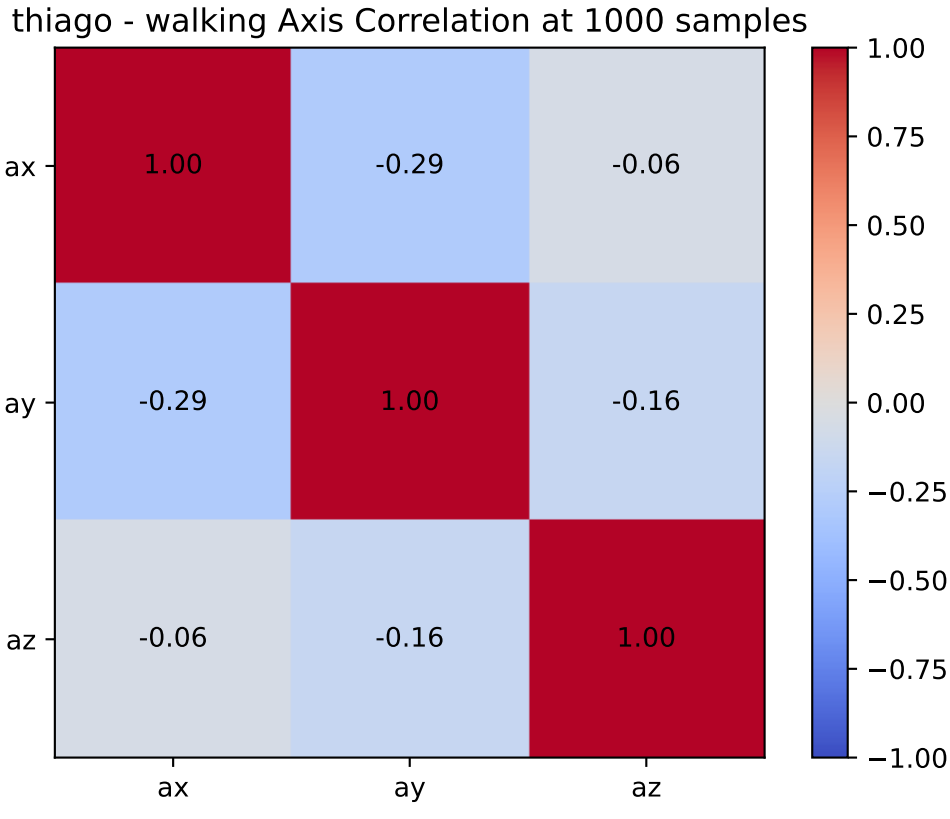

thiago - Walking and Jumping Acceleration at 5500 samples
walking
jumping
Acceleration Magnitude
Time (s)
0
5
10
15
20
25
30
35
40
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

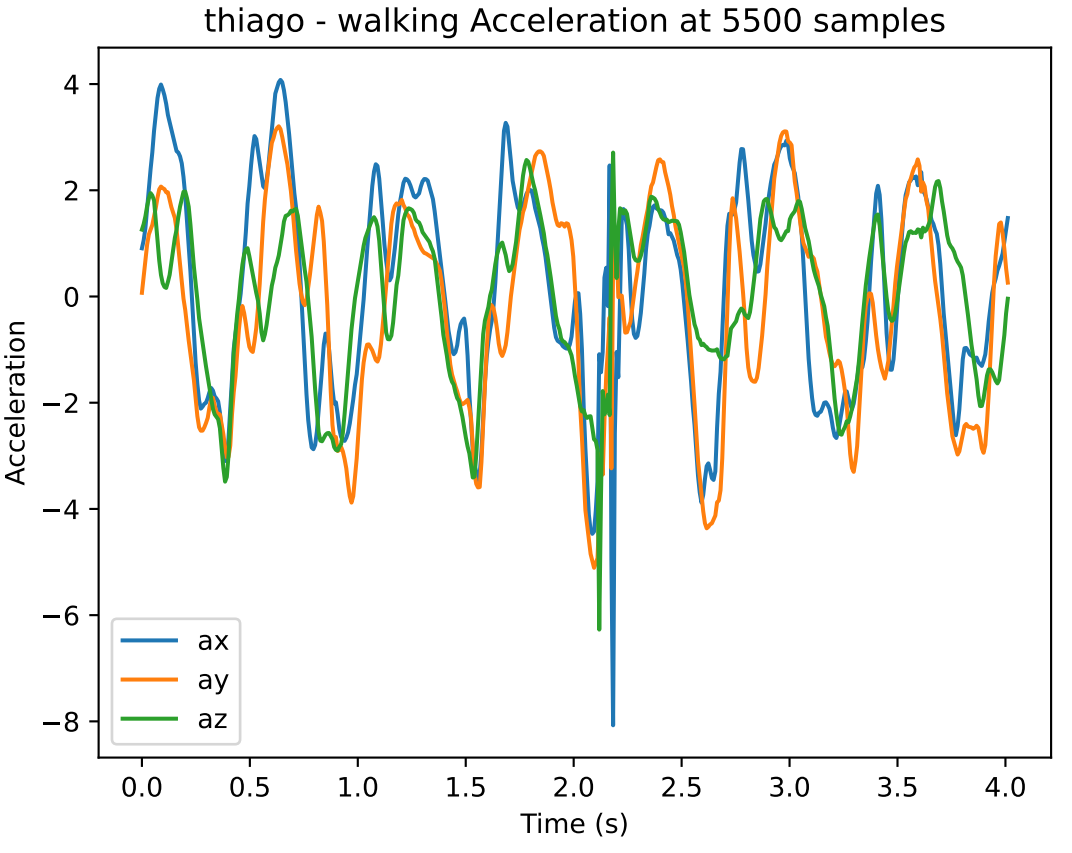

thiago - Walking and Jumping Acceleration at 10000 samples
walking
jumping
Acceleration Magnitude
0
10
20
30
40
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Time (s)

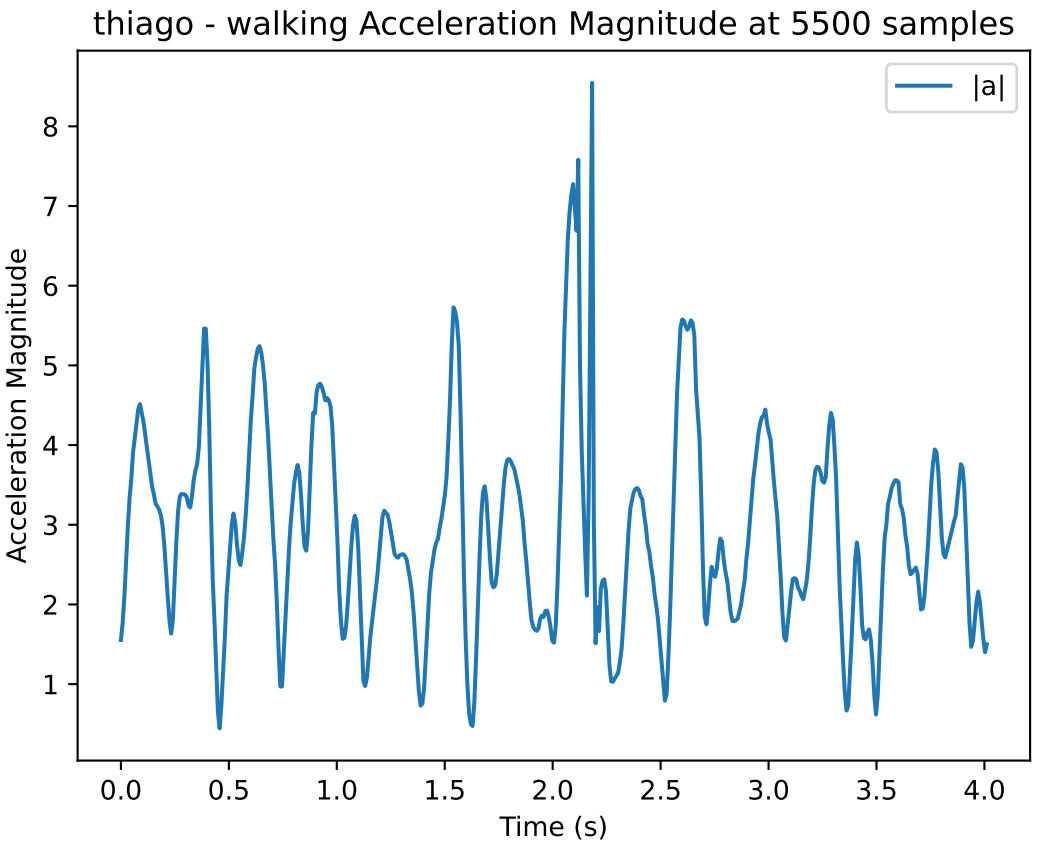

thiago - walking Acceleration at 1000 samples
ax
ay
az
Acceleration
Time (s)
15
10
5
0
−5
−10
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

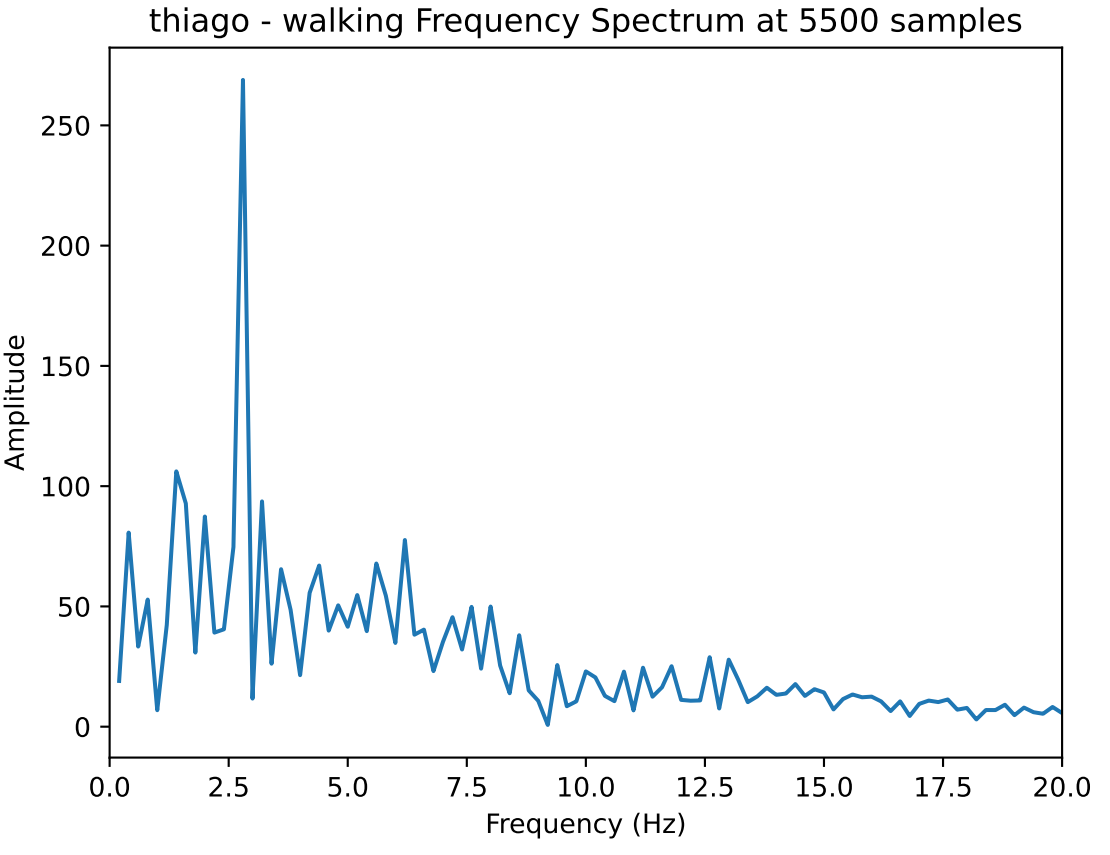

thiago - walking Acceleration Magnitude at 1000 samples
|a|
Acceleration Magnitude
Time (s)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

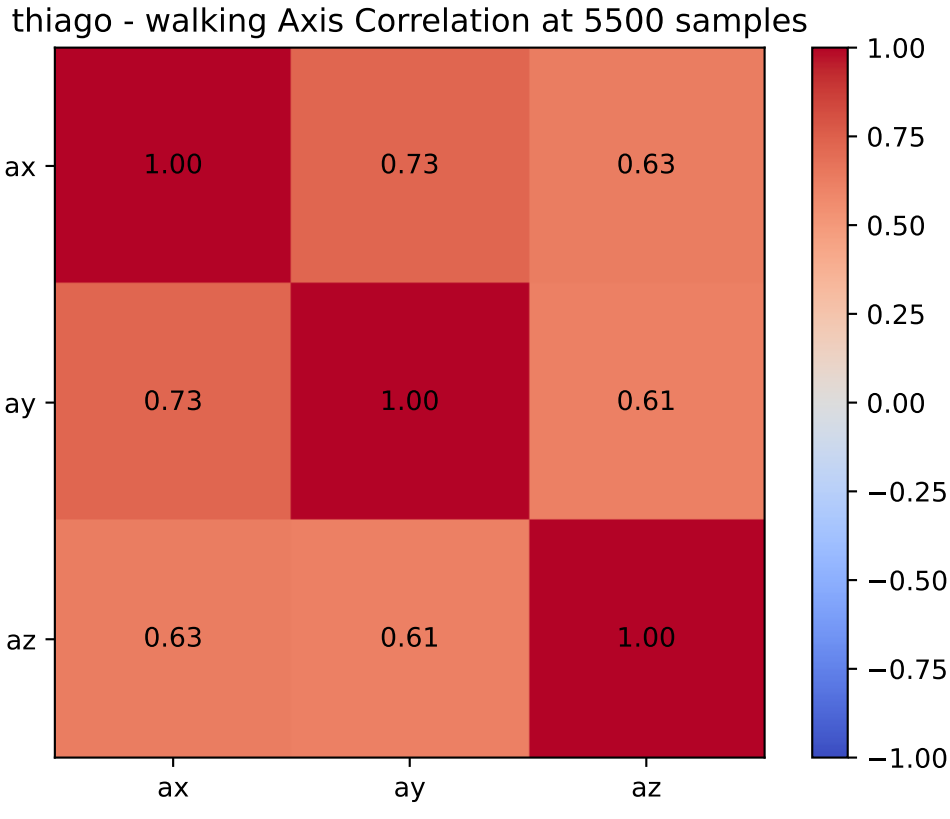

thiago - walking Frequency Spectrum at 1000 samples
Amplitude
500
400
300
200
100
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Frequency (Hz)

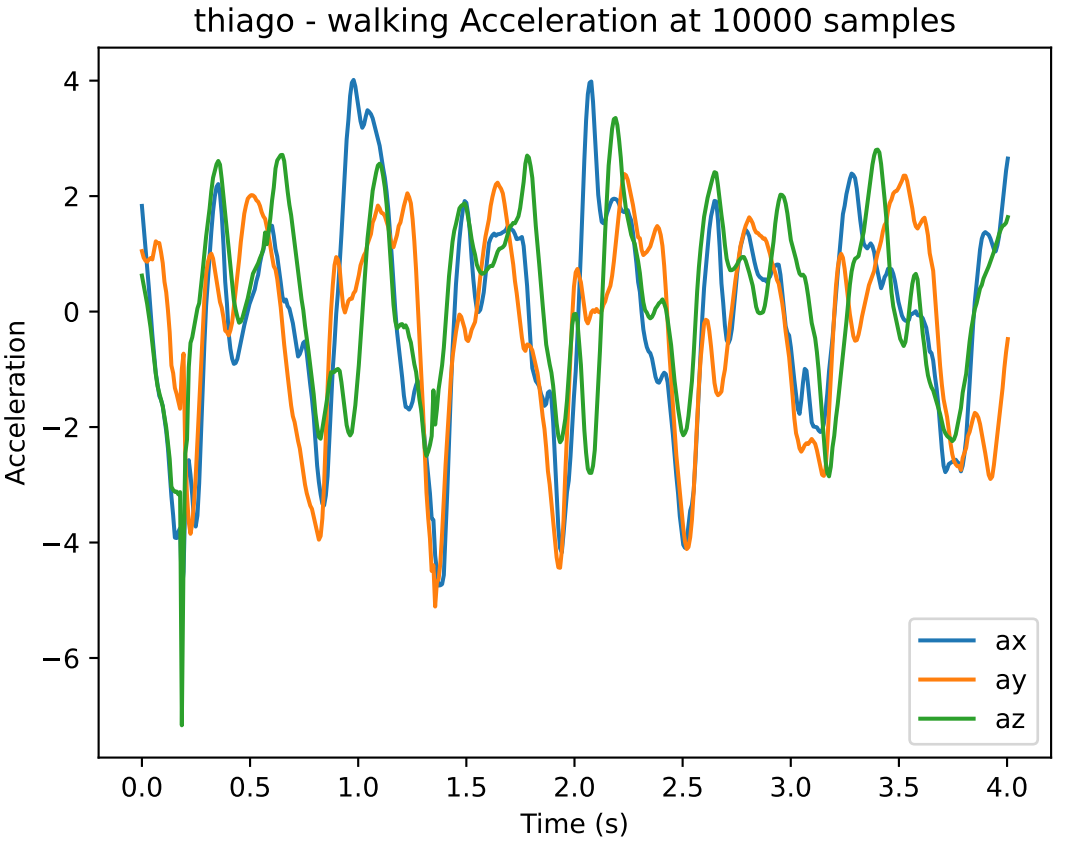

thiago - walking Axis Correlation at 1000 samples
ax
ay
az
1.00
-0.29
-0.06
-0.29
1.00
-0.16
-0.06
-0.16
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

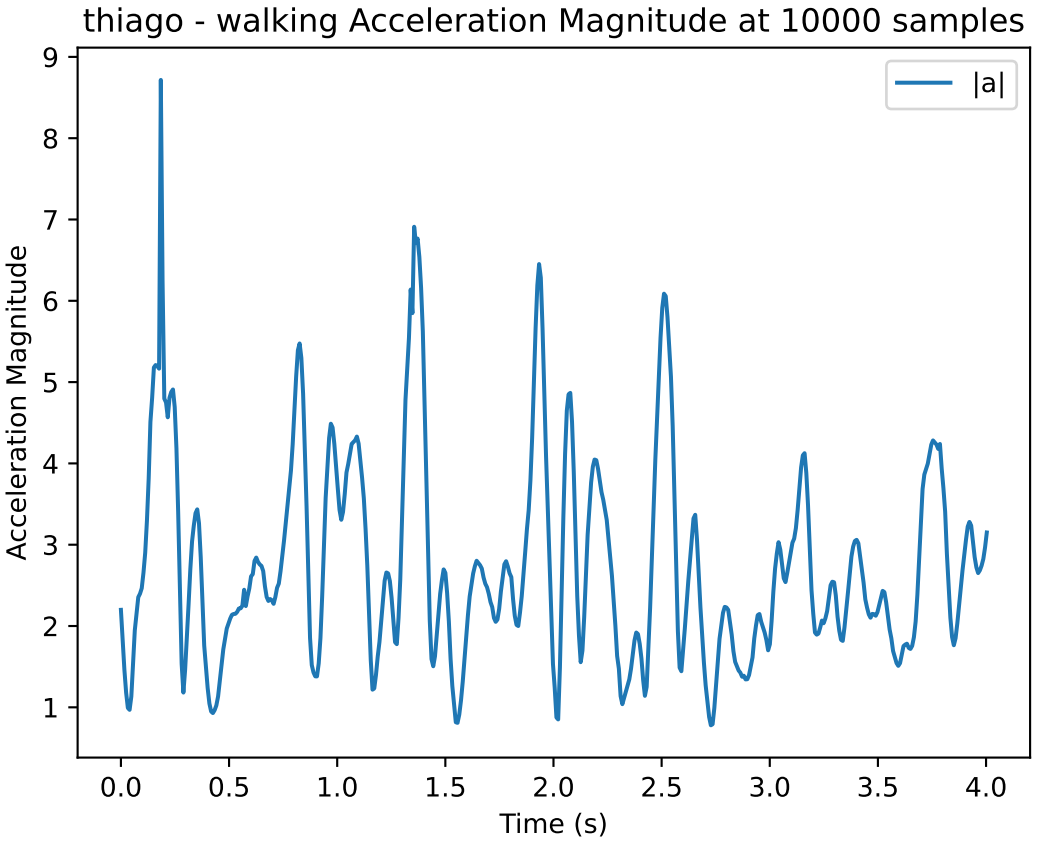

thiago - walking Acceleration at 5500 samples
Acceleration
Time (s)
ax
ay
az
4
2
0
−2
−4
−6
−8
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

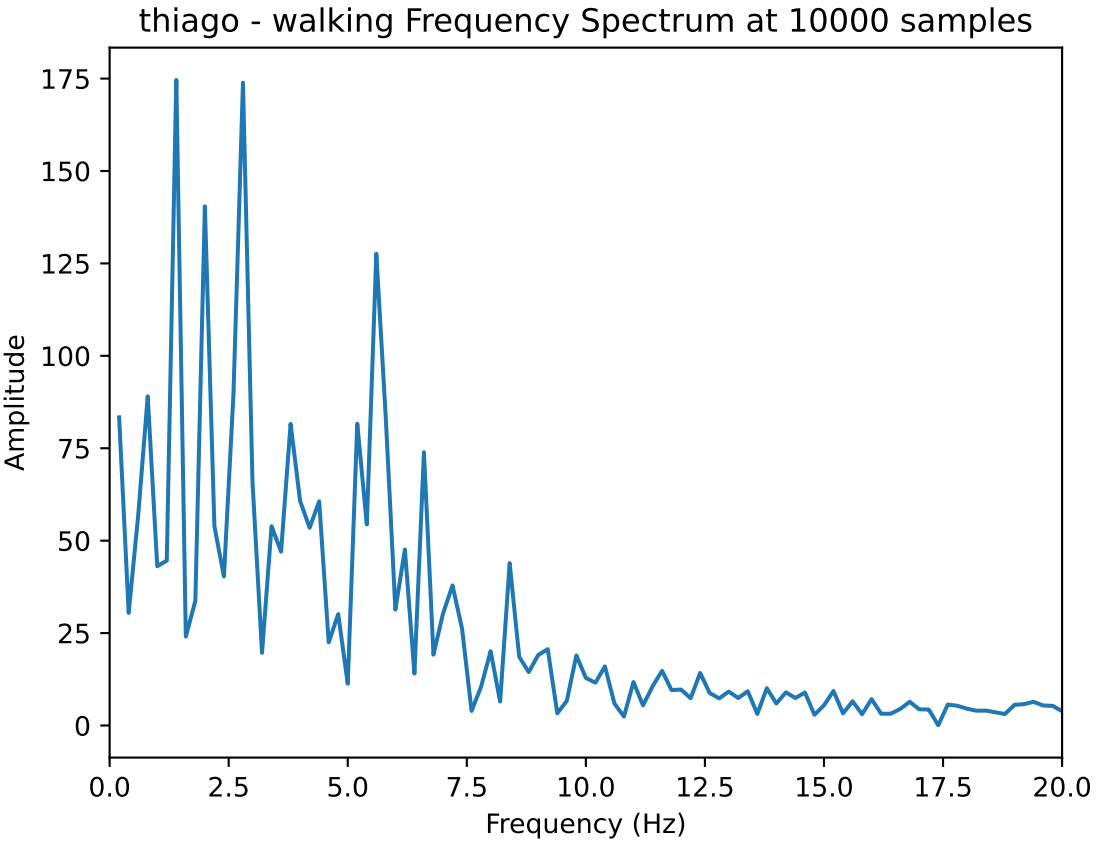

thiago - walking Acceleration Magnitude at 5500 samples
|a|
Acceleration Magnitude
Time (s)
1
2
3
4
5
6
7
8
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

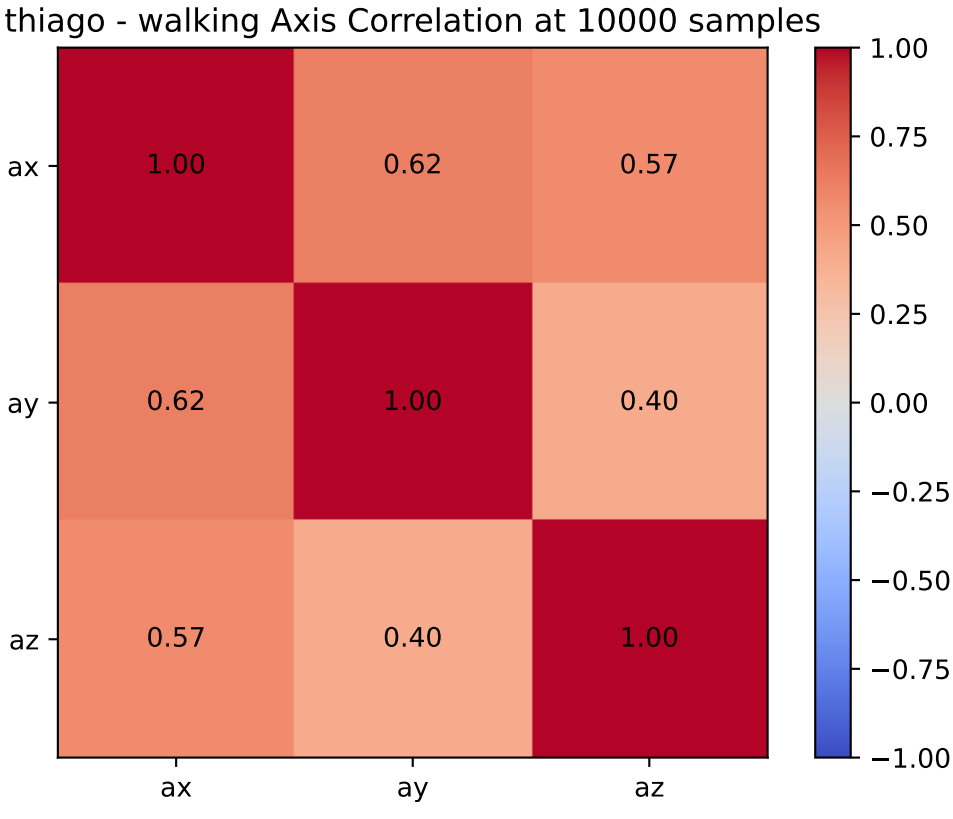

thiago - walking Frequency Spectrum at 5500 samples
Amplitude
250
200
150
100
50
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Frequency (Hz)

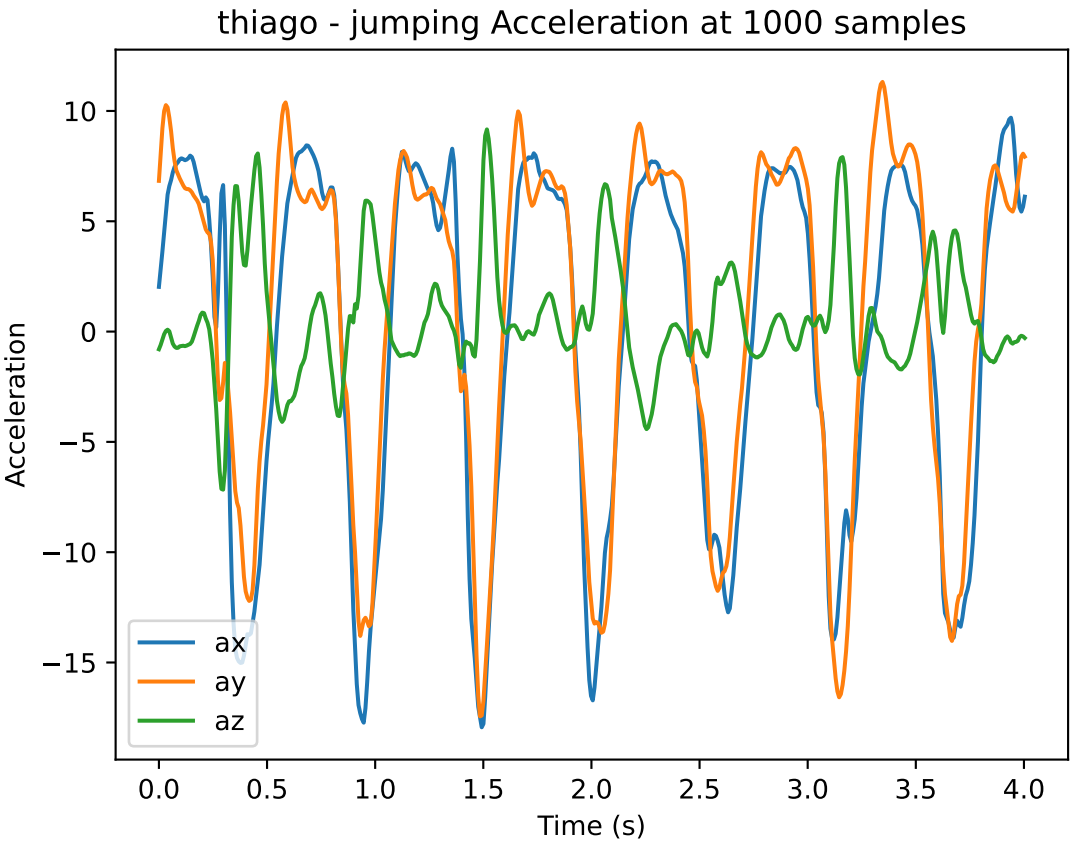

thiago - walking Axis Correlation at 5500 samples
ax
ay
az
1.00
0.73
0.63
0.73
1.00
0.61
0.63
0.61
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

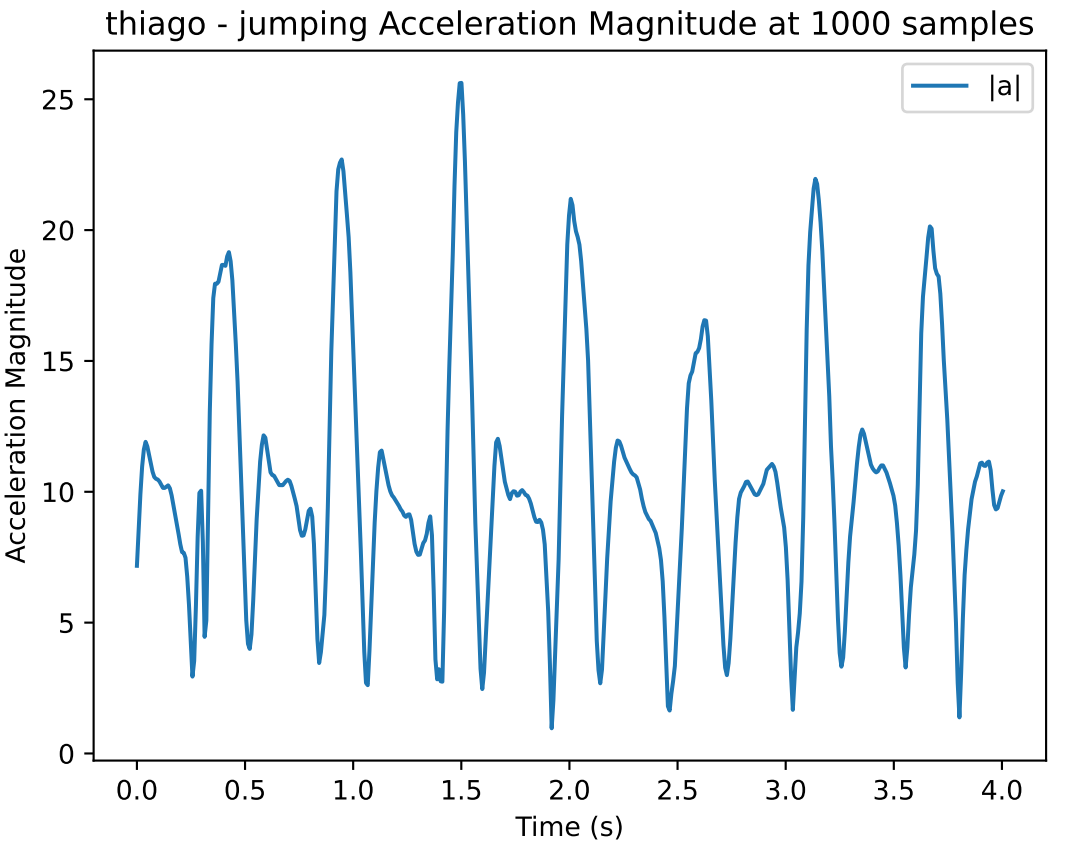

thiago - walking Acceleration at 10000 samples
Acceleration
Time (s)
4
2
0
−2
−4
−6
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
ax
ay
az

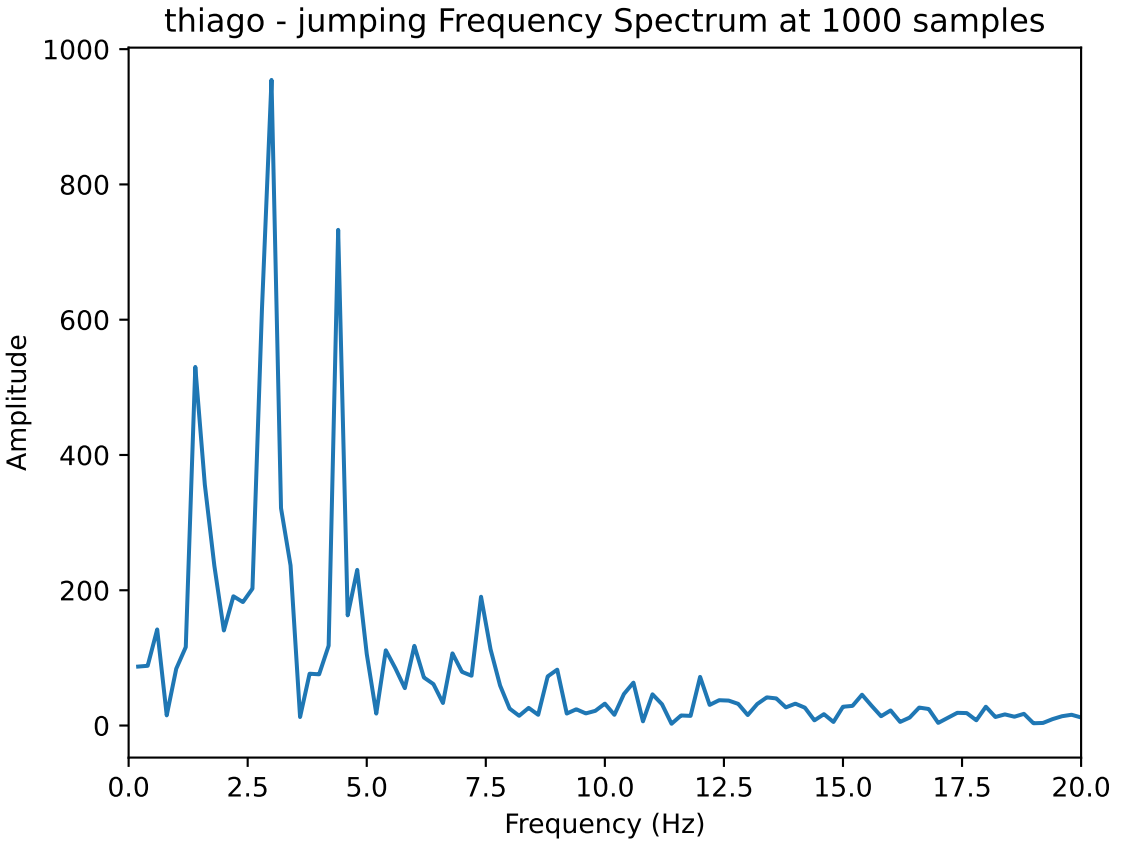

thiago - walking Acceleration Magnitude at 10000 samples
|a|
Acceleration Magnitude
Time (s)
9
8
7
6
5
4
3
2
1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

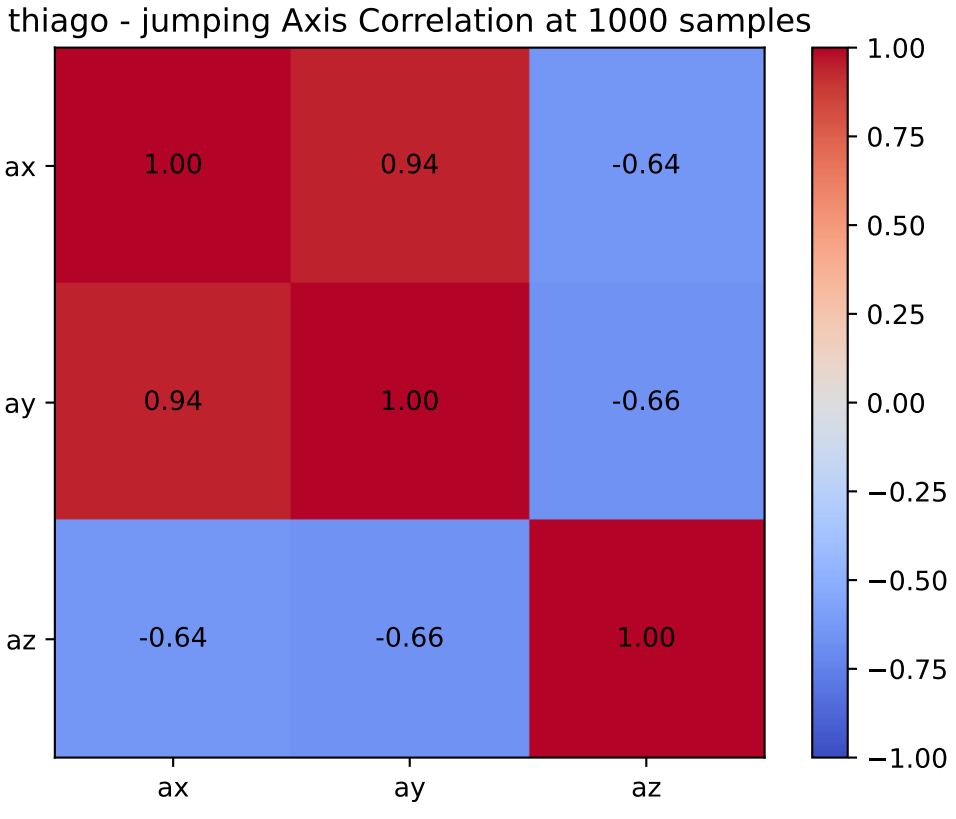

thiago - walking Frequency Spectrum at 10000 samples
Amplitude
175
150
125
100
75
50
25
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Frequency (Hz)

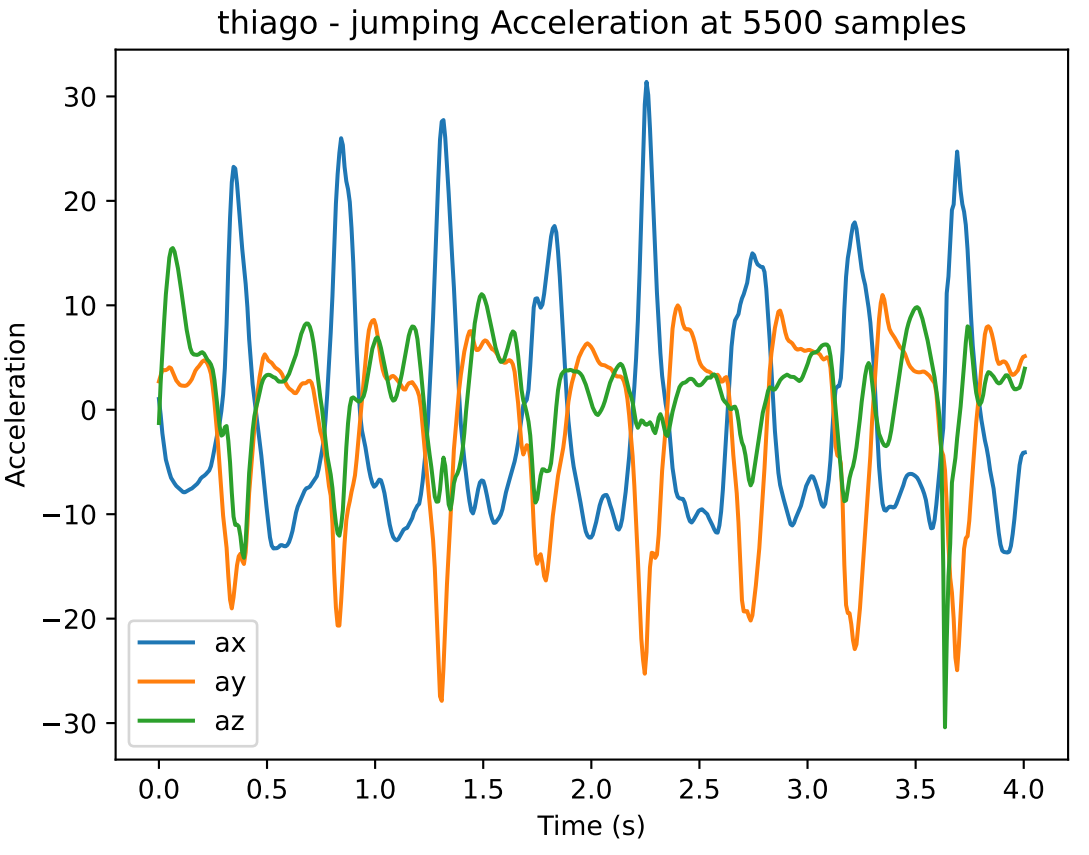

thiago - walking Axis Correlation at 10000 samples
ax
ay
az
1.00
0.62
0.57
0.62
1.00
0.40
0.57
0.40
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

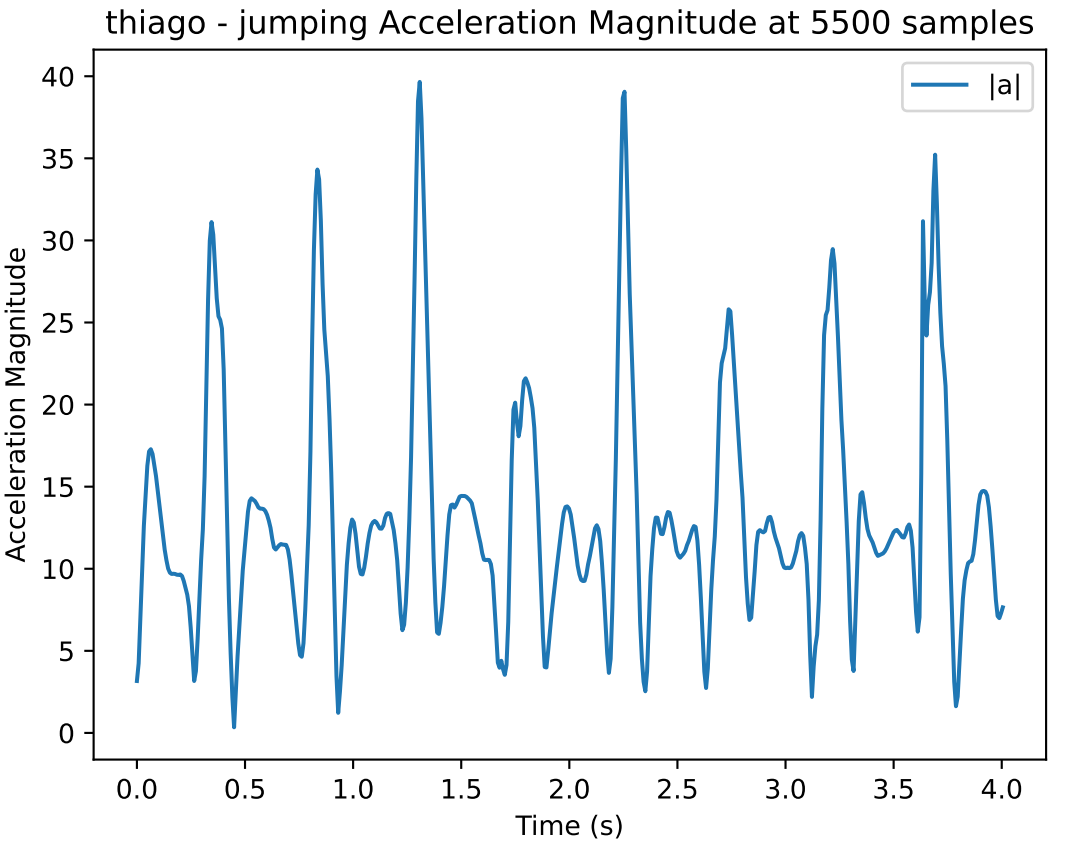

thiago - jumping Acceleration at 1000 samples
Acceleration
Time (s)
10
5
0
−5
−10
−15
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
ax
ay
az

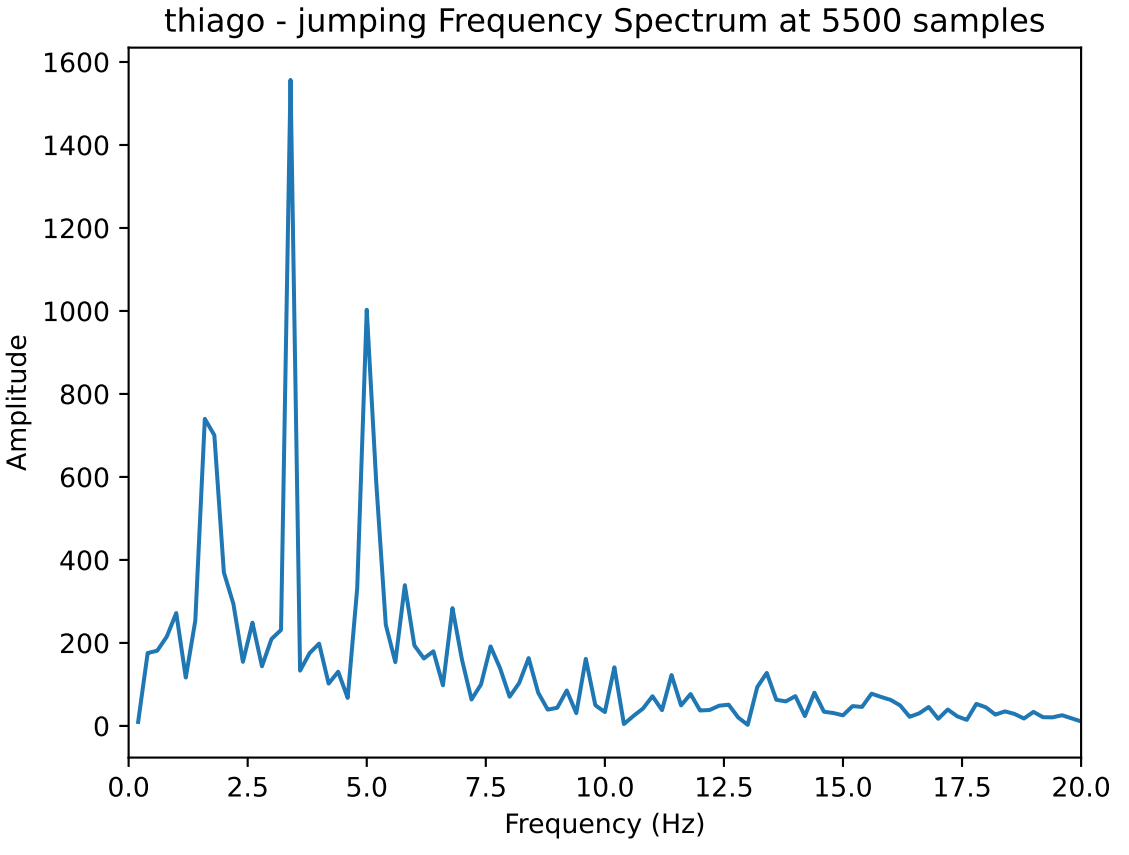

thiago - jumping Acceleration Magnitude at 1000 samples
|a|
Acceleration Magnitude
Time (s)
0
5
10
15
20
25
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

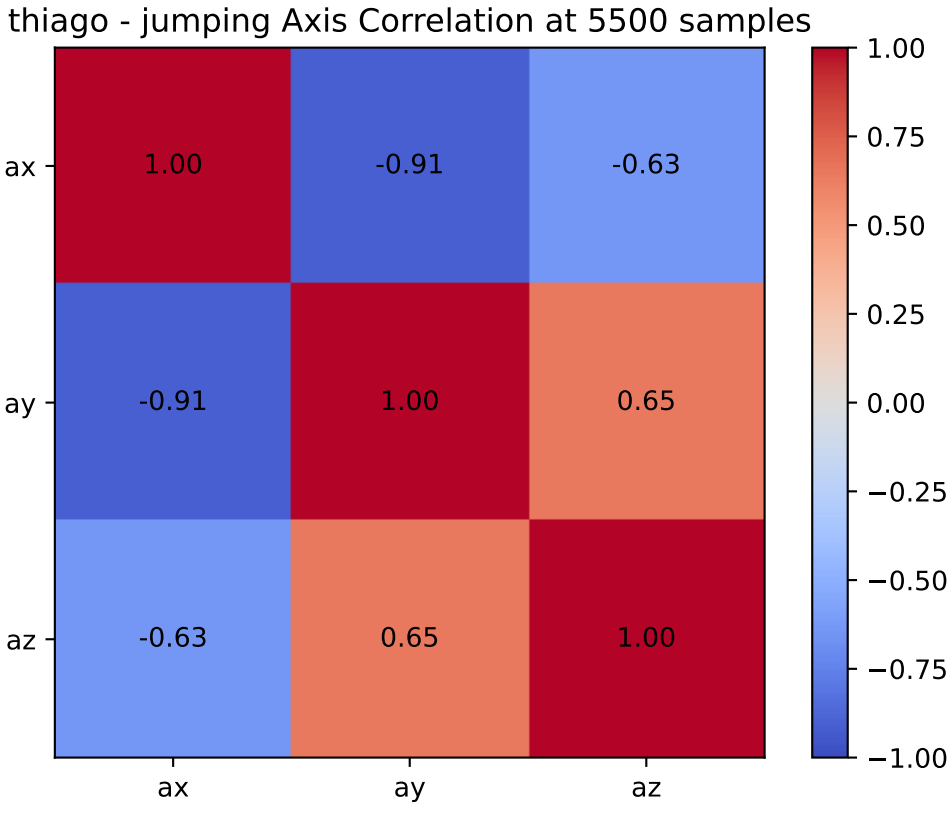

thiago - jumping Frequency Spectrum at 1000 samples
Amplitude
Frequency (Hz)
1000
800
600
400
200
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

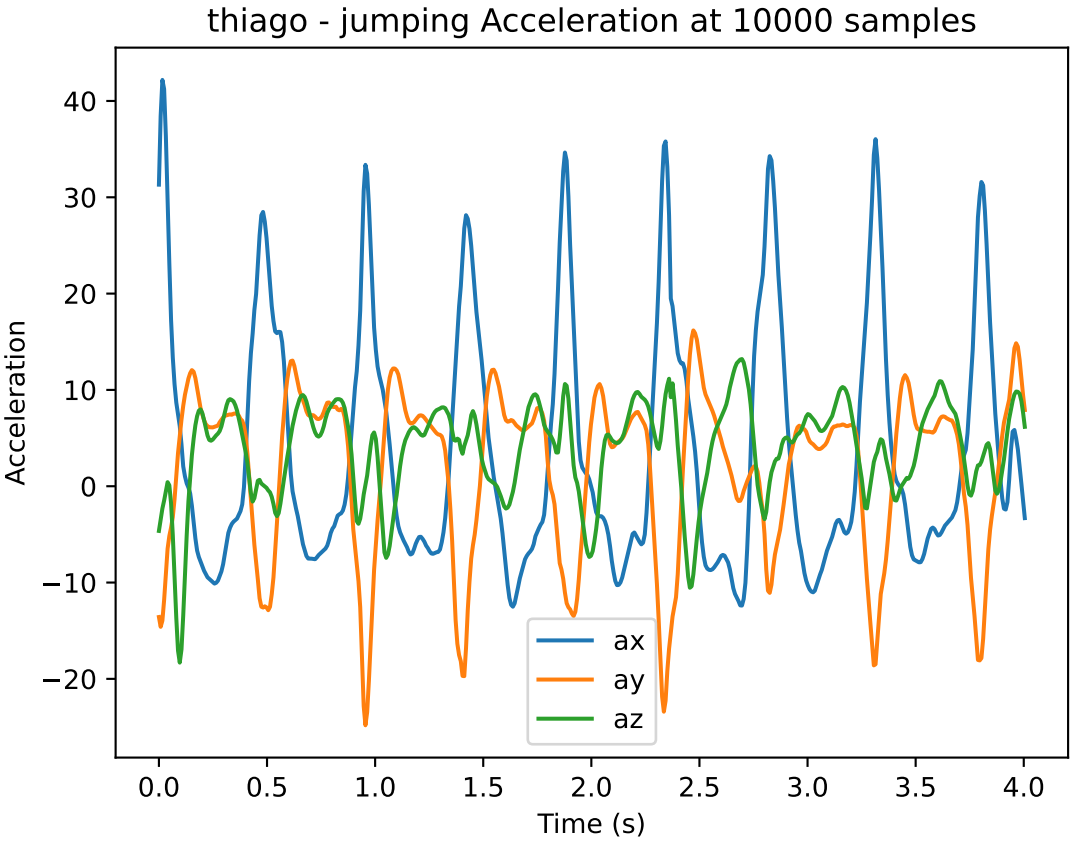

thiago - jumping Axis Correlation at 1000 samples
ax
ay
az
1.00
0.94
-0.64
0.94
1.00
-0.66
-0.64
-0.66
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

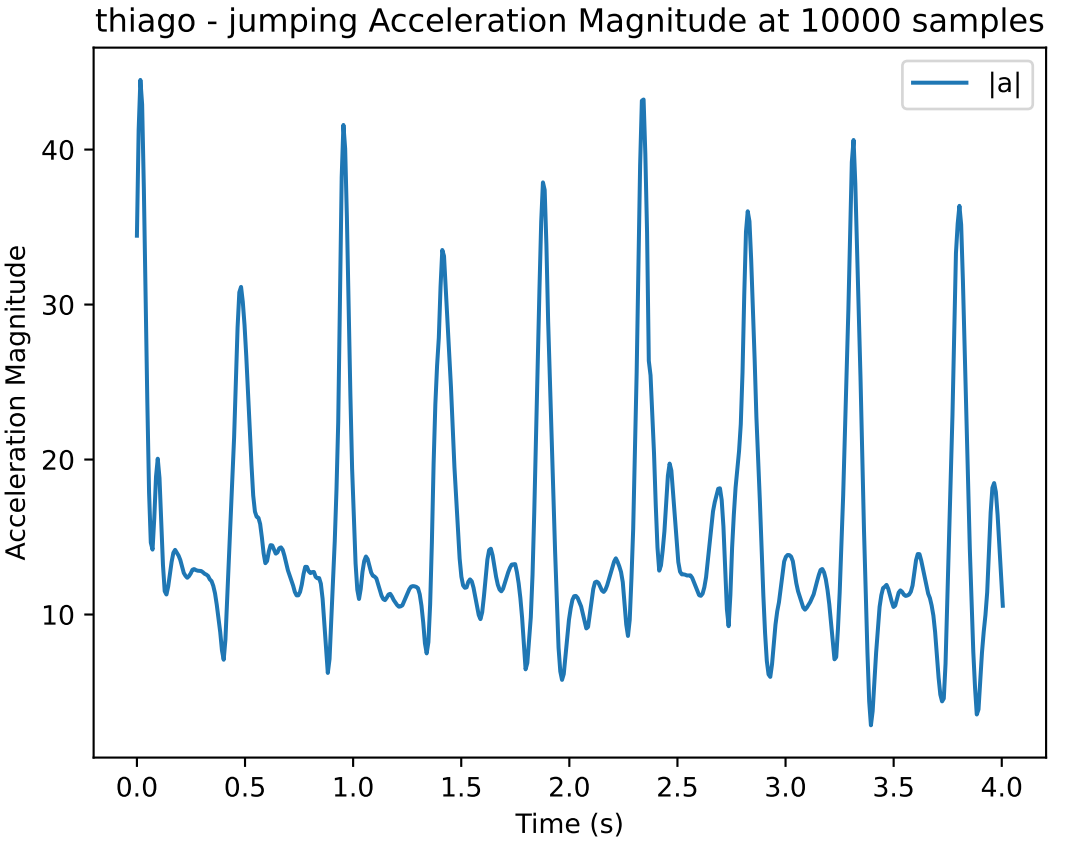

thiago - jumping Acceleration at 5500 samples
Acceleration
Time (s)
ax
ay
az
30
20
10
0
−10
−20
−30
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

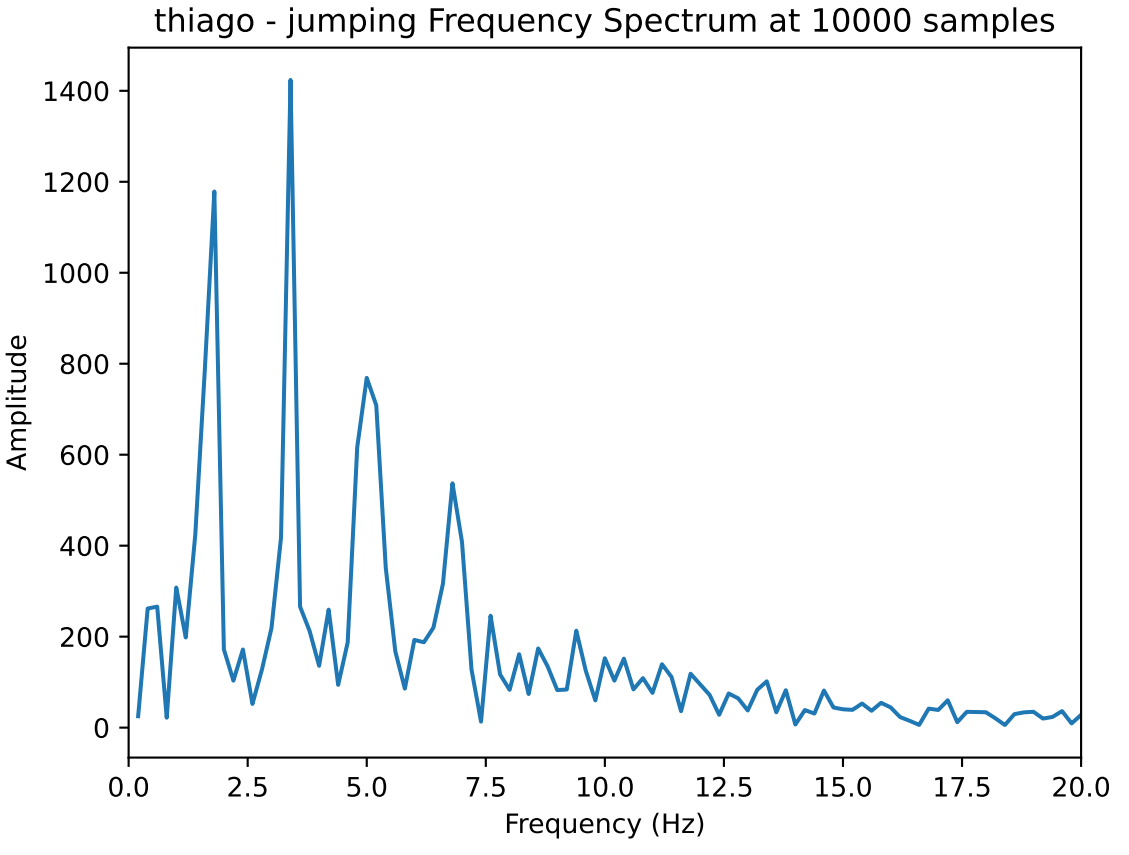

thiago - jumping Acceleration Magnitude at 5500 samples
|a|
Acceleration Magnitude
Time (s)
0
5
10
15
20
25
30
35
40
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

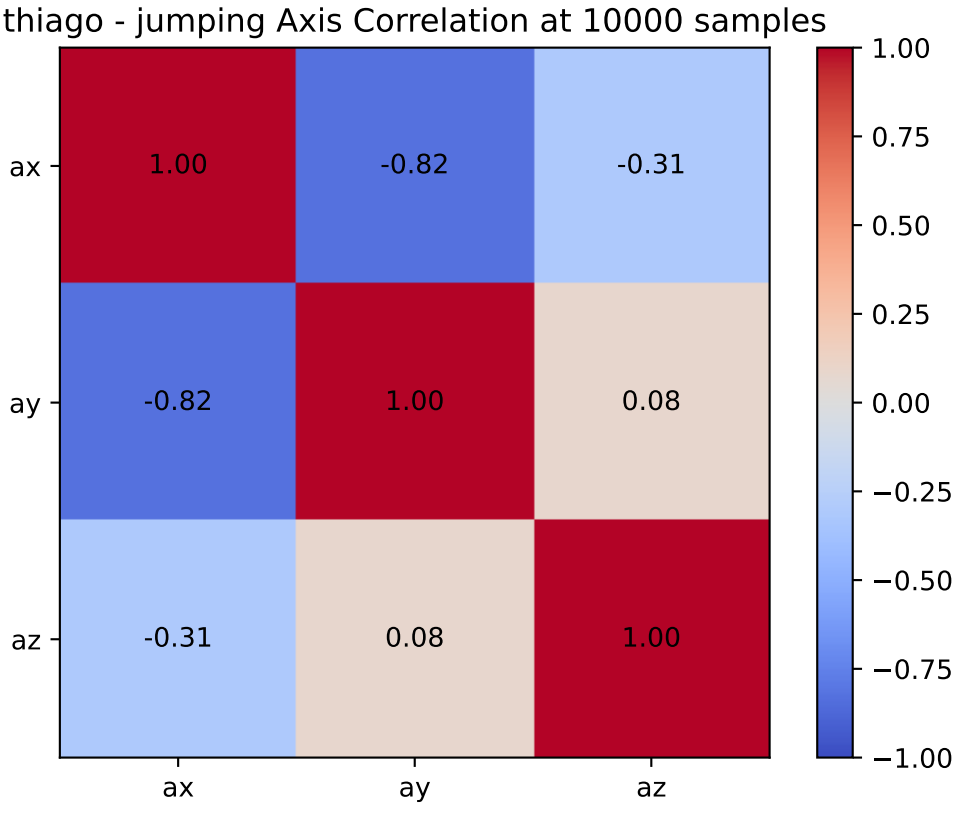

thiago - jumping Frequency Spectrum at 5500 samples
Amplitude
Frequency (Hz)
1600
1400
1200
1000
800
600
400
200
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

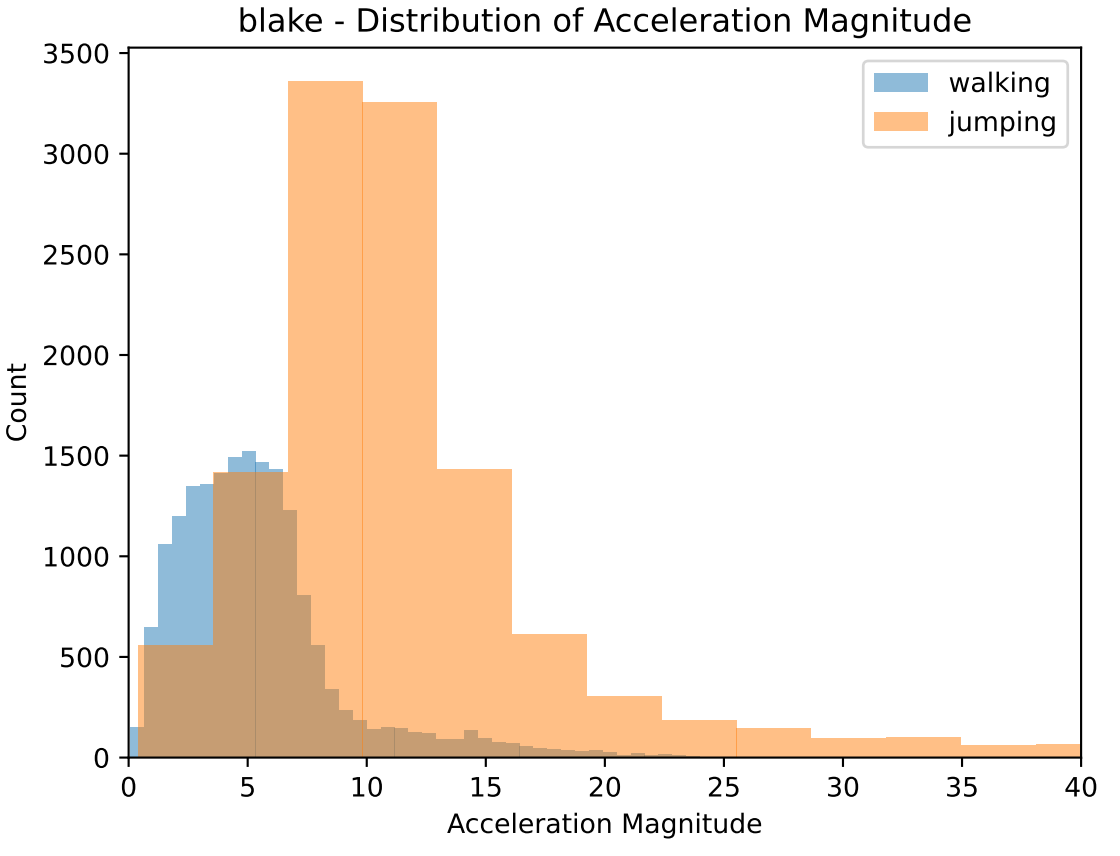

thiago - jumping Axis Correlation at 5500 samples
ax
ay
az
1.00
-0.91
-0.63
-0.91
1.00
0.65
-0.63
0.65
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

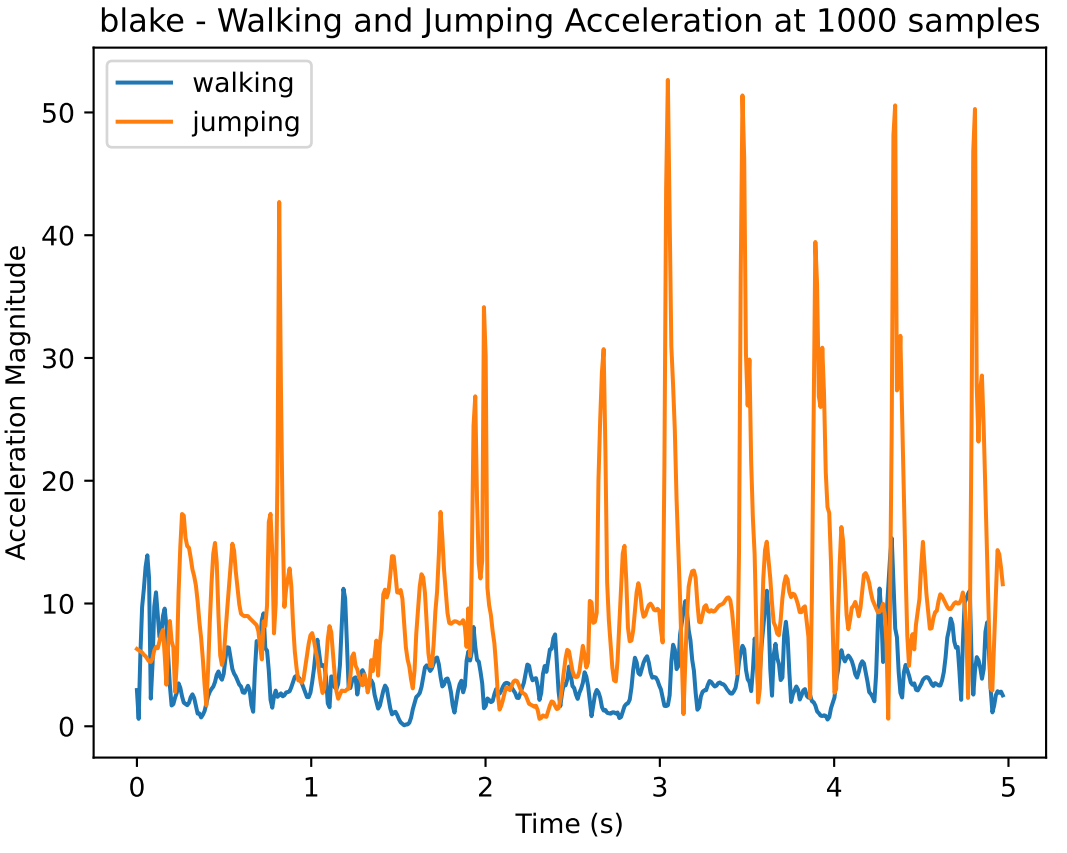

thiago - jumping Acceleration at 10000 samples
Acceleration
Time (s)
ax
ay
az
40
30
20
10
0
−10
−20
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

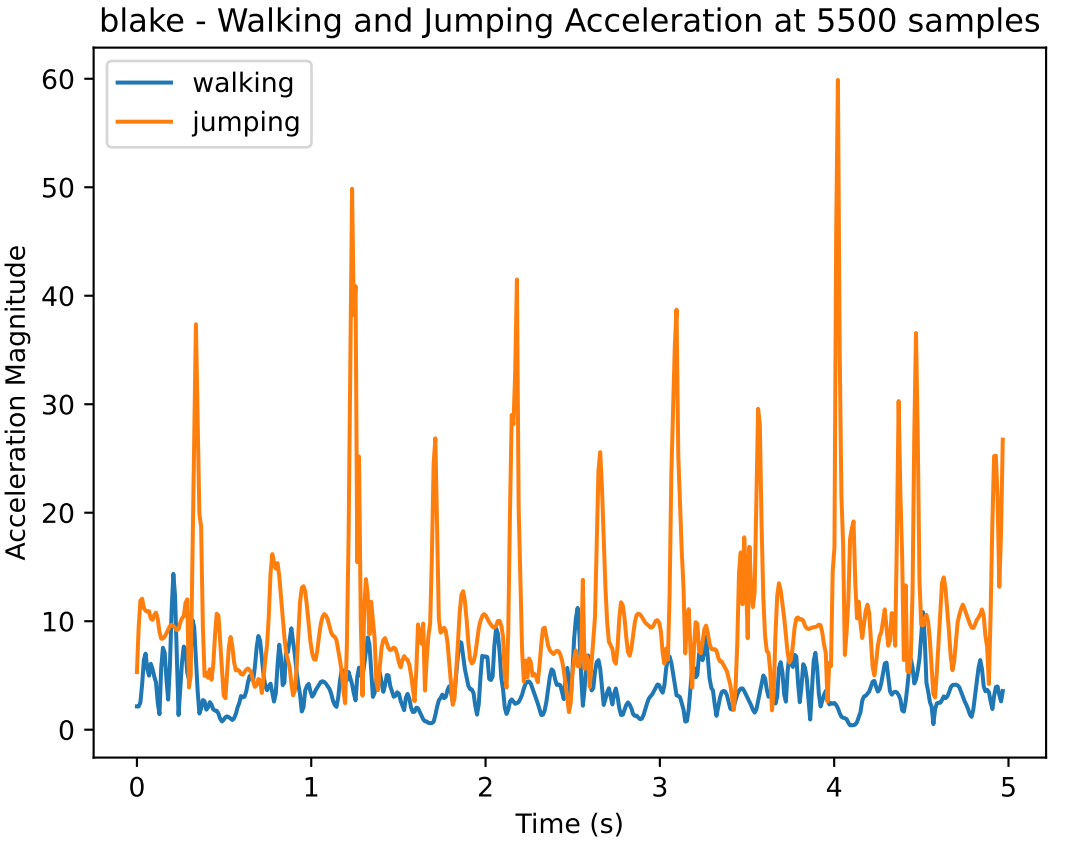

thiago - jumping Acceleration Magnitude at 10000 samples
|a|
Acceleration Magnitude
Time (s)
10
20
30
40
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

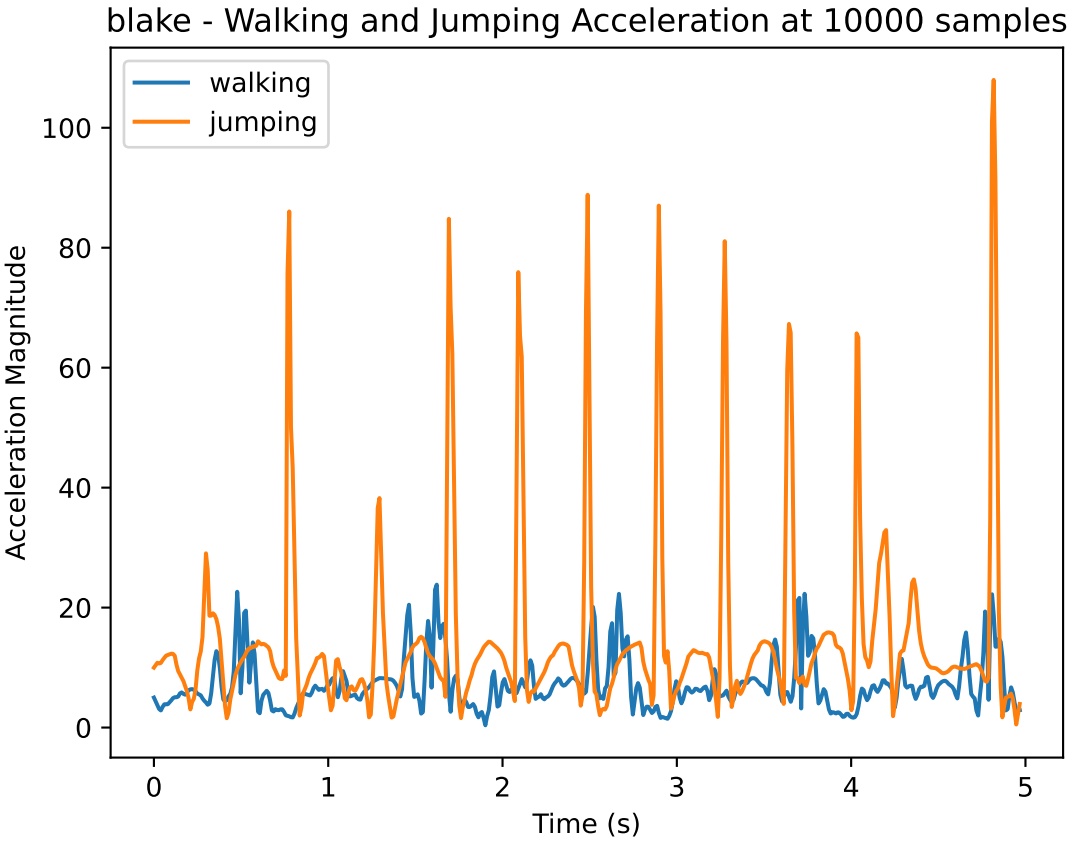

thiago - jumping Frequency Spectrum at 10000 samples
Amplitude
Frequency (Hz)
0
200
400
600
800
1000
1200
1400
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

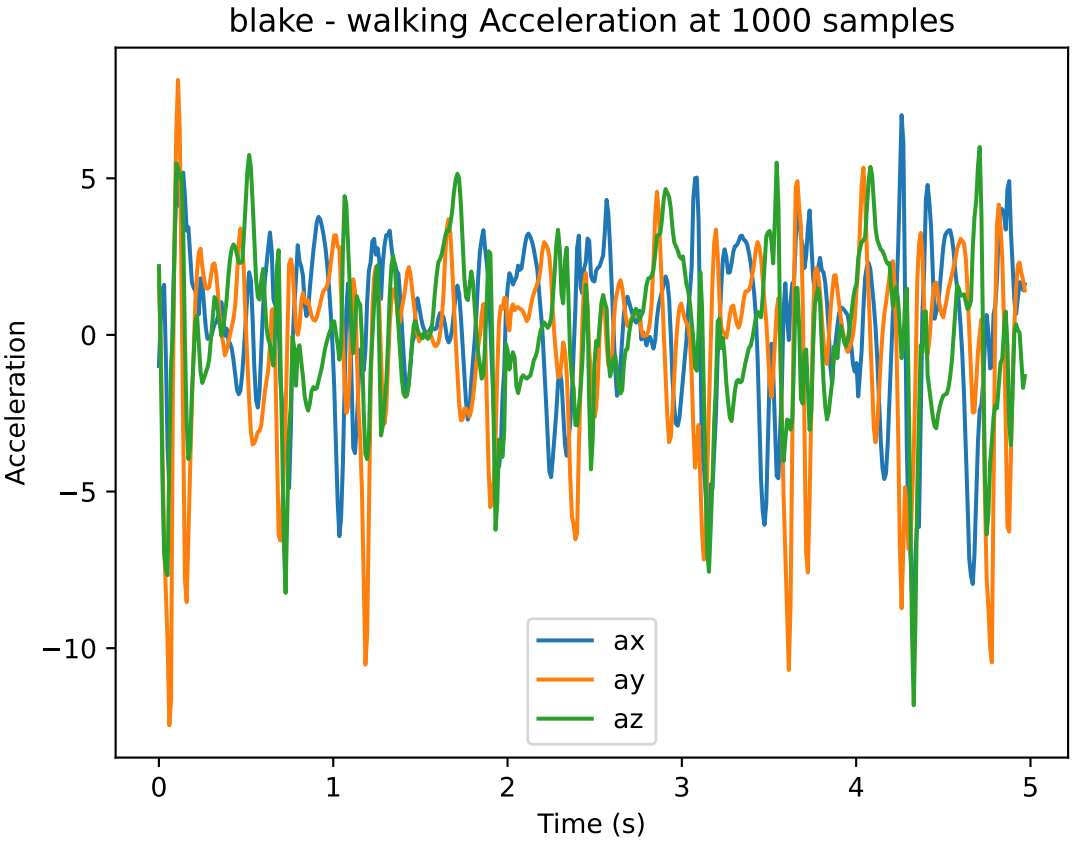

thiago - jumping Axis Correlation at 10000 samples
ax
ay
az
1.00
-0.82
-0.31
-0.82
1.00
0.08
-0.31
0.08
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

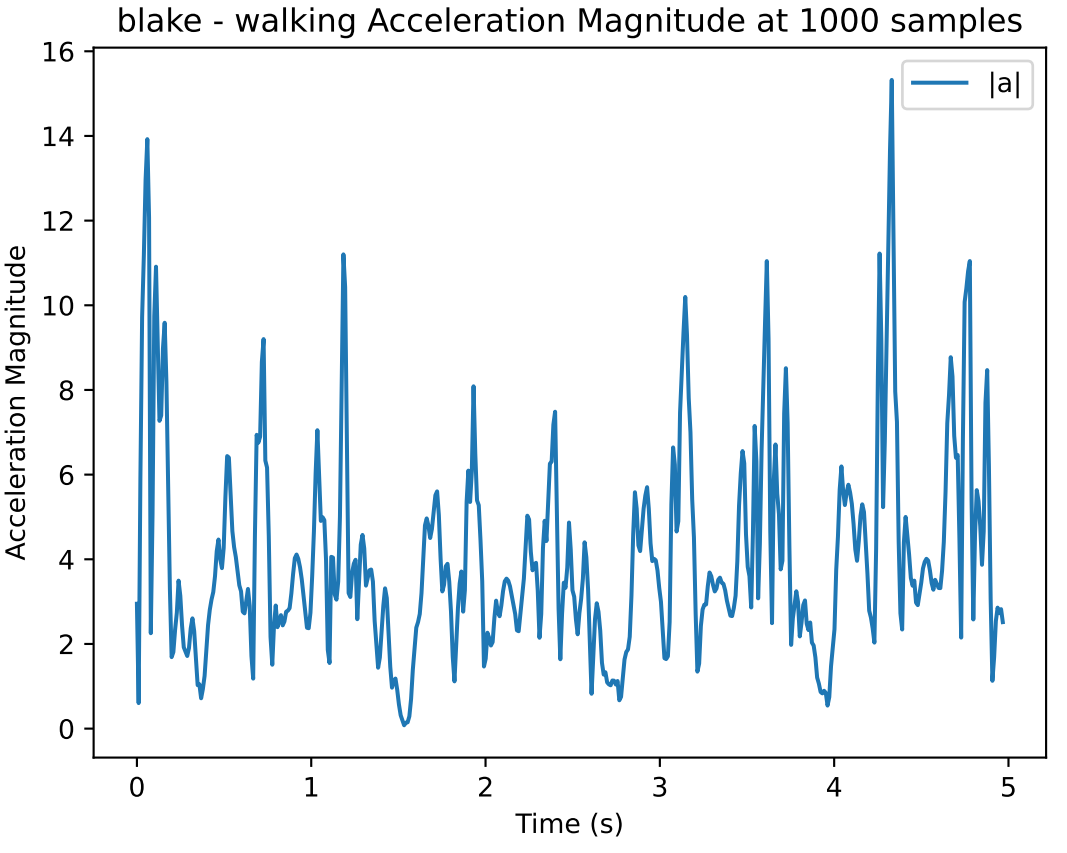

blake - Distribution of Acceleration Magnitude
walking
jumping
Count
Acceleration Magnitude
0
500
1000
1500
2000
2500
3000
3500
0
5
10
15
20
25
30
35
40

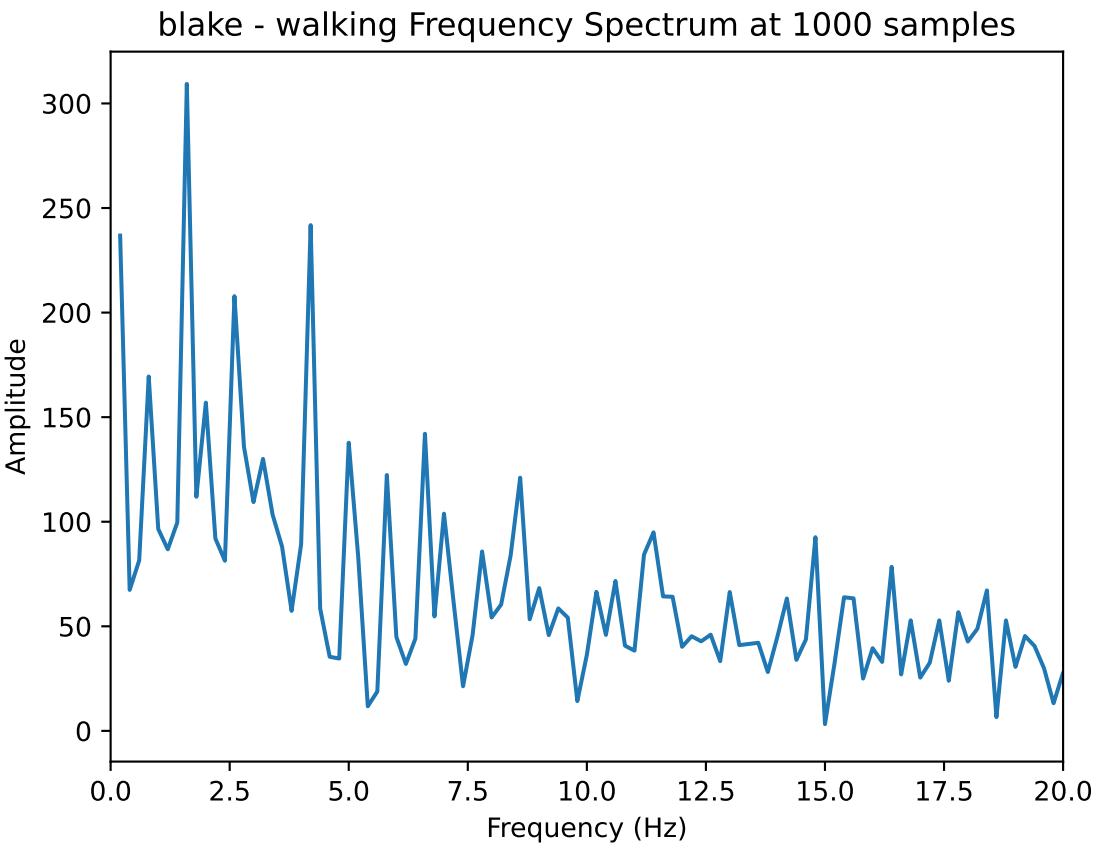

blake - Walking and Jumping Acceleration at 1000 samples
walking
jumping
Acceleration Magnitude
Time (s)
0
10
20
30
40
50
0
1
2
3
4
5

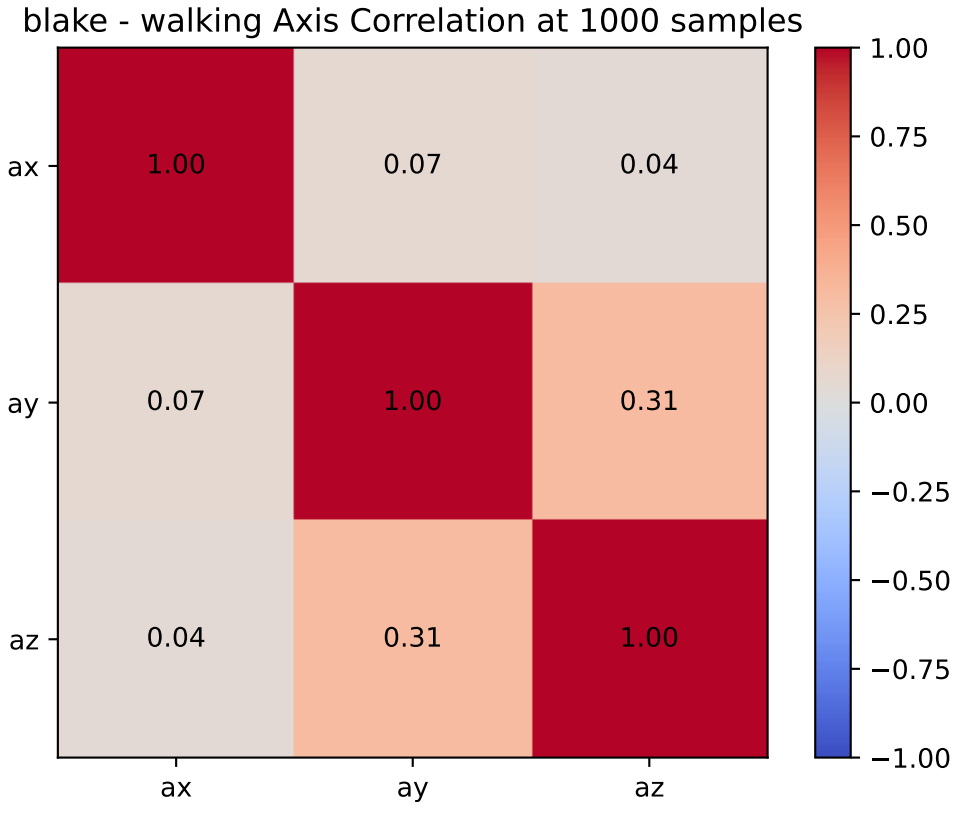

blake - Walking and Jumping Acceleration at 5500 samples
walking
jumping
Acceleration Magnitude
Time (s)
0
10
20
30
40
50
60
0
1
2
3
4
5

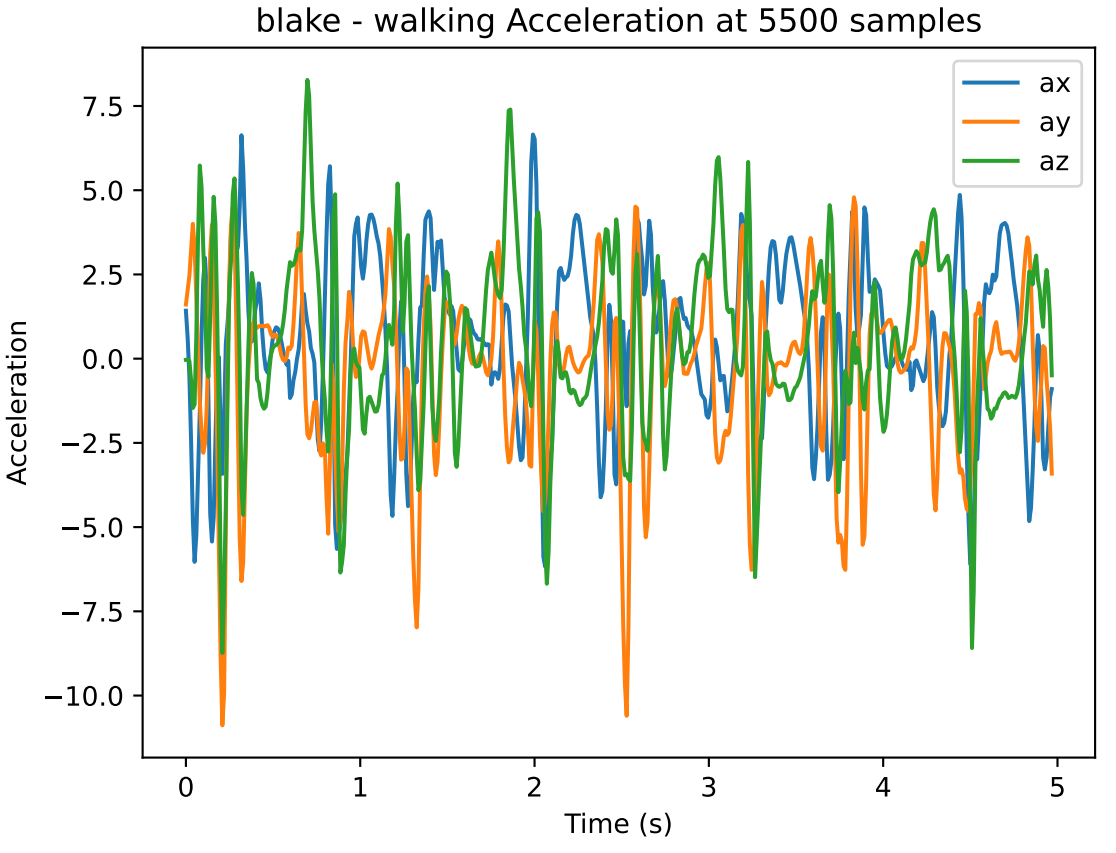

blake - Walking and Jumping Acceleration at 10000 samples
walking
jumping
Acceleration Magnitude
Time (s)
0
20
40
60
80
100
0
1
2
3
4
5

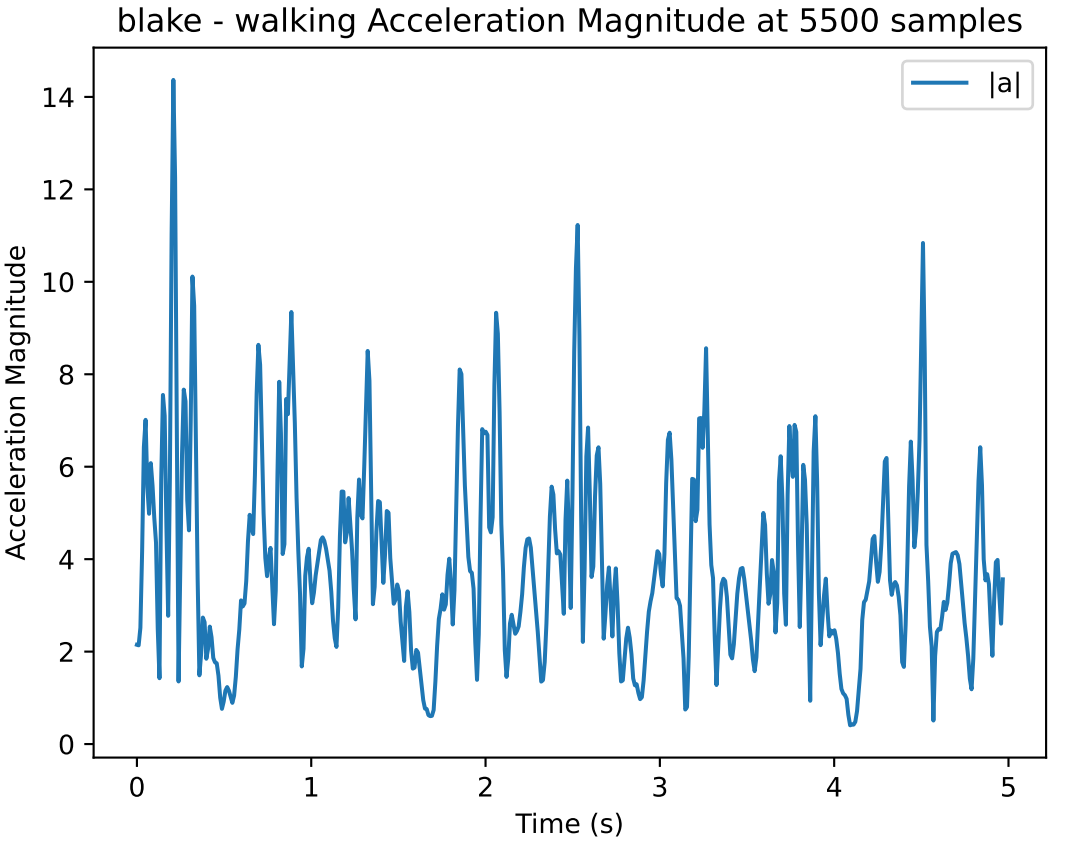

blake - walking Acceleration at 1000 samples
Acceleration
Time (s)
5
0
−5
−10
0
1
2
3
4
5
ax
ay
az

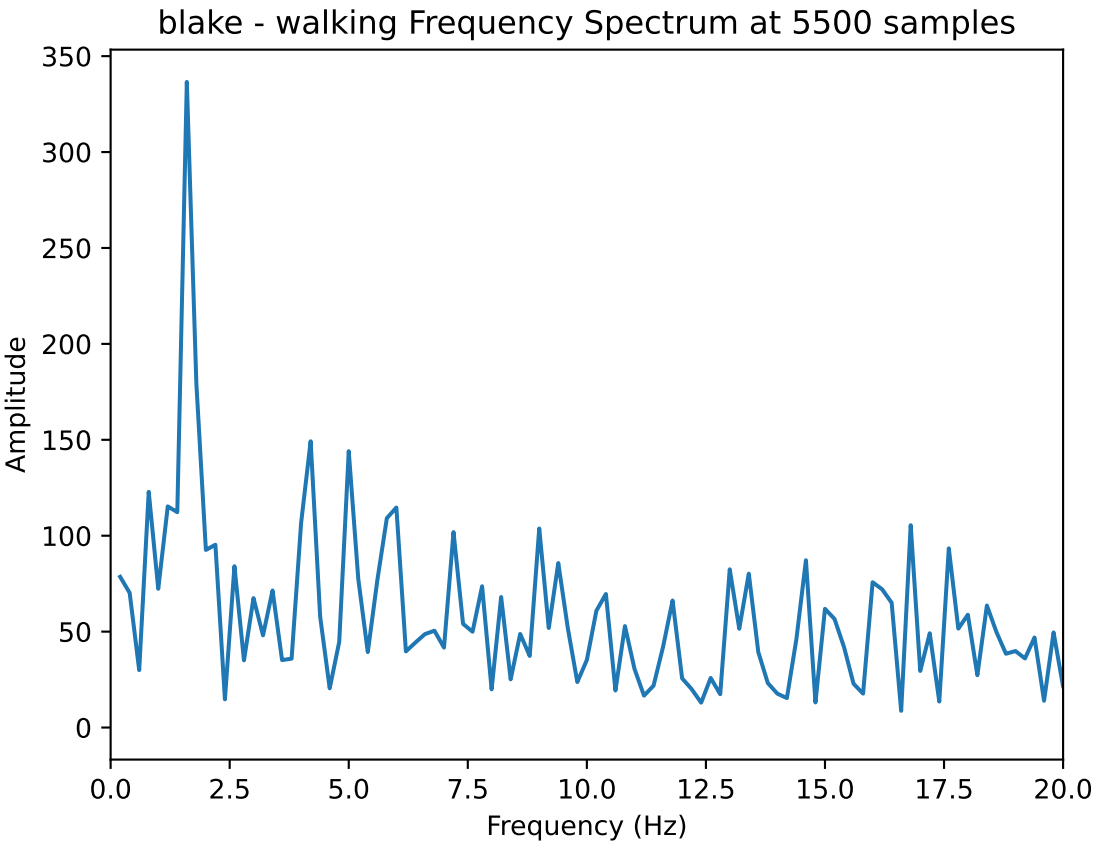

blake - walking Acceleration Magnitude at 1000 samples
|a|
Acceleration Magnitude
Time (s)
0
2
4
6
8
10
12
14
16
0
1
2
3
4
5

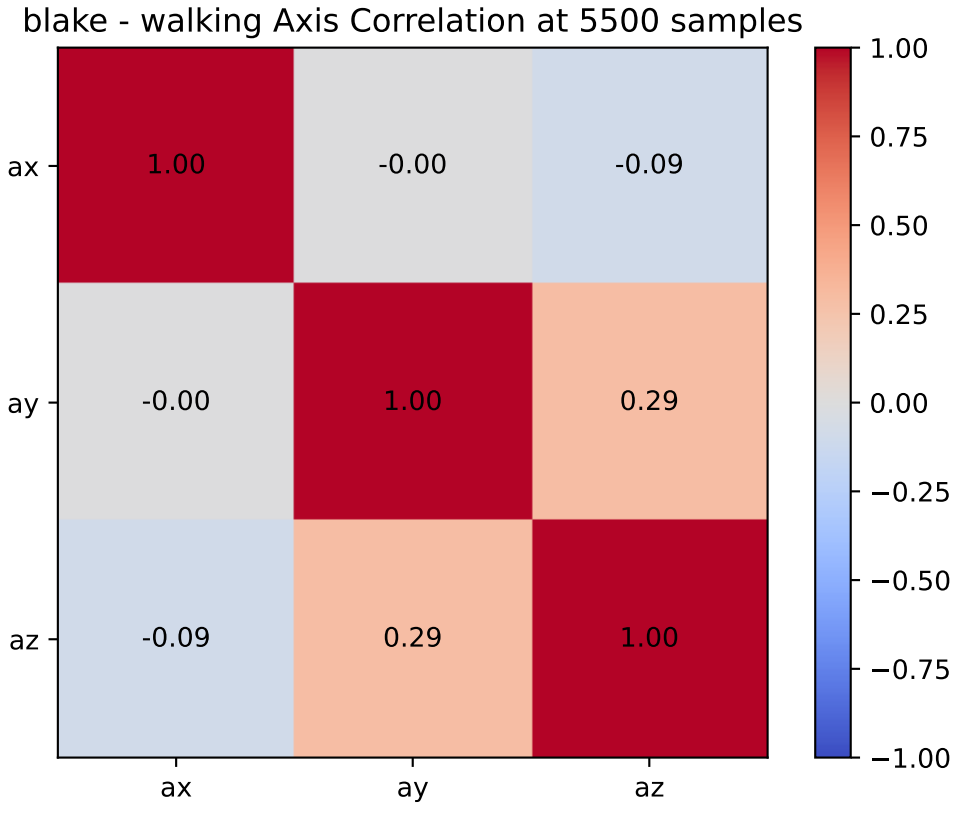

blake - walking Frequency Spectrum at 1000 samples
Amplitude
Frequency (Hz)
0
50
100
150
200
250
300
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

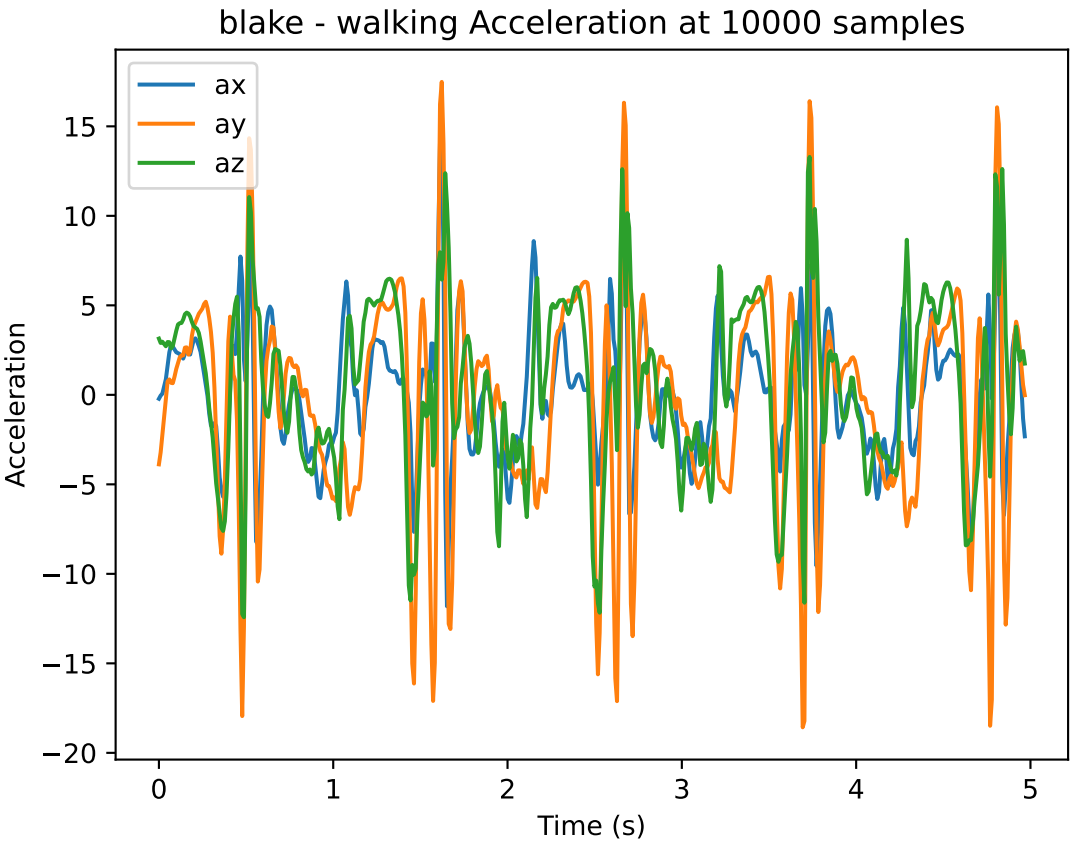

blake - walking Axis Correlation at 1000 samples
ax
ay
az
1.00
0.07
0.04
0.07
1.00
0.31
0.04
0.31
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

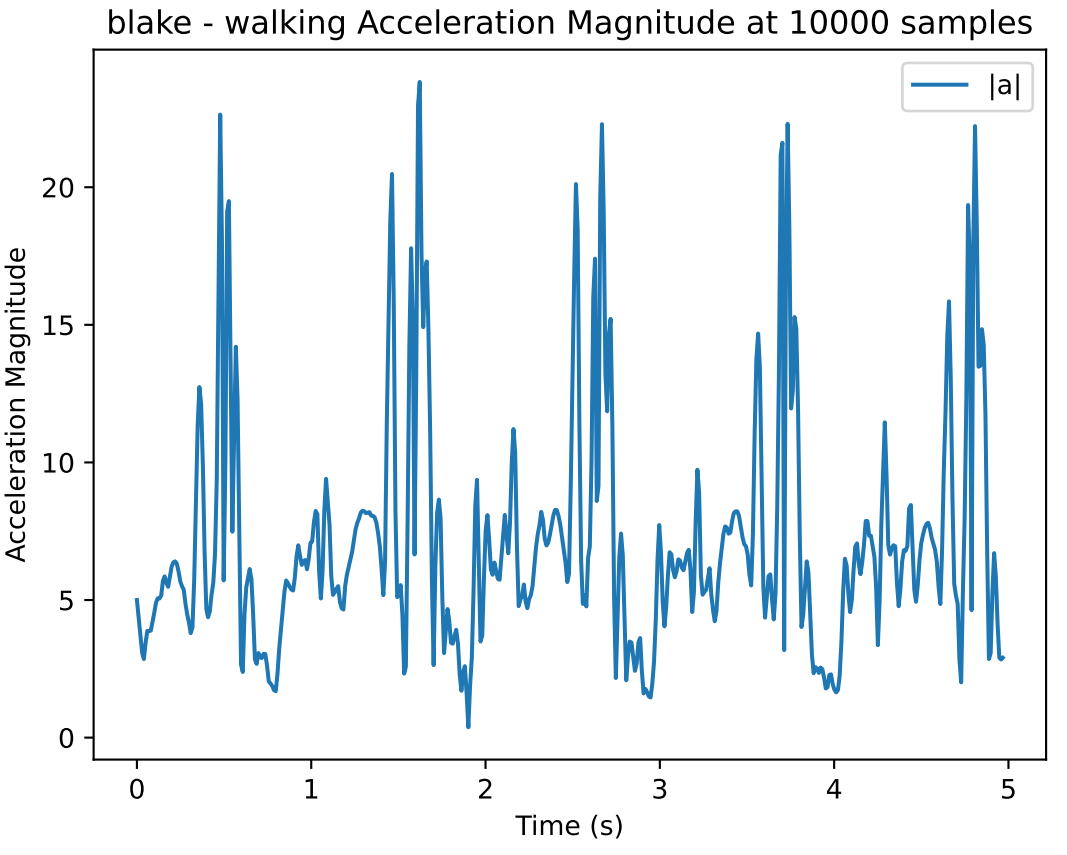

blake - walking Acceleration at 5500 samples
ax
ay
az
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
Acceleration
0
1
2
3
4
5
Time (s)

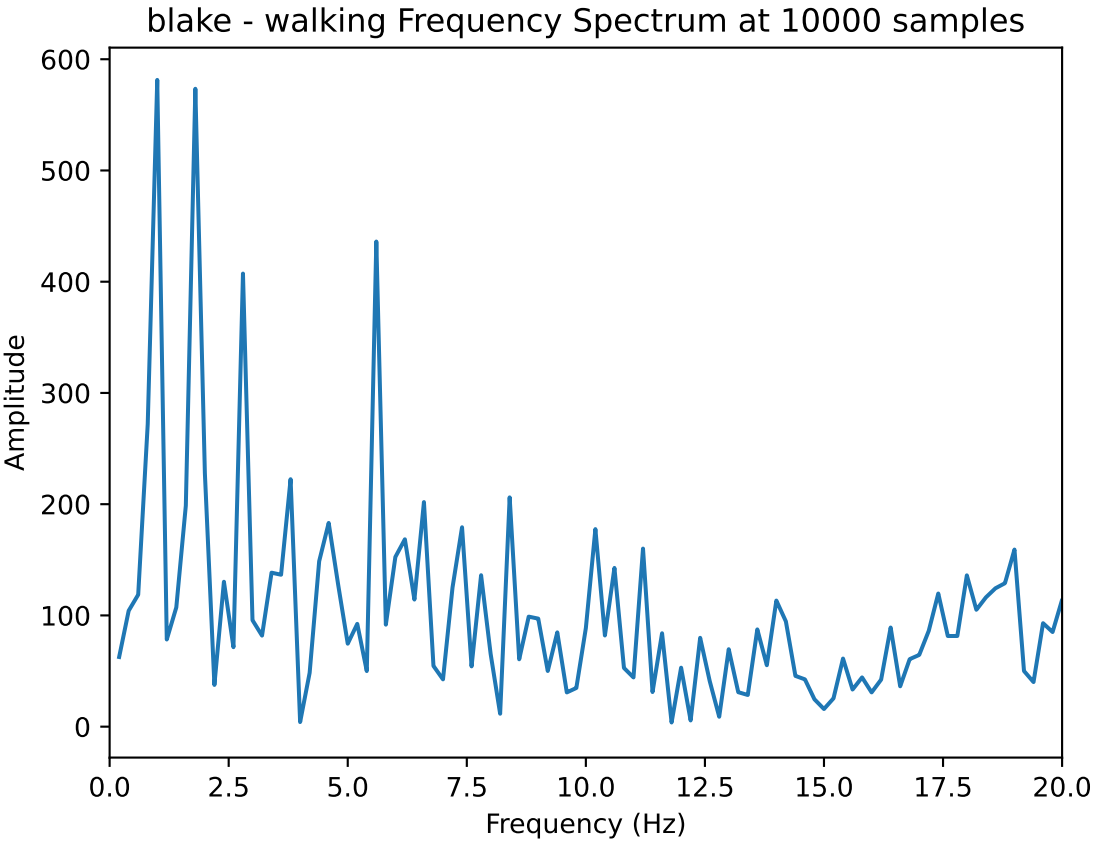

blake - walking Acceleration Magnitude at 5500 samples
|a|
Acceleration Magnitude
Time (s)
0
2
4
6
8
10
12
14
0
1
2
3
4
5

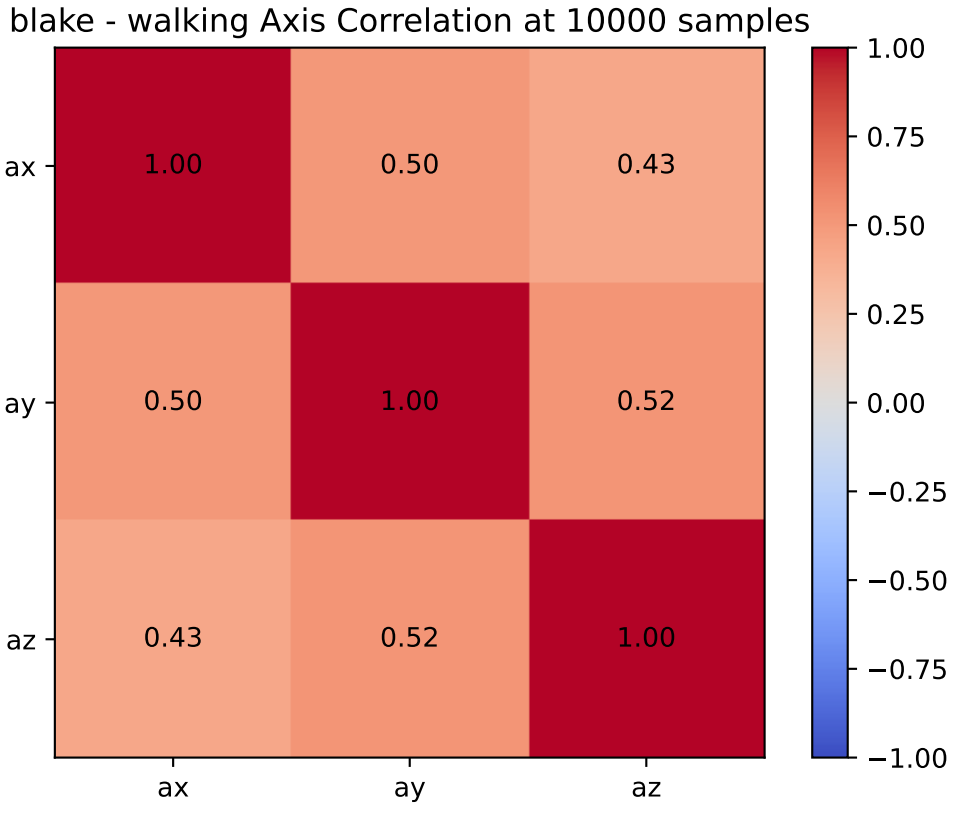

blake - walking Frequency Spectrum at 5500 samples
Amplitude
Frequency (Hz)
350
300
250
200
150
100
50
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

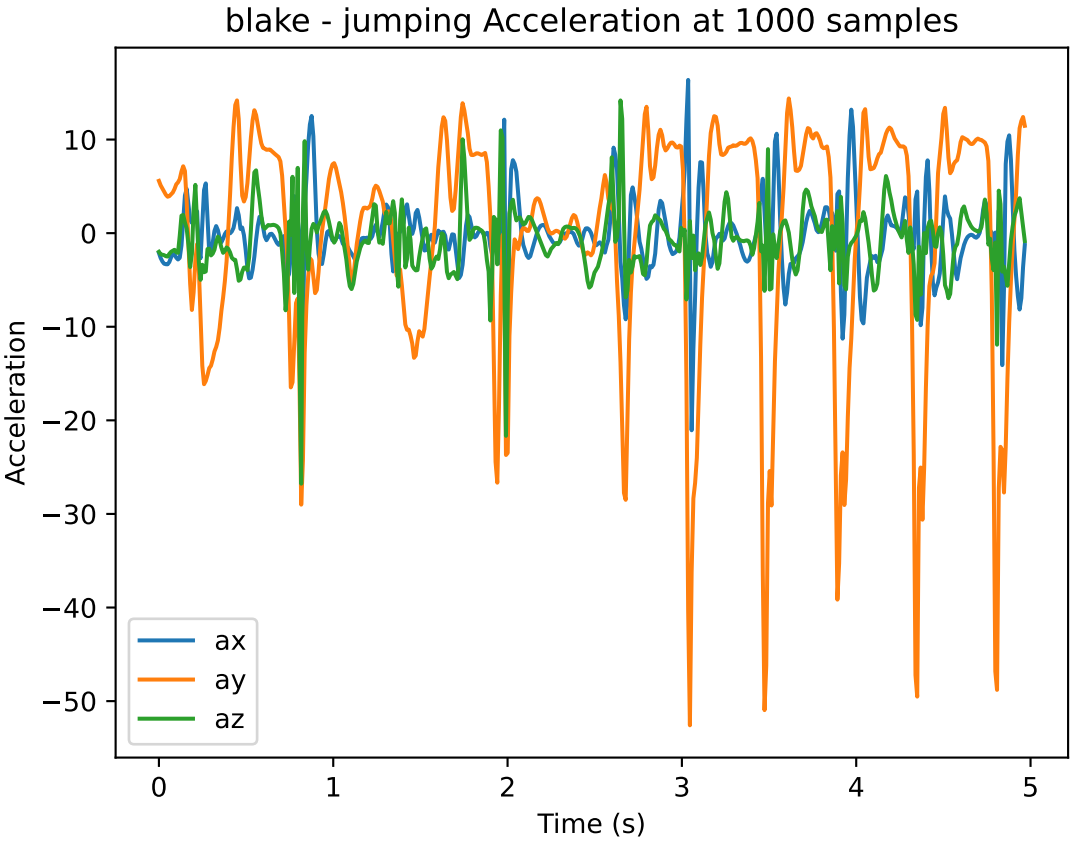

blake - walking Axis Correlation at 5500 samples
ax
ay
az
1.00
-0.00
-0.09
-0.00
1.00
0.29
-0.09
0.29
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

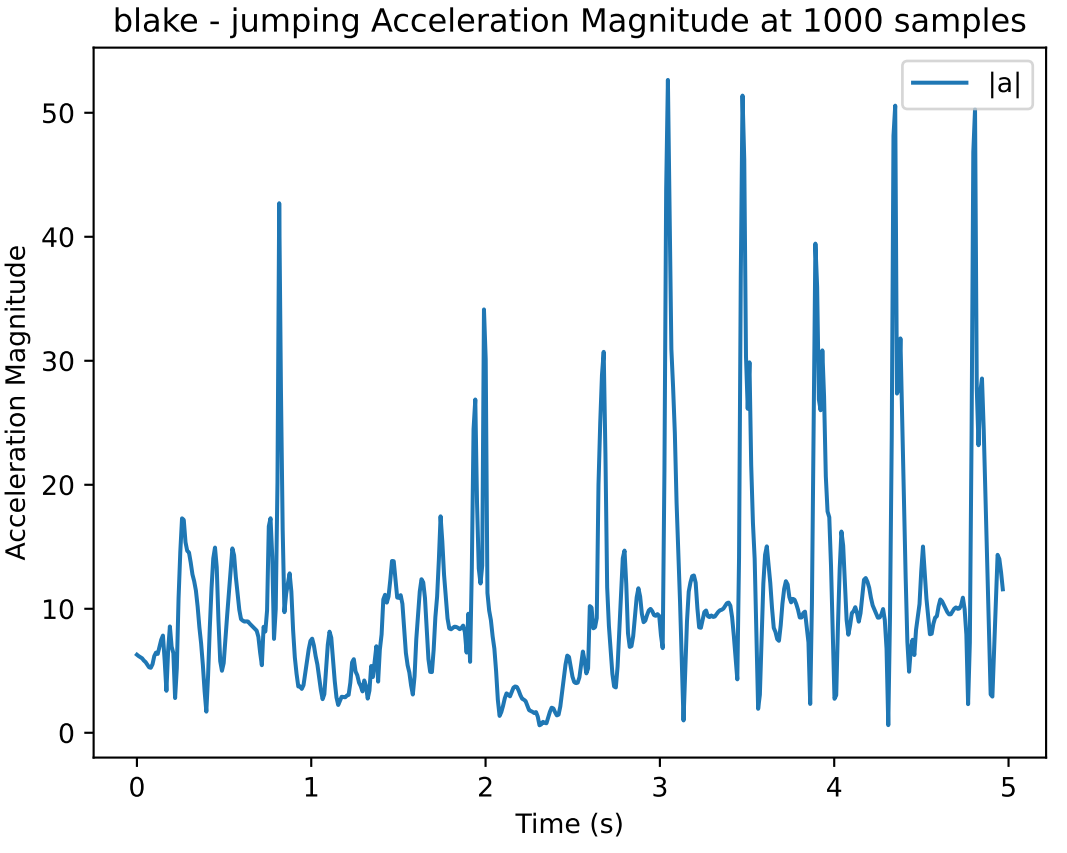

blake - walking Acceleration at 10000 samples
ax
ay
az
Acceleration
Time (s)
15
10
5
0
−5
−10
−15
−20
0
1
2
3
4
5

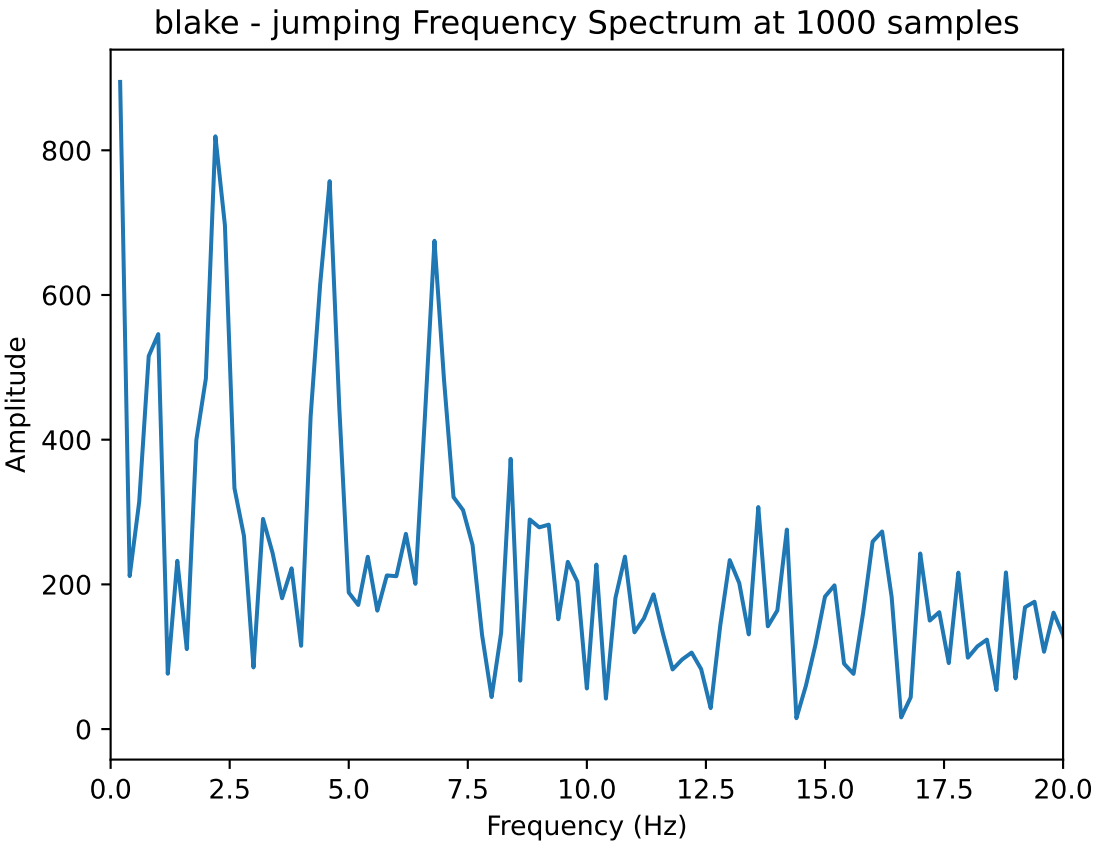

blake - walking Acceleration Magnitude at 10000 samples
|a|
Acceleration Magnitude
Time (s)
0
5
10
15
20
0
1
2
3
4
5

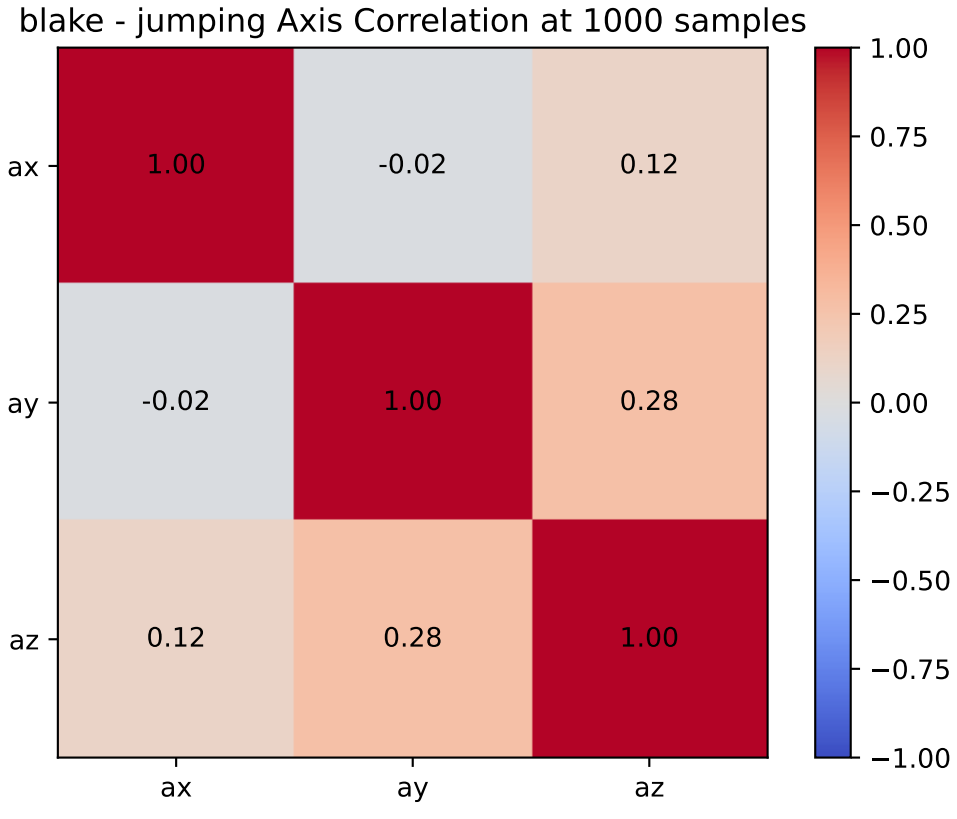

blake - walking Frequency Spectrum at 10000 samples
Amplitude
Frequency (Hz)
600
500
400
300
200
100
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

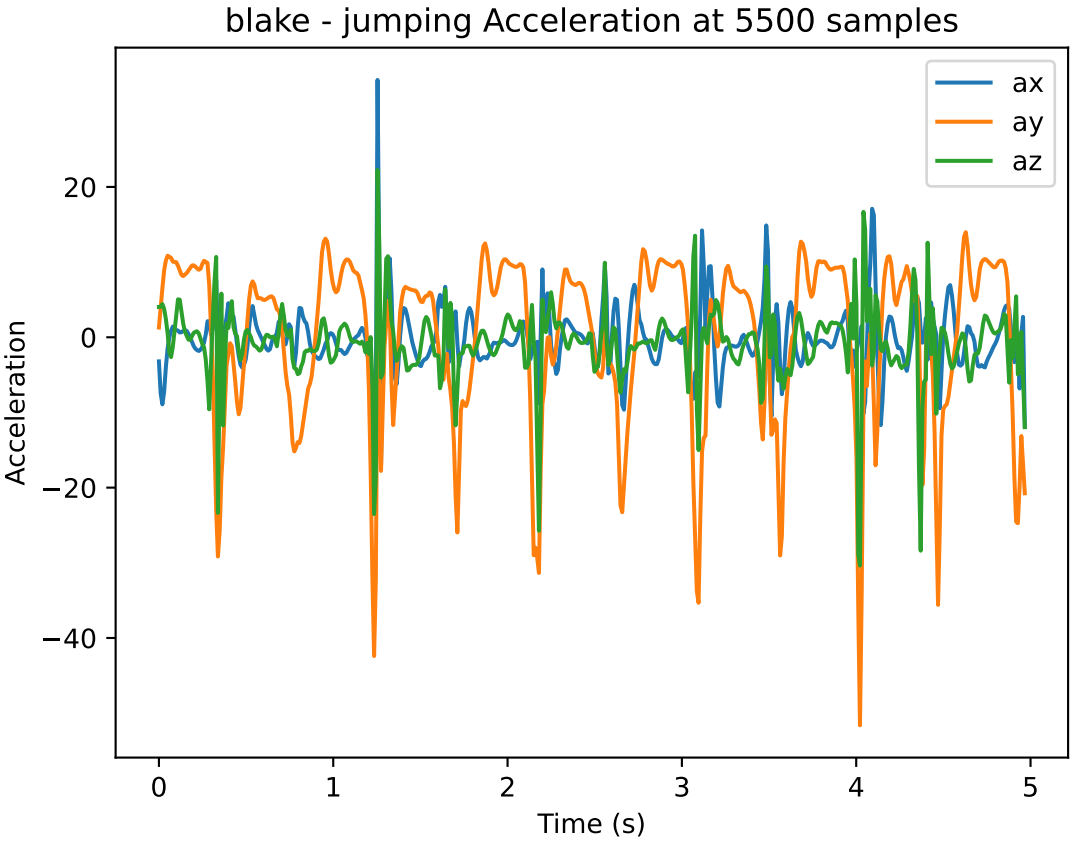

blake - walking Axis Correlation at 10000 samples
ax
ay
az
1.00
0.50
0.43
0.50
1.00
0.52
0.43
0.52
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

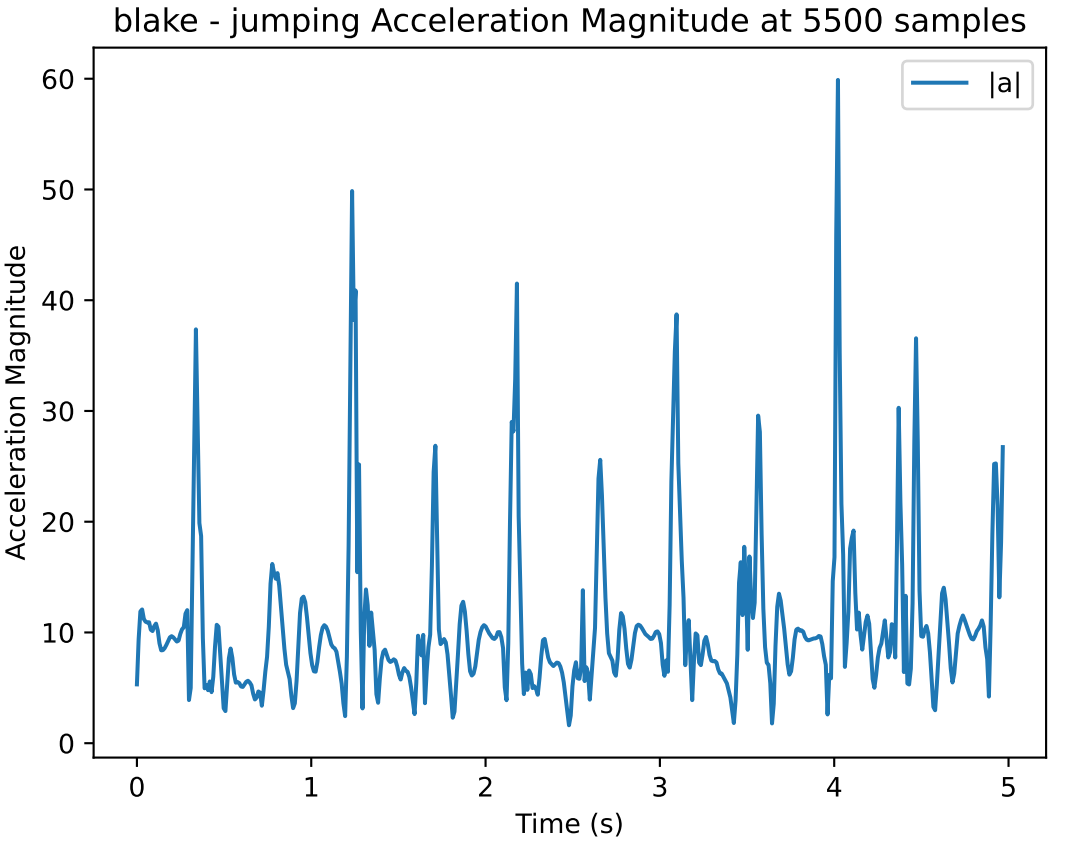

blake - jumping Acceleration at 1000 samples
Acceleration
Time (s)
10
0
−10
−20
−30
−40
−50
0
1
2
3
4
5
ax
ay
az

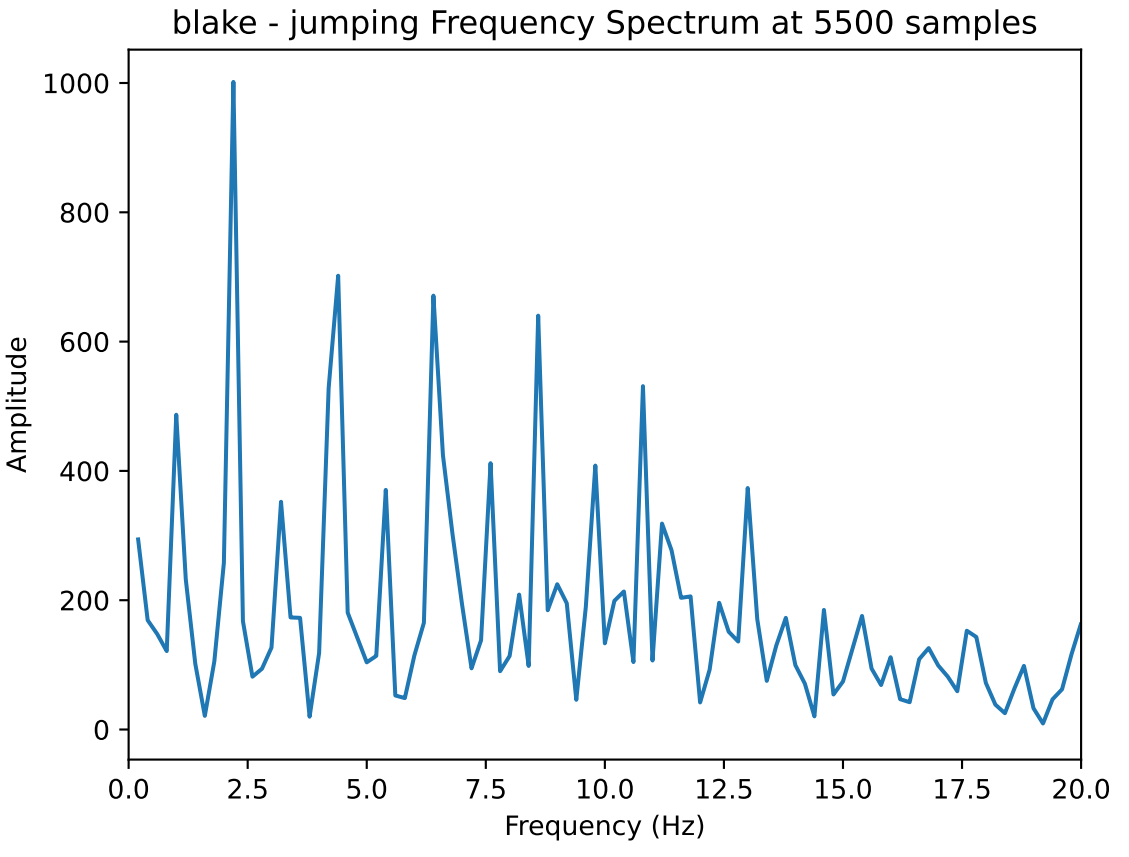

blake - jumping Acceleration Magnitude at 1000 samples
|a|
Acceleration Magnitude
Time (s)
0
10
20
30
40
50
0
1
2
3
4
5

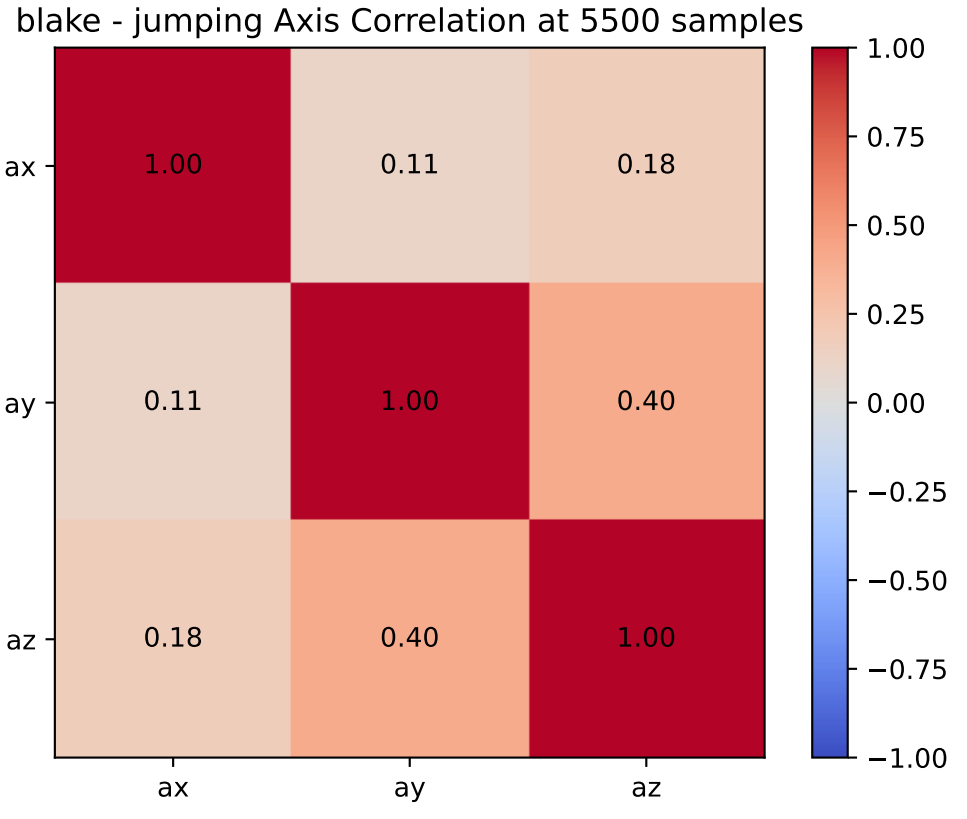

blake - jumping Frequency Spectrum at 1000 samples
Amplitude
0
200
400
600
800
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Frequency (Hz)

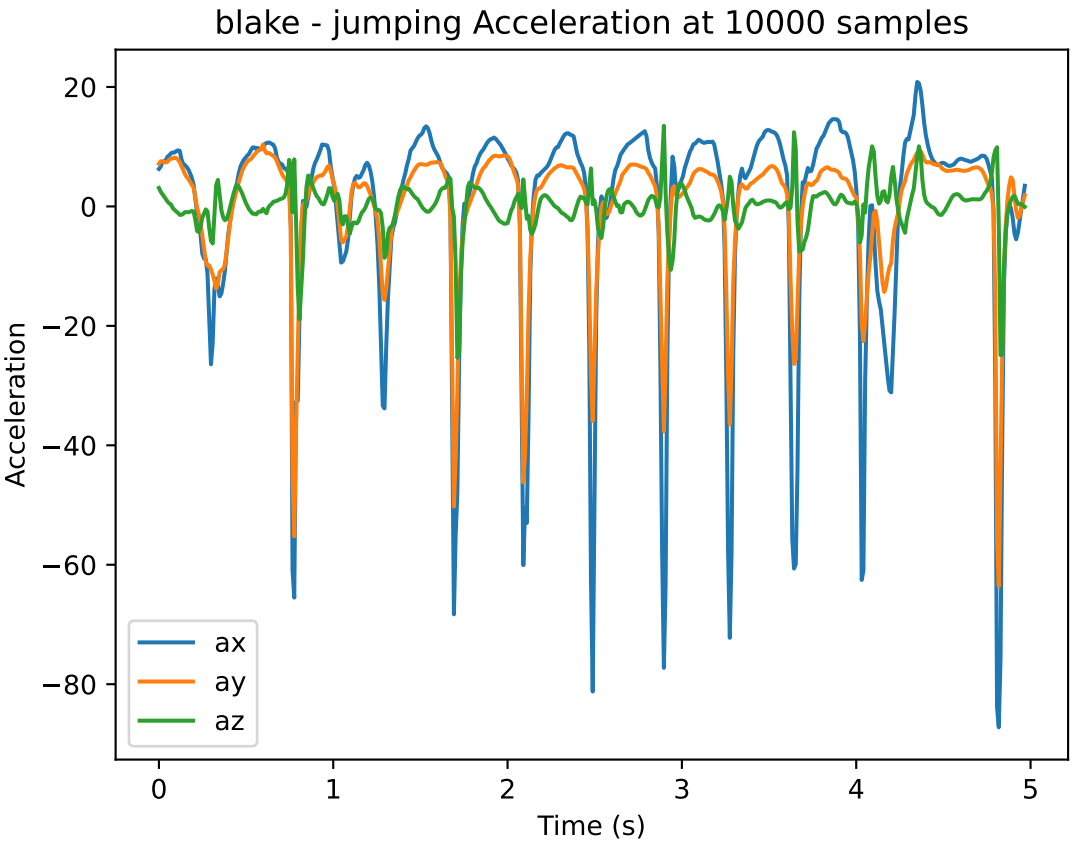

blake - jumping Axis Correlation at 1000 samples
ax
ay
az
1.00
-0.02
0.12
-0.02
1.00
0.28
0.12
0.28
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

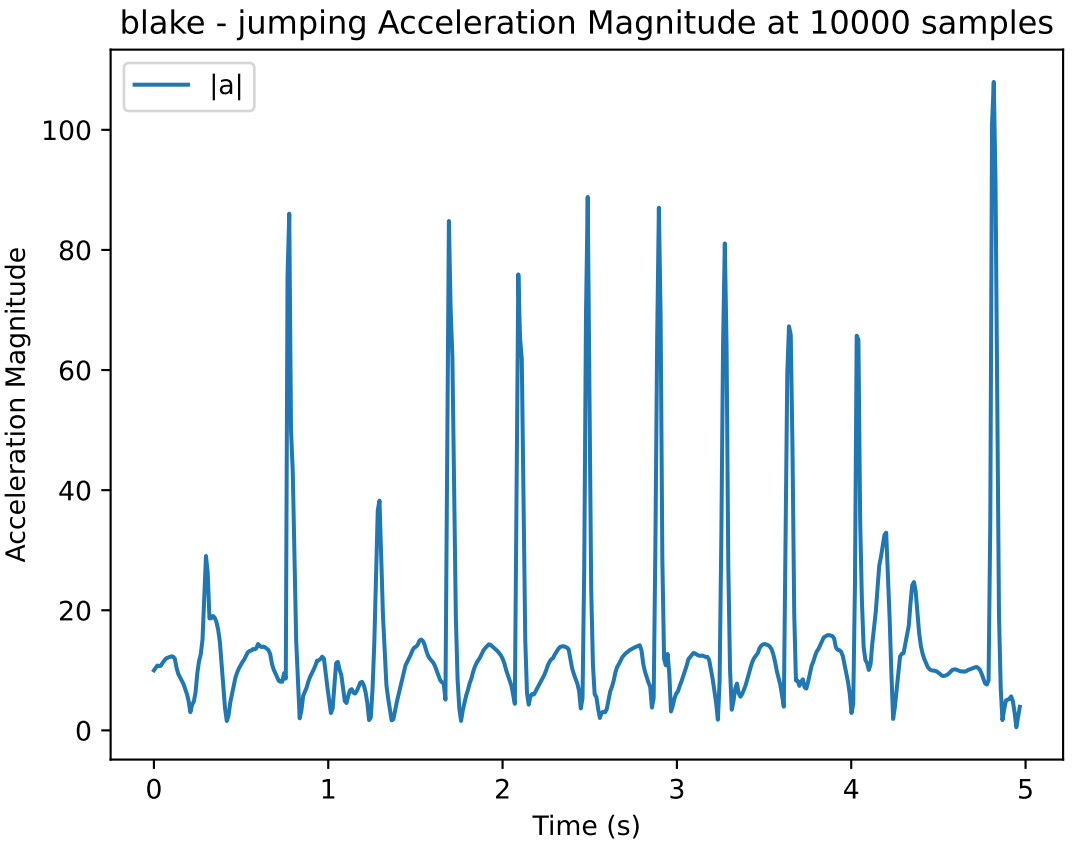

blake - jumping Acceleration at 5500 samples
ax
ay
az
20
0
−20
−40
Acceleration
0
1
2
3
4
5
Time (s)

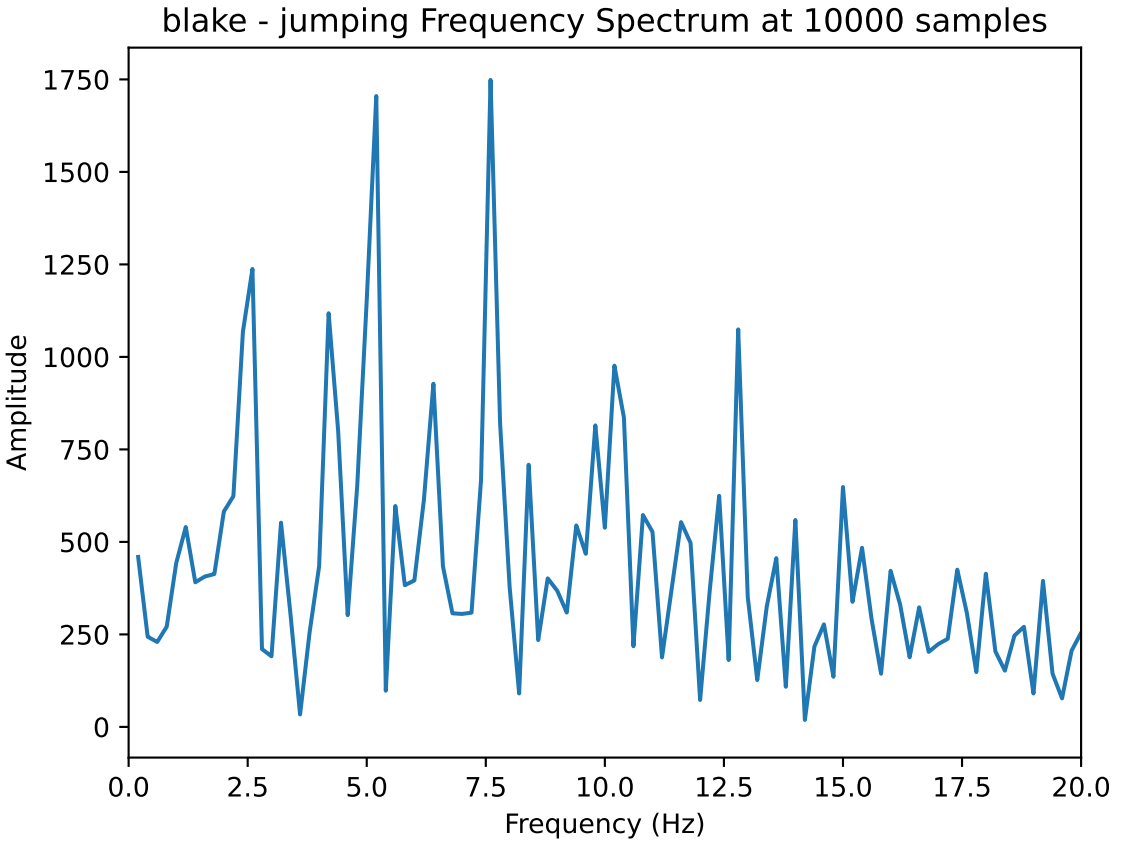

blake - jumping Acceleration Magnitude at 5500 samples
|a|
Acceleration Magnitude
Time (s)
0
10
20
30
40
50
60
0
1
2
3
4
5

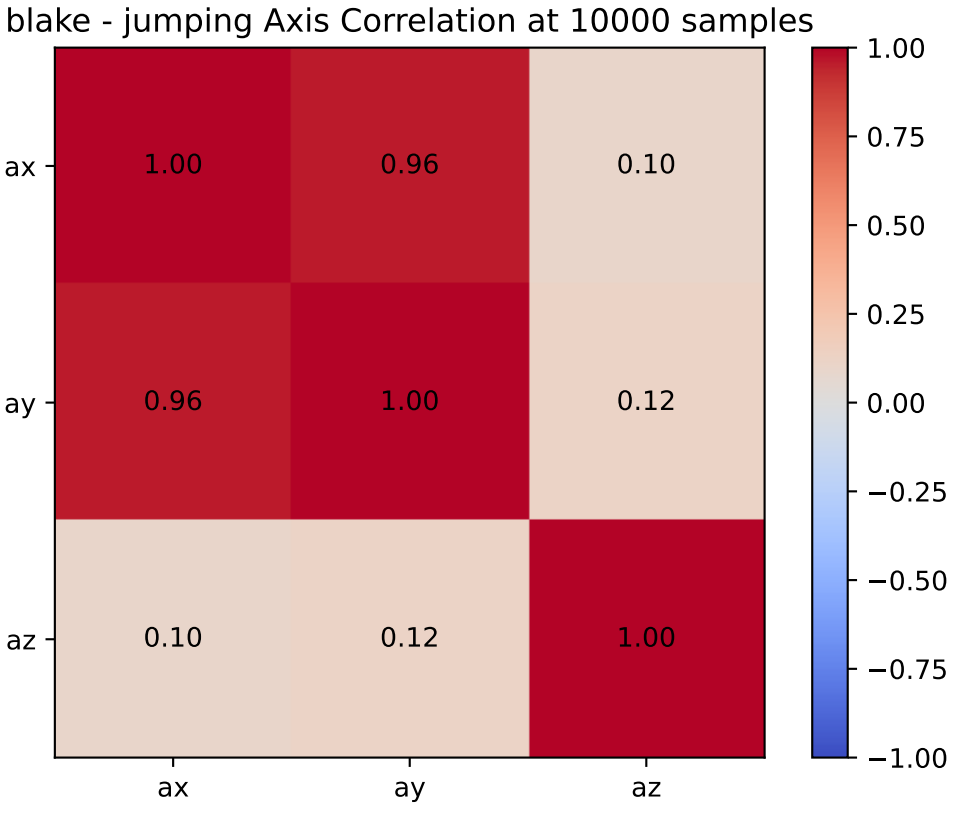

blake - jumping Frequency Spectrum at 5500 samples
Amplitude
1000
800
600
400
200
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Frequency (Hz)

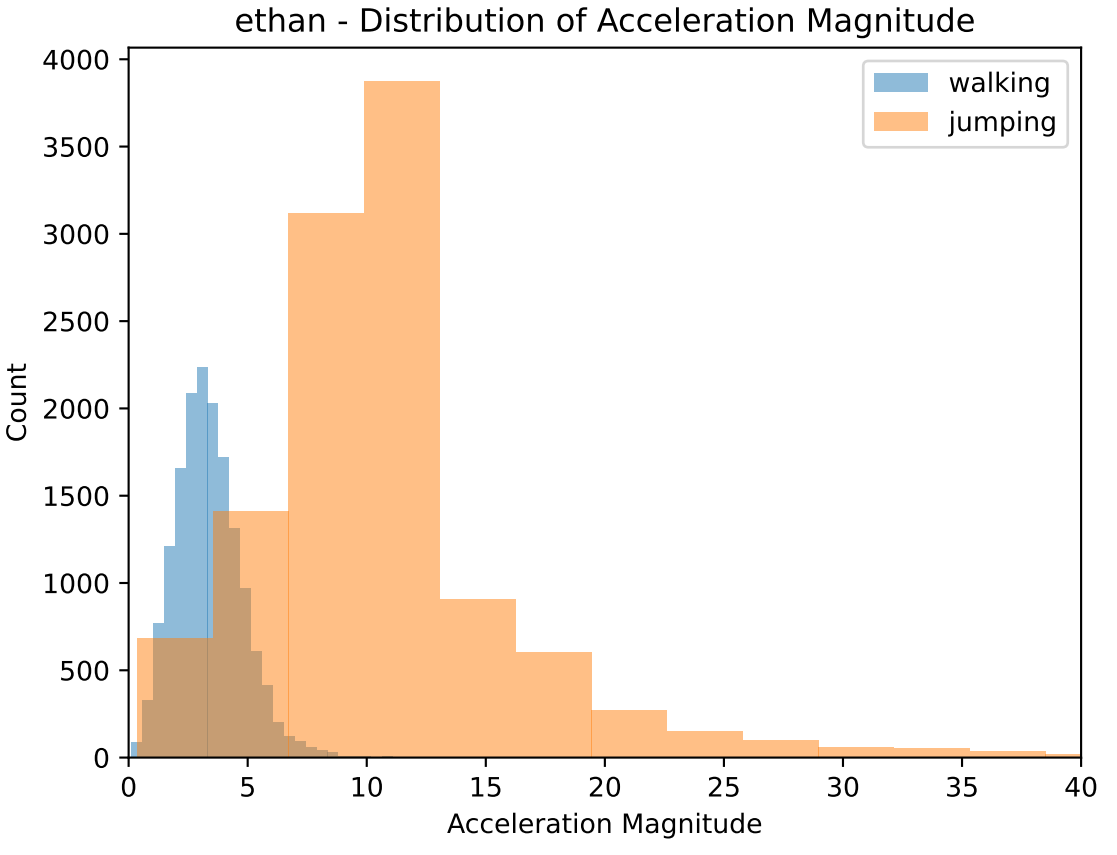

blake - jumping Axis Correlation at 5500 samples
ax
ay
az
1.00
0.11
0.18
0.11
1.00
0.40
0.18
0.40
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

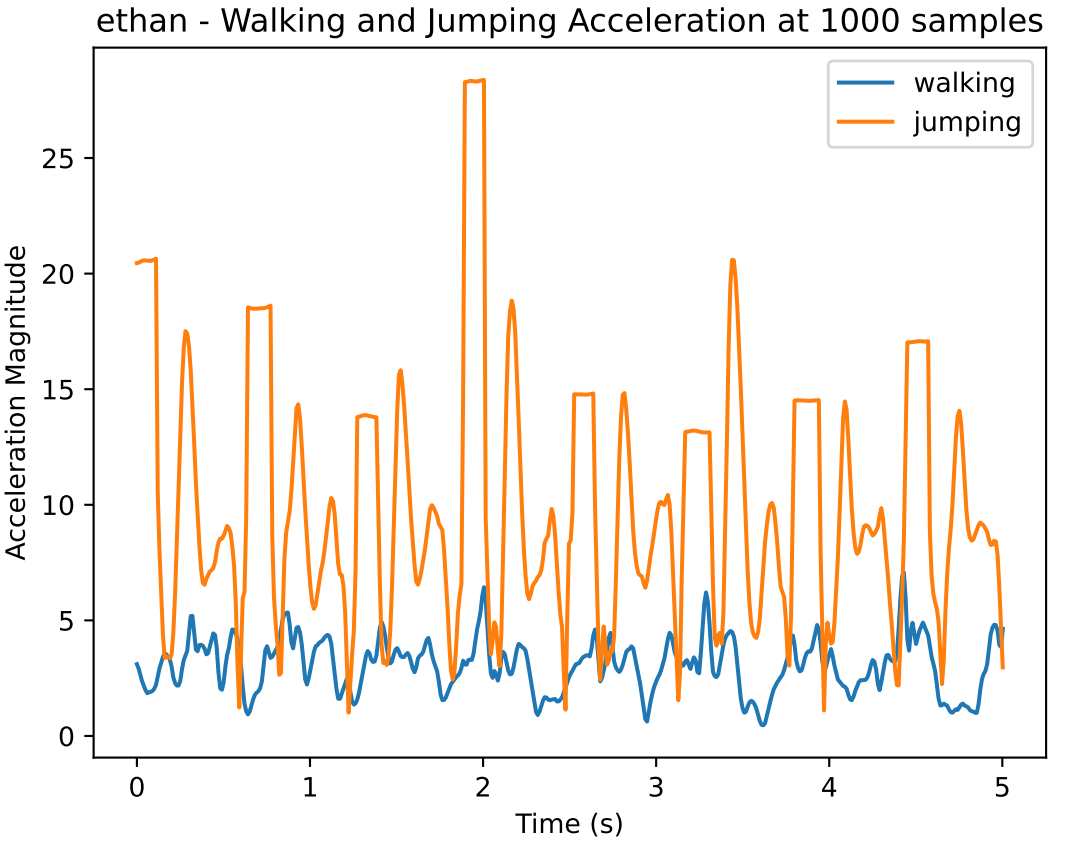

blake - jumping Acceleration at 10000 samples
Acceleration
Time (s)
ax
ay
az
20
0
−20
−40
−60
−80
0
1
2
3
4
5

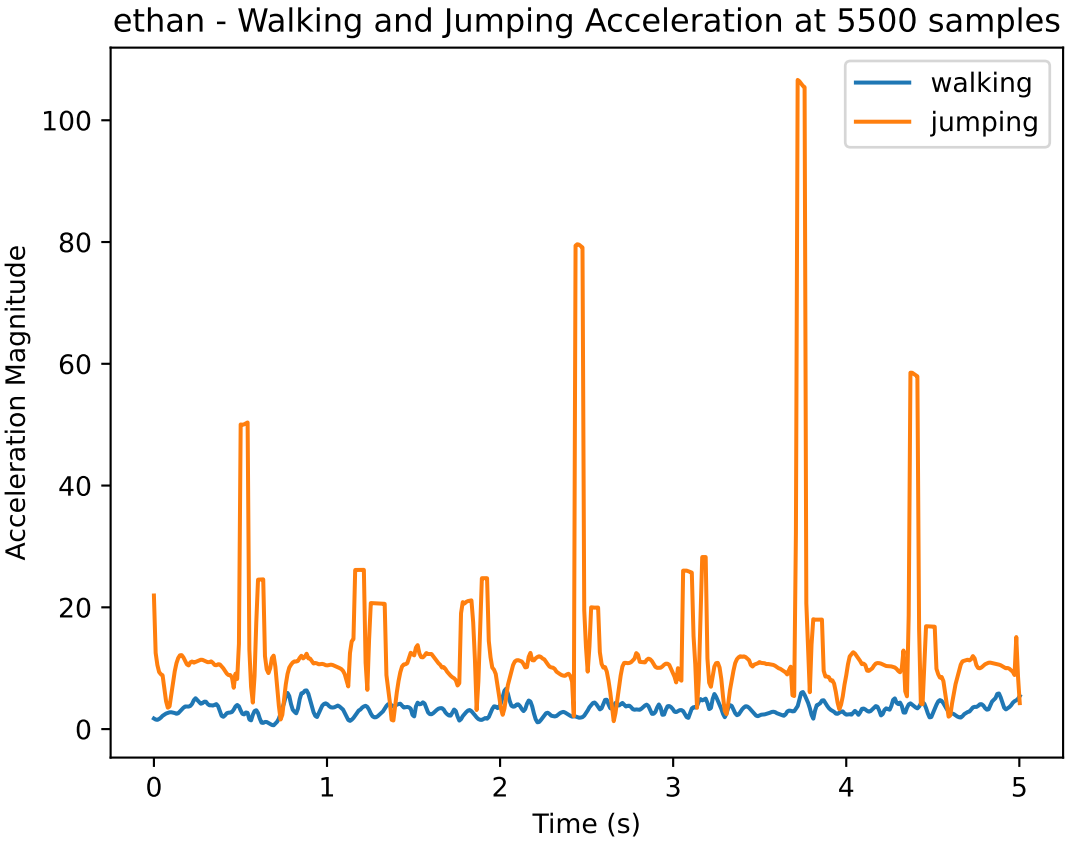

blake - jumping Acceleration Magnitude at 10000 samples
|a|
Acceleration Magnitude
Time (s)
0
20
40
60
80
100
0
1
2
3
4
5

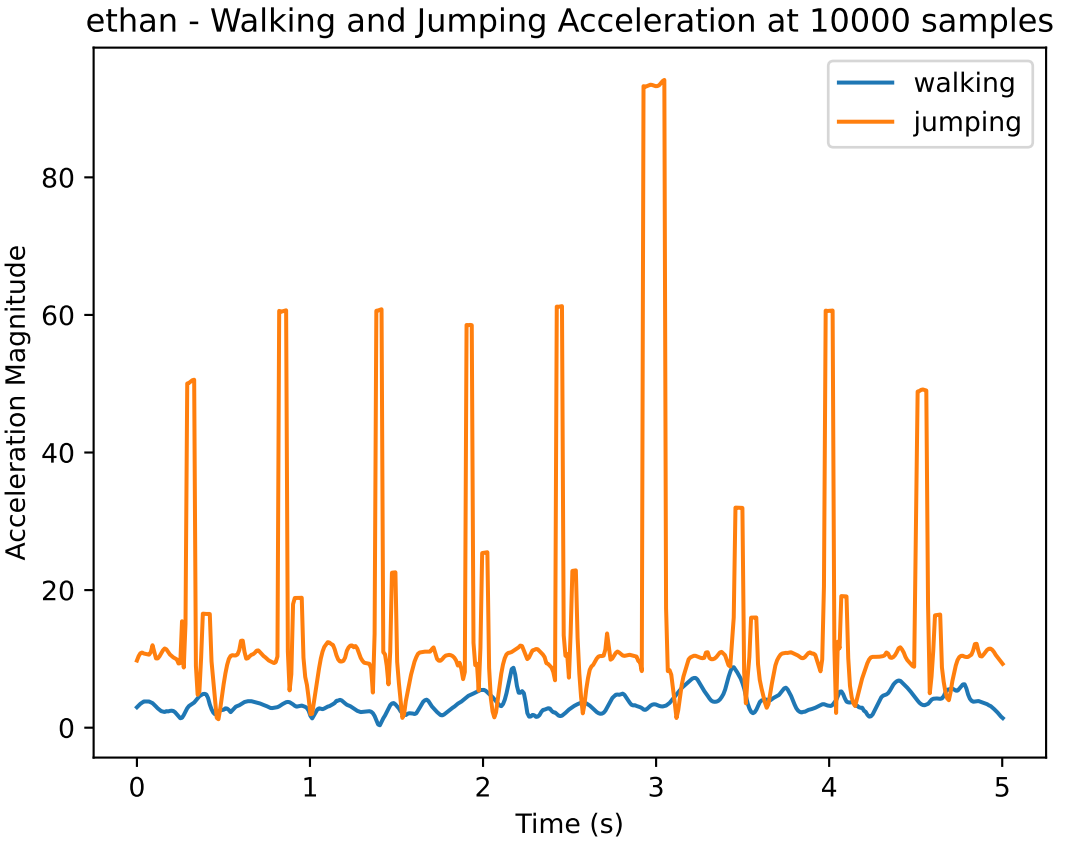

blake - jumping Frequency Spectrum at 10000 samples
Amplitude
Frequency (Hz)
0
250
500
750
1000
1250
1500
1750
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

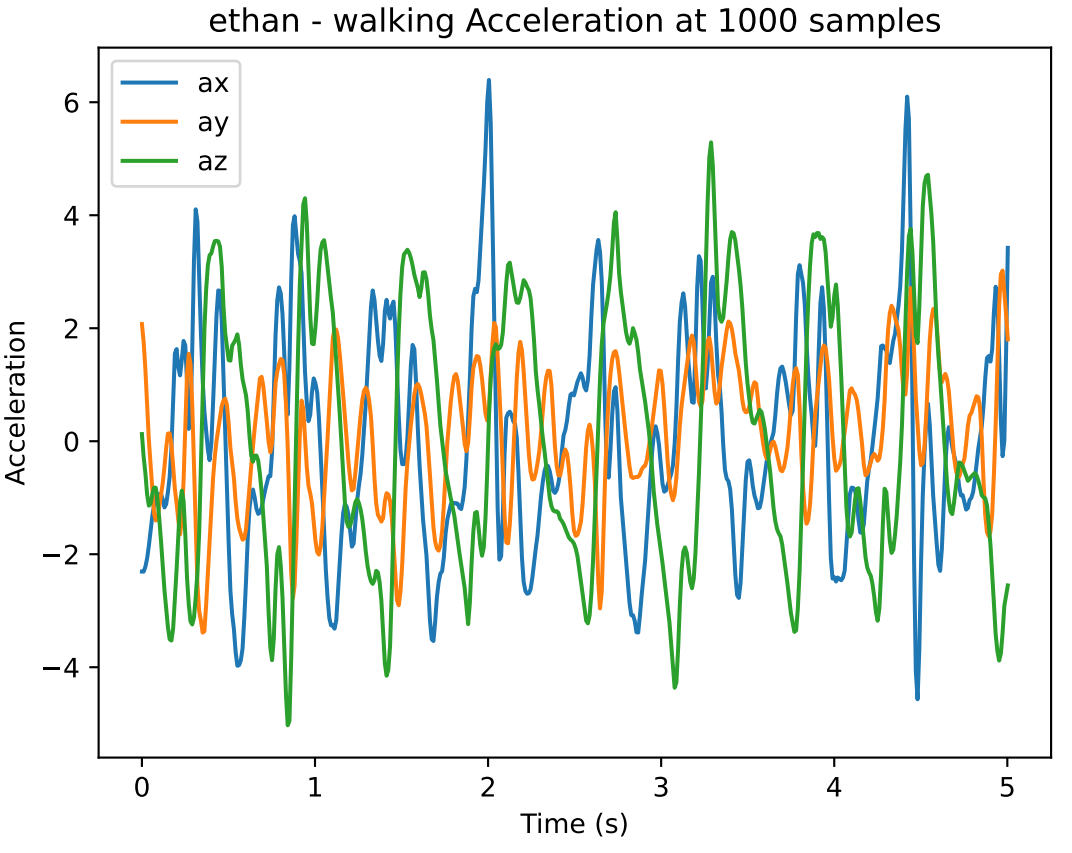

blake - jumping Axis Correlation at 10000 samples
ax
ay
az
1.00
0.96
0.10
0.96
1.00
0.12
0.10
0.12
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

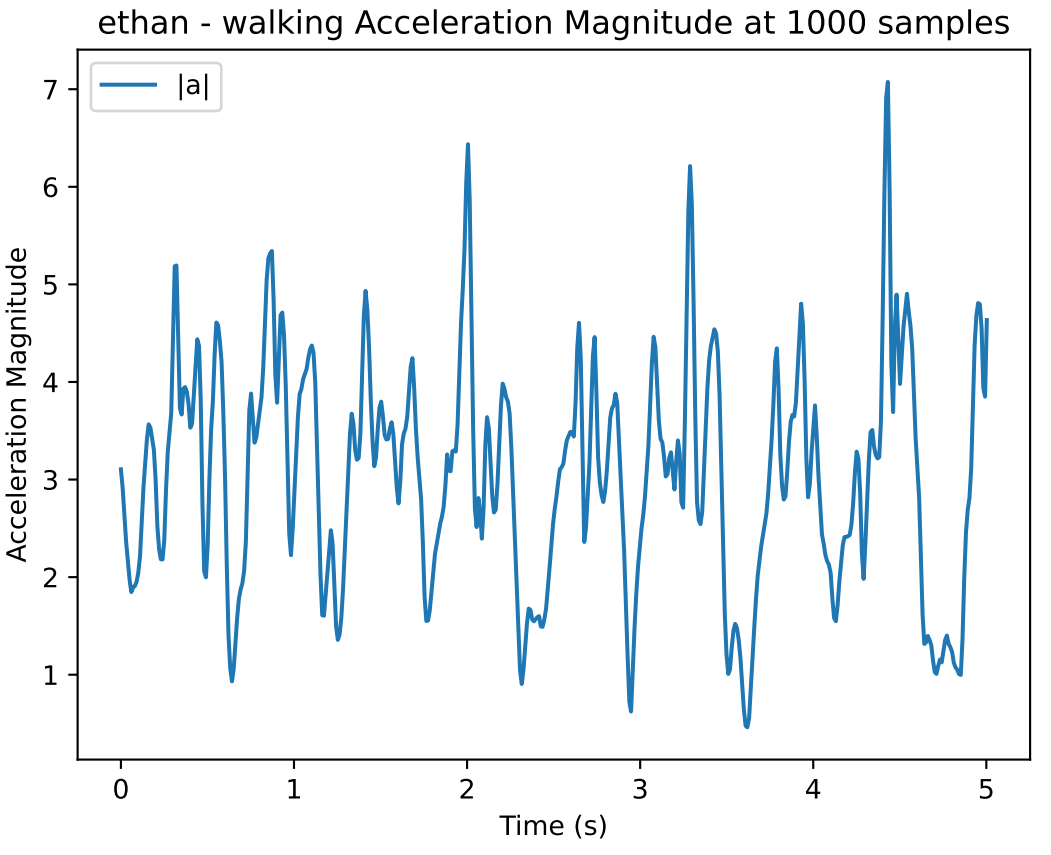

ethan - Distribution of Acceleration Magnitude
walking
jumping
Count
Acceleration Magnitude
0
500
1000
1500
2000
2500
3000
3500
4000
0
5
10
15
20
25
30
35
40

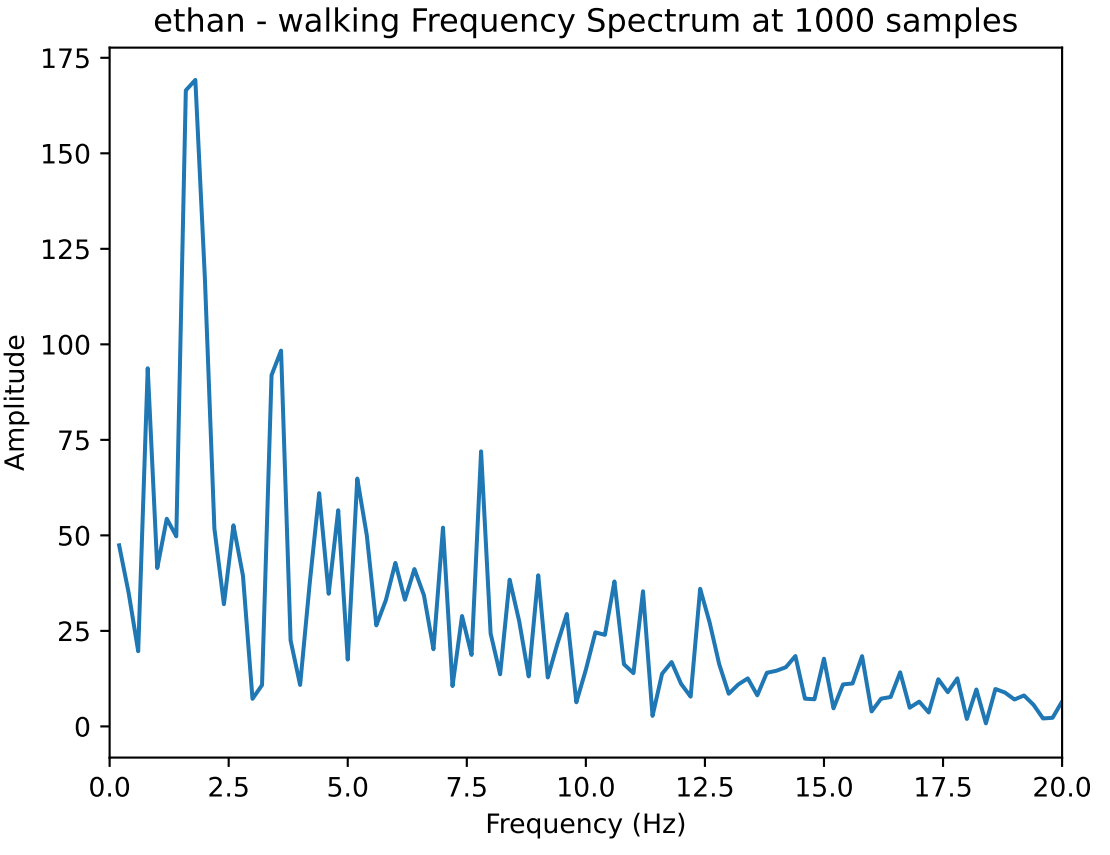

ethan - Walking and Jumping Acceleration at 1000 samples
walking
jumping
Acceleration Magnitude
Time (s)
0
5
10
15
20
25
0
1
2
3
4
5

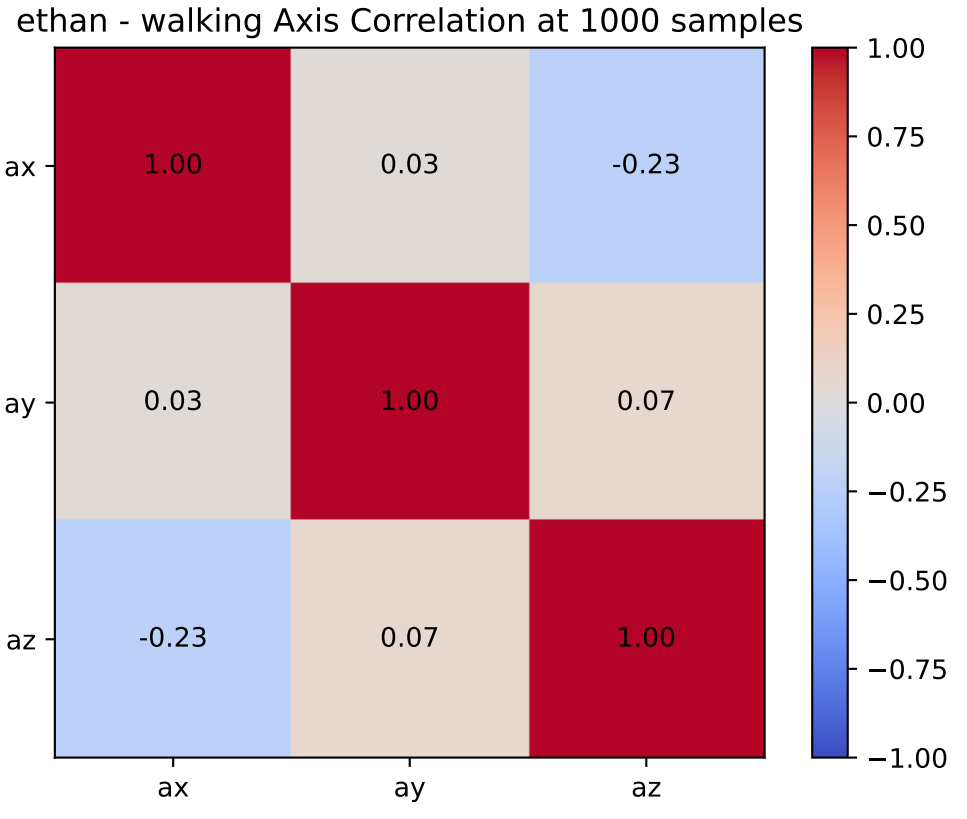

ethan - Walking and Jumping Acceleration at 5500 samples
walking
jumping
Acceleration Magnitude
Time (s)
0
20
40
60
80
100
0
1
2
3
4
5

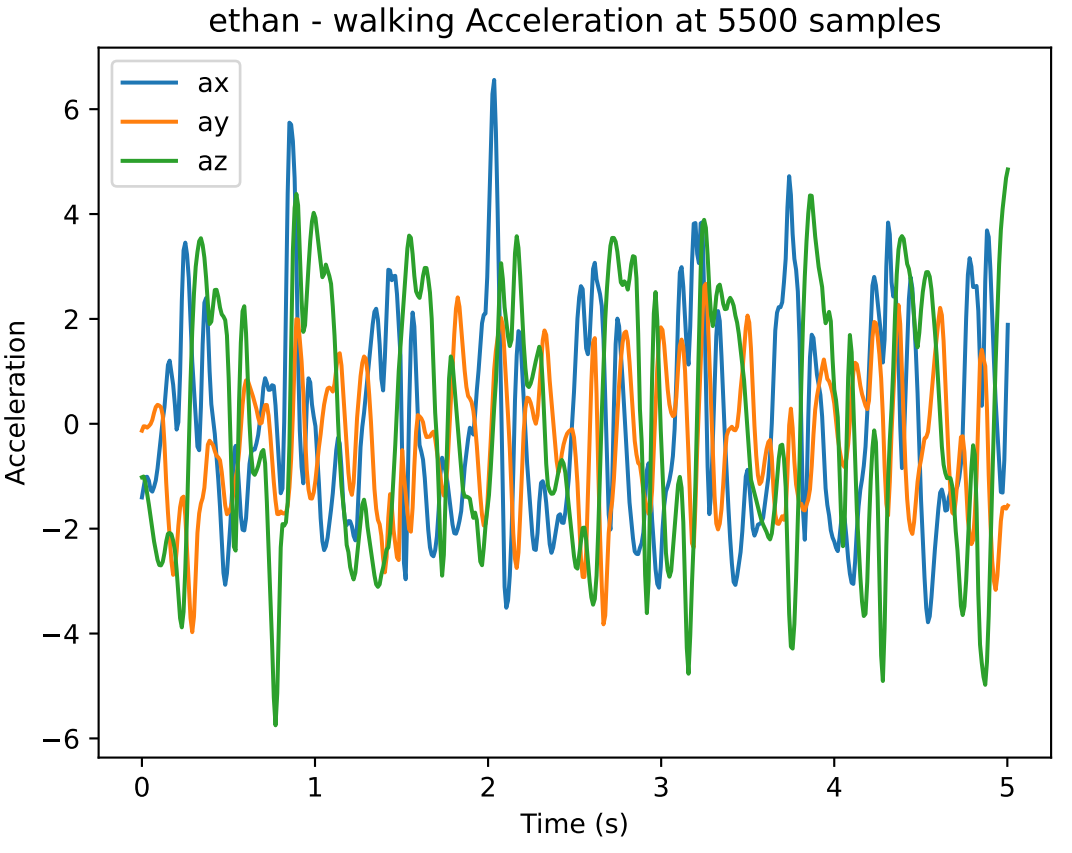

ethan - Walking and Jumping Acceleration at 10000 samples
walking
jumping
Acceleration Magnitude
0
20
40
60
80
0
1
2
3
4
5
Time (s)

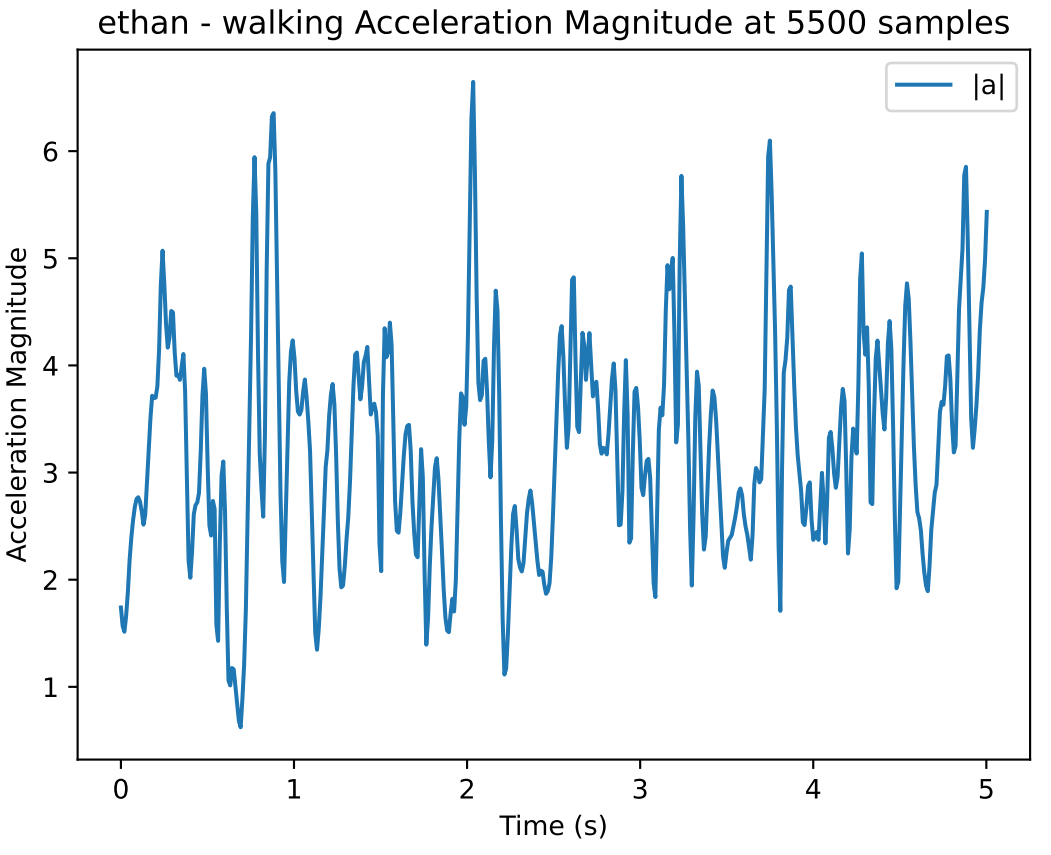

ethan - walking Acceleration at 1000 samples
ax
ay
az
Acceleration
Time (s)
6
4
2
0
−2
−4
0
1
2
3
4
5

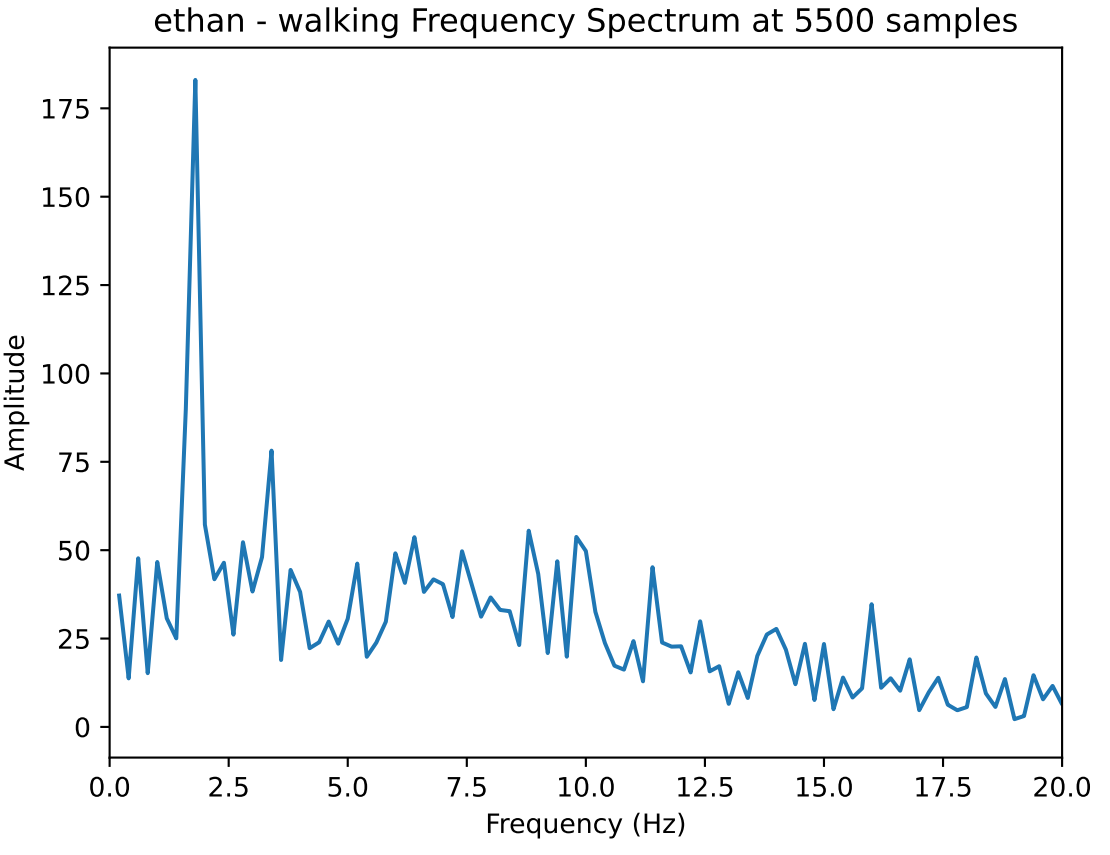

ethan - walking Acceleration Magnitude at 1000 samples
|a|
Acceleration Magnitude
Time (s)
1
2
3
4
5
6
7
0
1
2
3
4
5

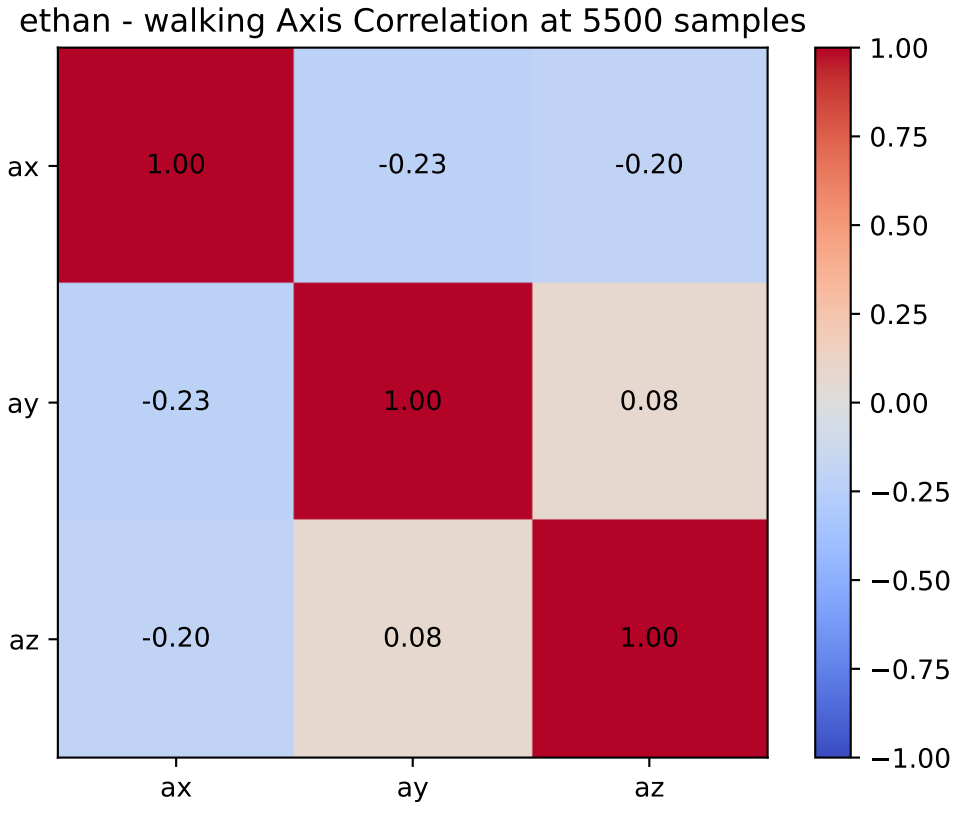

ethan - walking Frequency Spectrum at 1000 samples
Amplitude
Frequency (Hz)
175
150
125
100
75
50
25
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

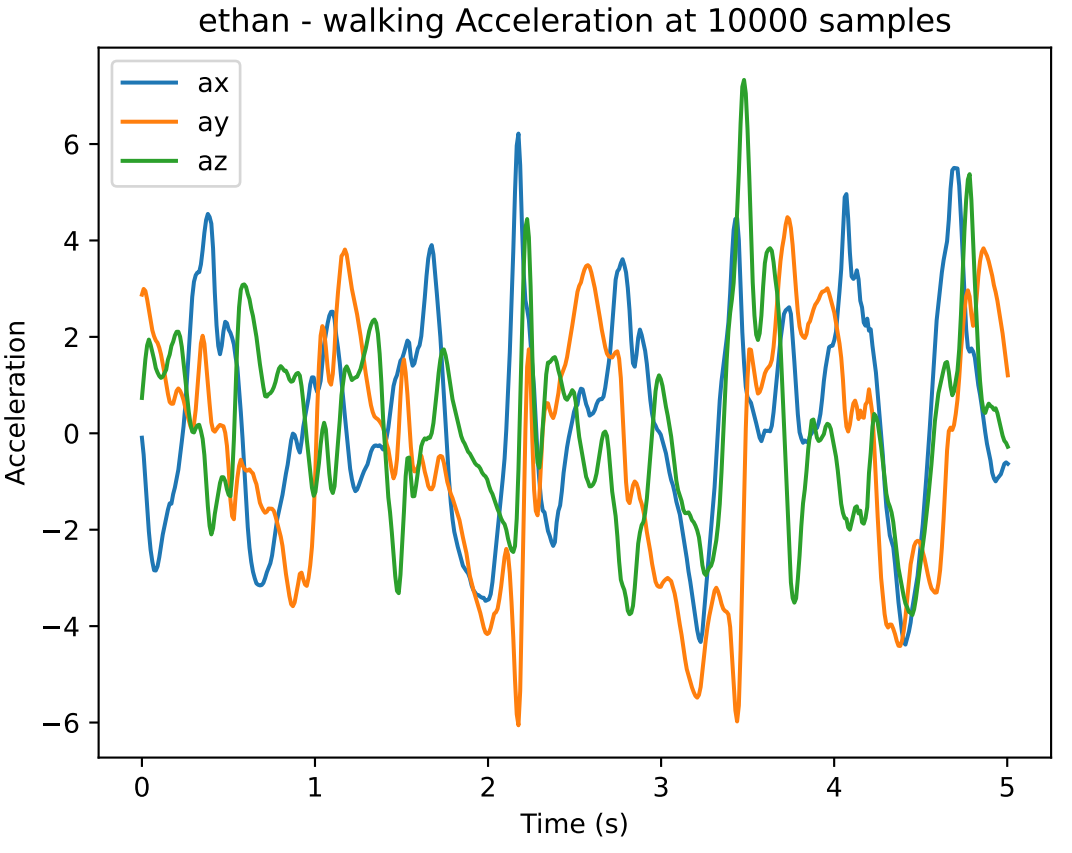

ethan - walking Axis Correlation at 1000 samples
ax
ay
az
1.00
0.03
-0.23
0.03
1.00
0.07
-0.23
0.07
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

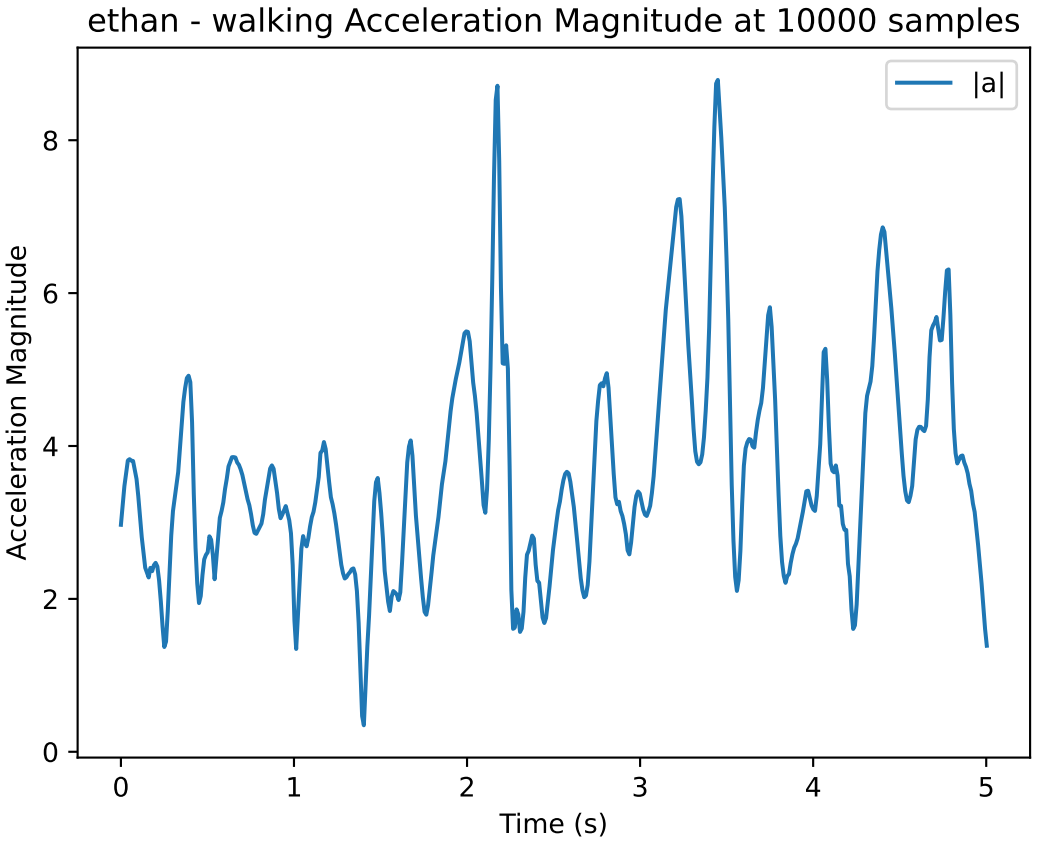

ethan - walking Acceleration at 5500 samples
ax
ay
az
Acceleration
Time (s)
6
4
2
0
−2
−4
−6
0
1
2
3
4
5

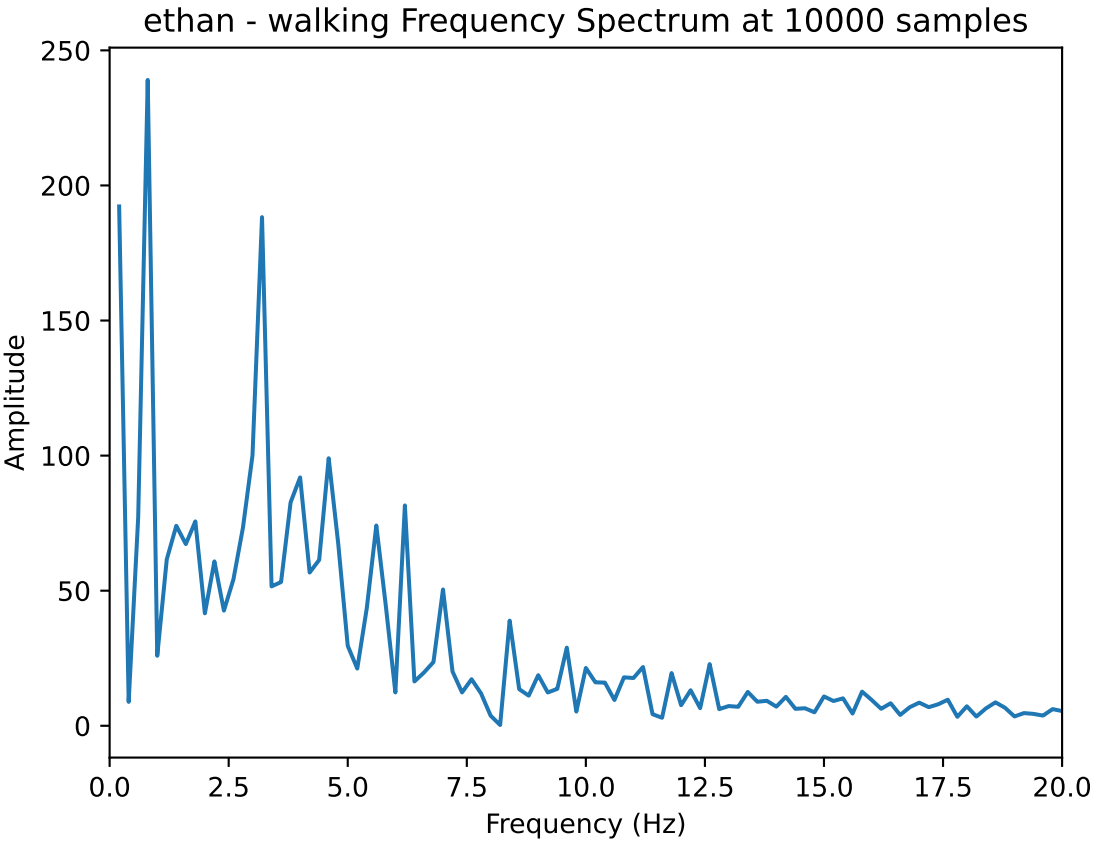

ethan - walking Acceleration Magnitude at 5500 samples
|a|
Acceleration Magnitude
Time (s)
1
2
3
4
5
6
0
1
2
3
4
5

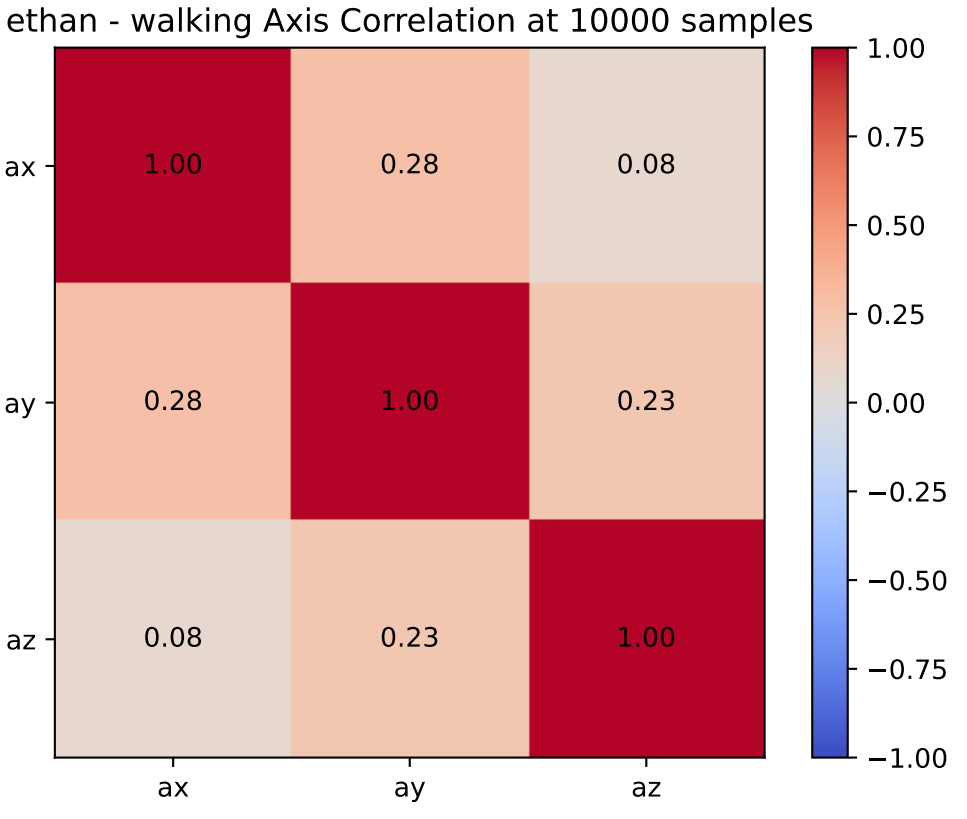

ethan - walking Frequency Spectrum at 5500 samples
Amplitude
175
150
125
100
75
50
25
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Frequency (Hz)

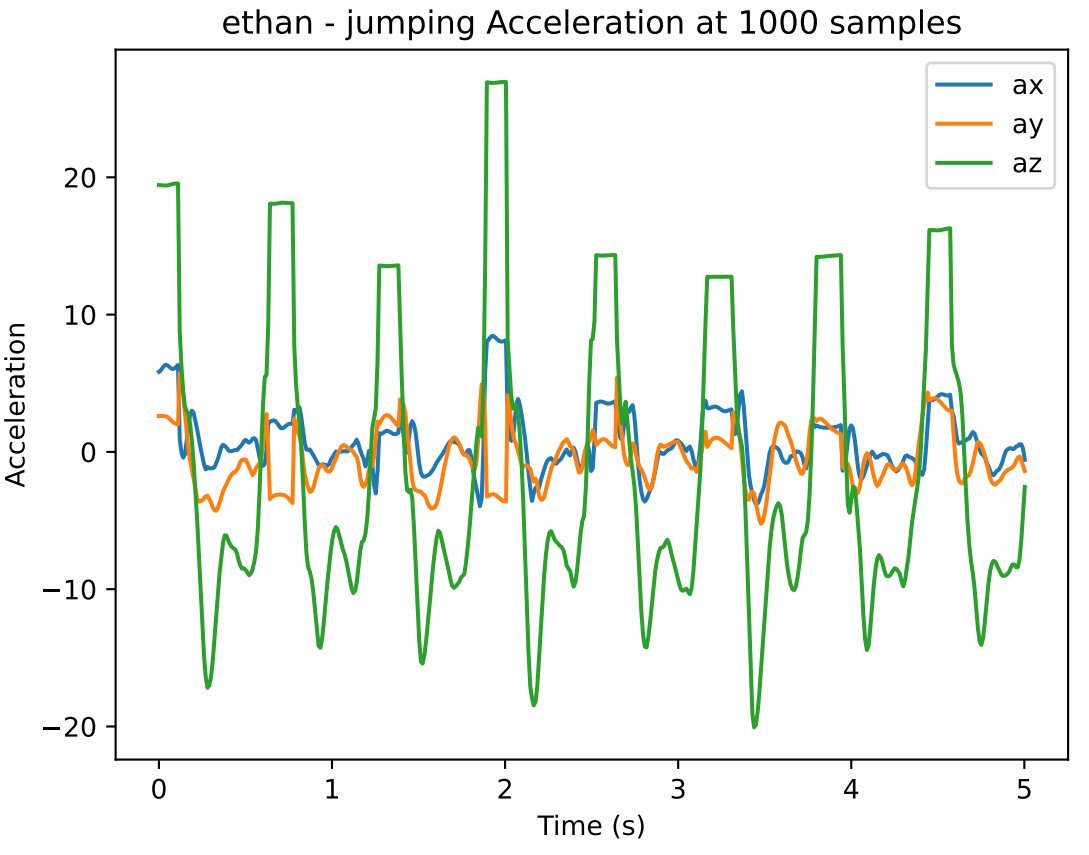

ethan - walking Axis Correlation at 5500 samples
ax
ay
az
1.00
-0.23
-0.20
-0.23
1.00
0.08
-0.20
0.08
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

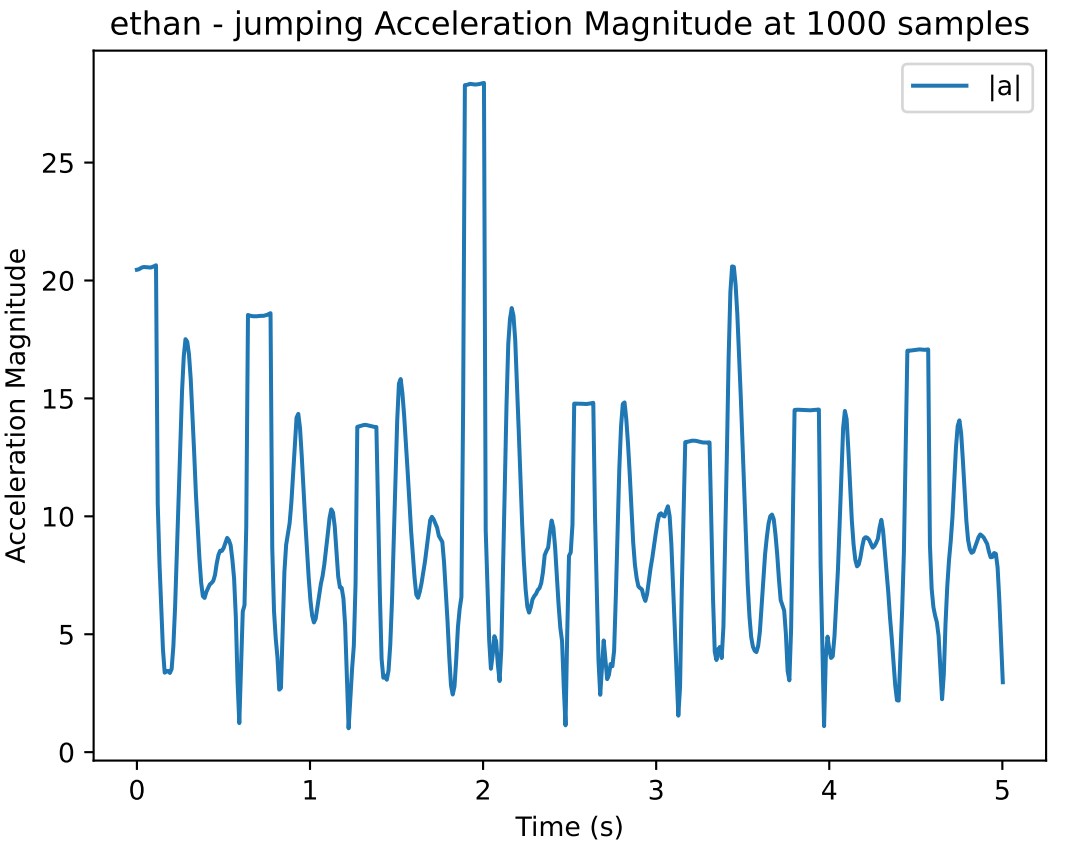

ethan - walking Acceleration at 10000 samples
ax
ay
az
Acceleration
Time (s)
6
4
2
0
−2
−4
−6
0
1
2
3
4
5

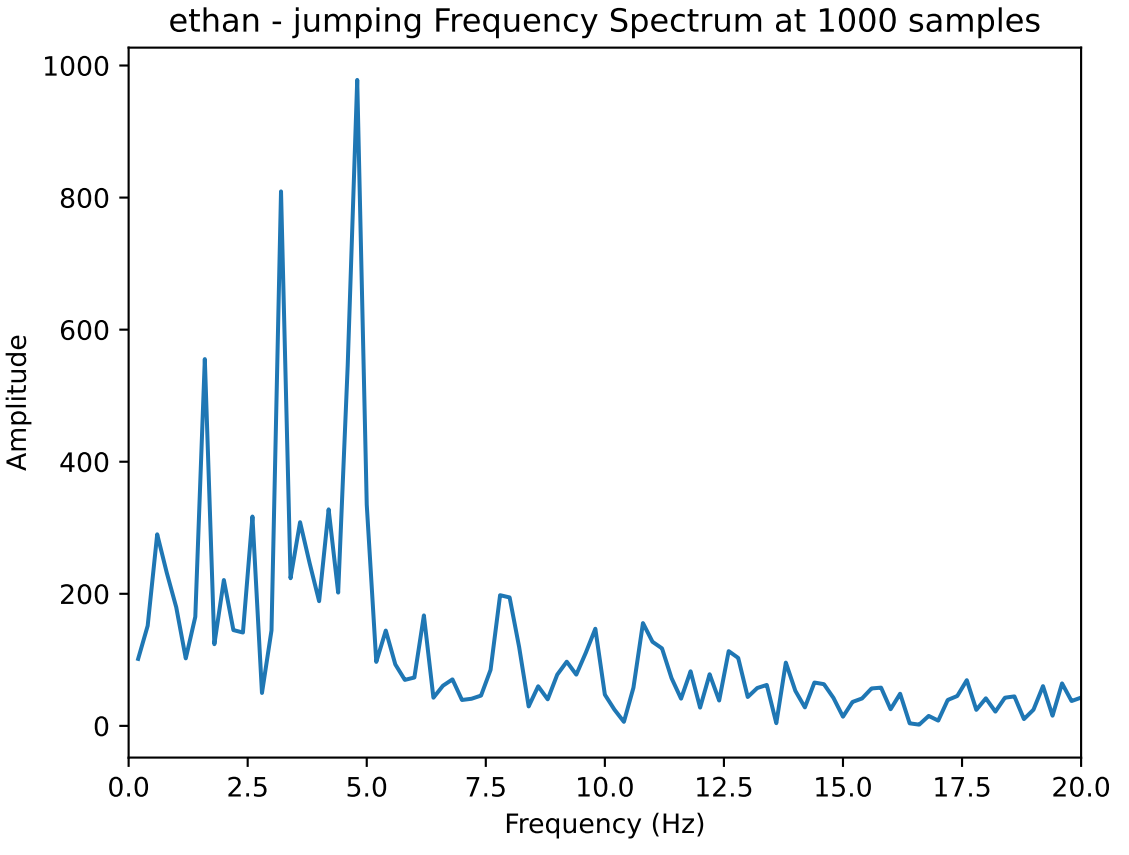

ethan - walking Acceleration Magnitude at 10000 samples
|a|
Acceleration Magnitude
Time (s)
0
2
4
6
8
0
1
2
3
4
5

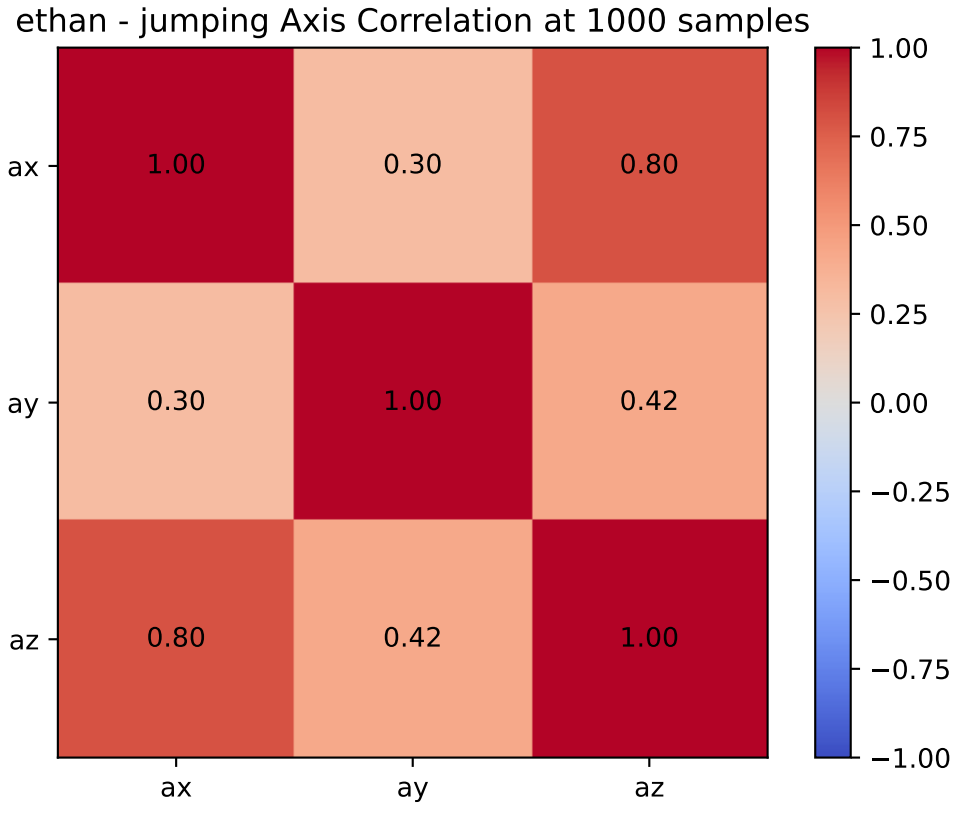

ethan - walking Frequency Spectrum at 10000 samples
Amplitude
Frequency (Hz)
250
200
150
100
50
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

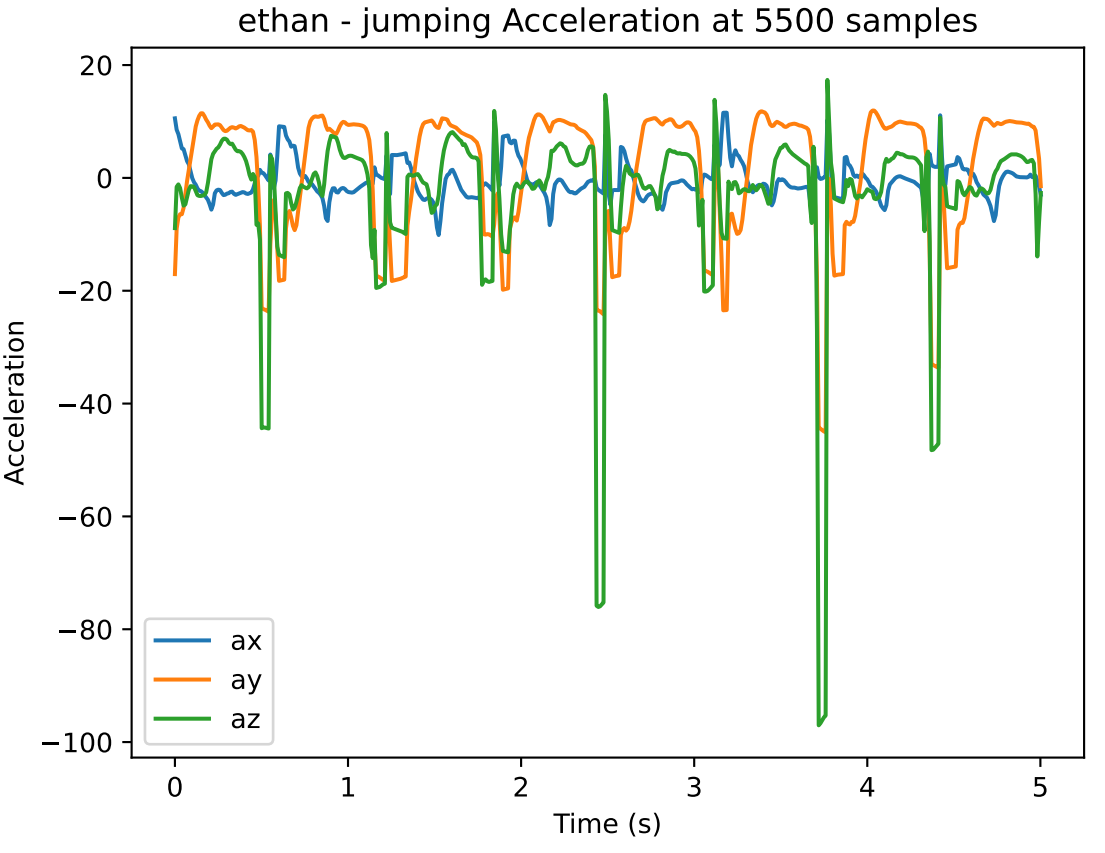

ethan - walking Axis Correlation at 10000 samples
ax
ay
az
1.00
0.28
0.08
0.28
1.00
0.23
0.08
0.23
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

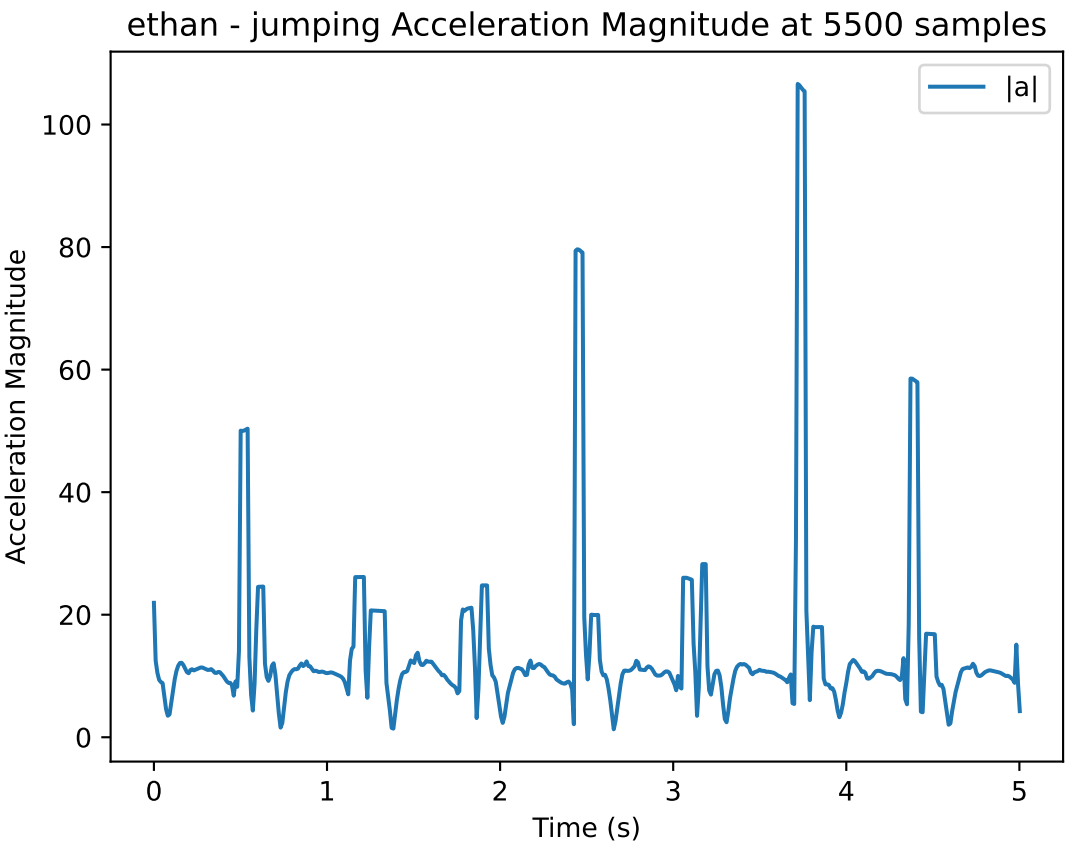

ethan - jumping Acceleration at 1000 samples
ax
ay
az
Acceleration
20
10
0
−10
−20
0
1
2
3
4
5
Time (s)

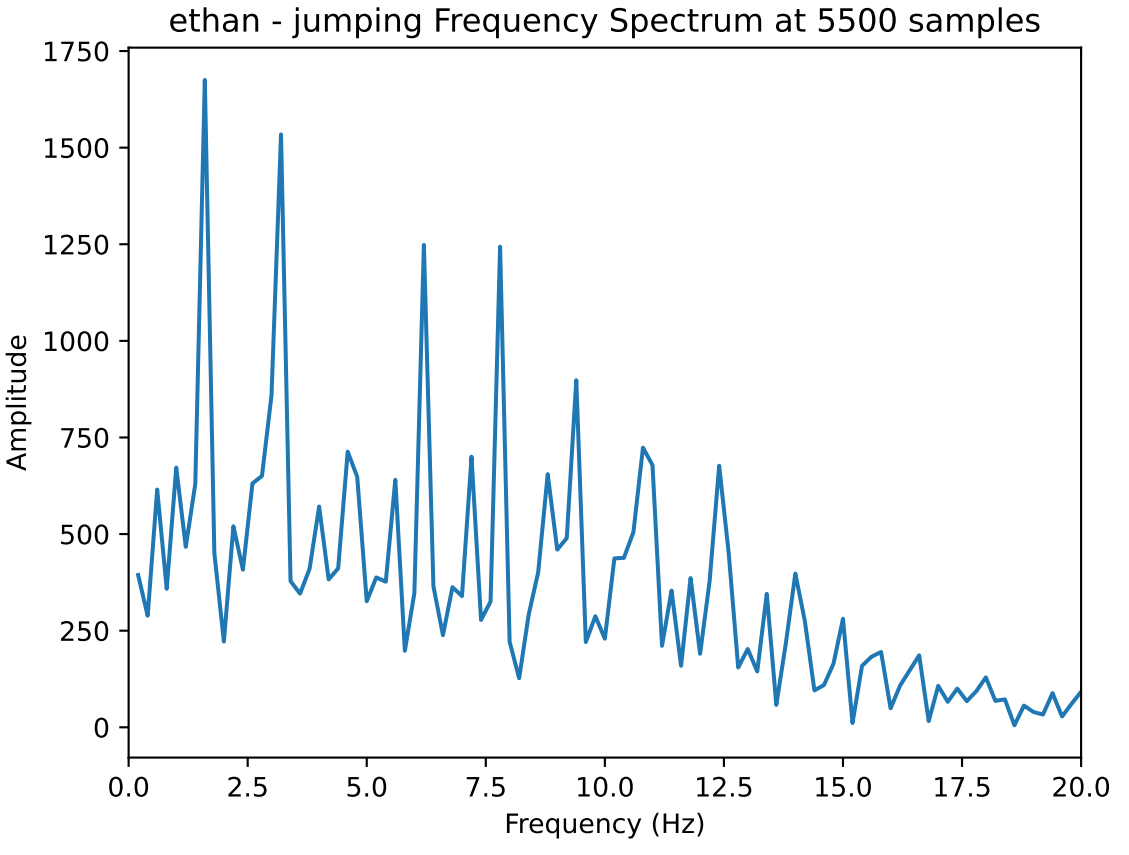

ethan - jumping Acceleration Magnitude at 1000 samples
|a|
Acceleration Magnitude
Time (s)
0
5
10
15
20
25
0
1
2
3
4
5

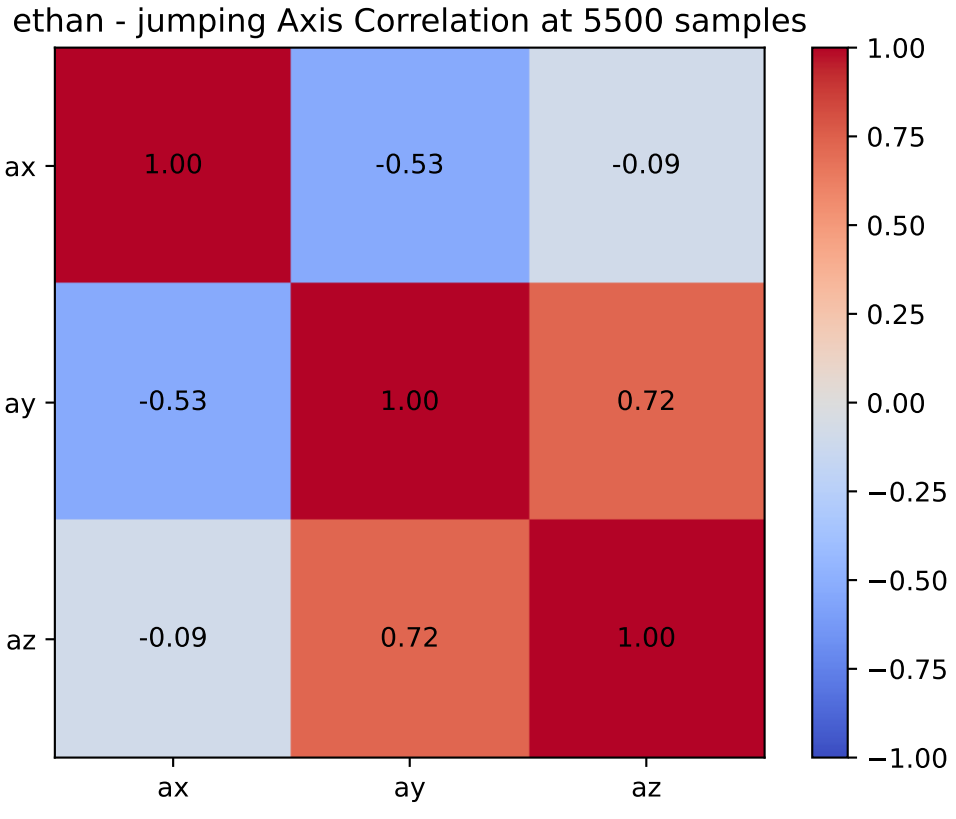

ethan - jumping Frequency Spectrum at 1000 samples
Amplitude
Frequency (Hz)
1000
800
600
400
200
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

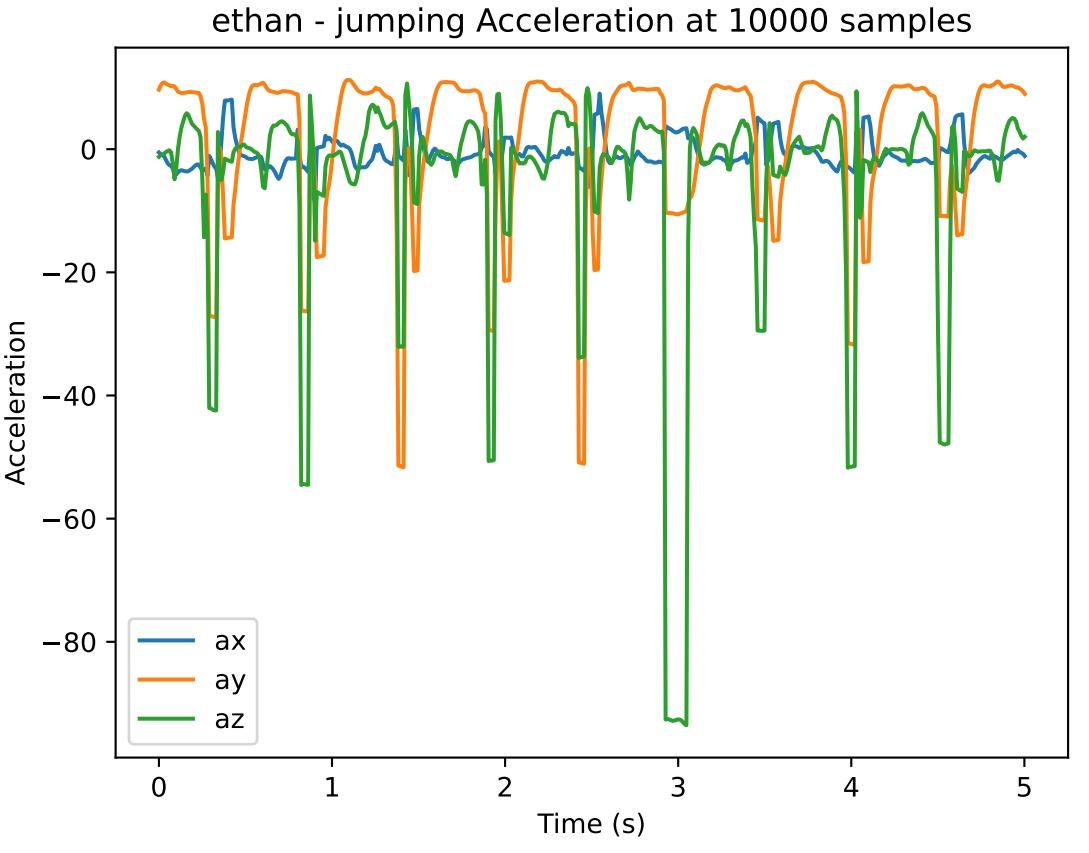

ethan - jumping Axis Correlation at 1000 samples
ax
ay
az
1.00
0.30
0.80
0.30
1.00
0.42
0.80
0.42
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

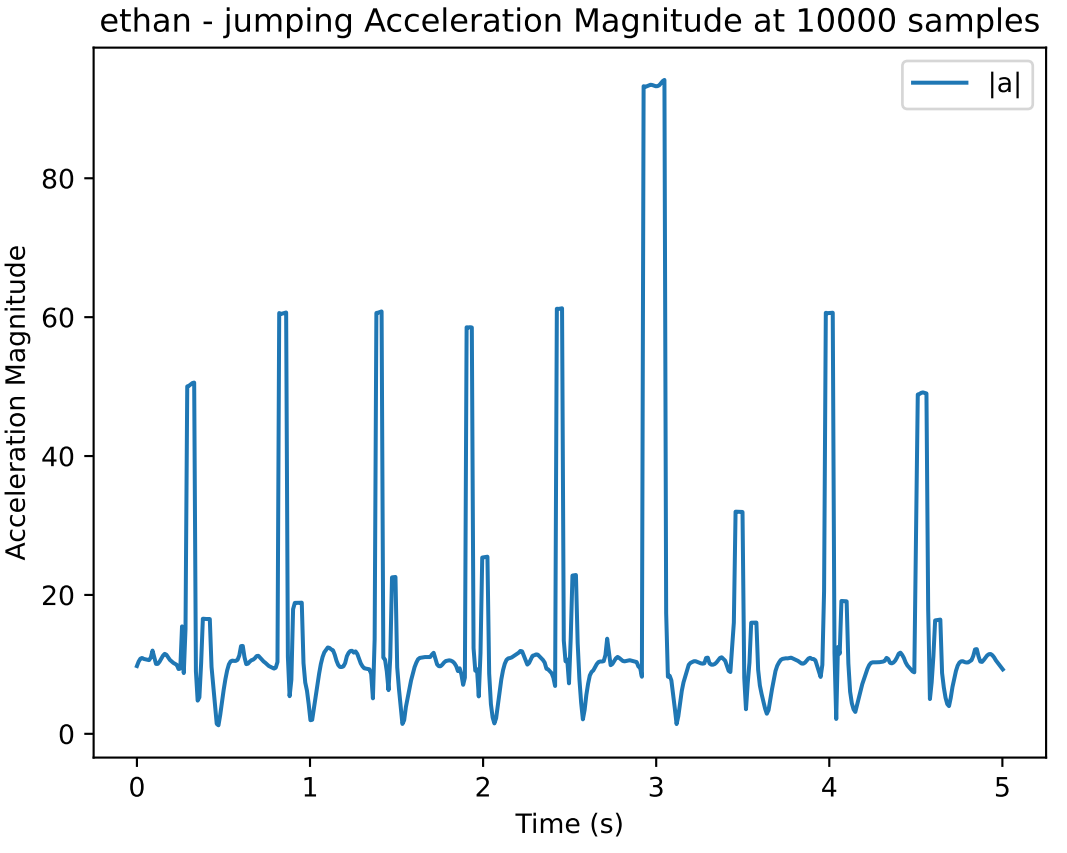

ethan - jumping Acceleration at 5500 samples
Acceleration
Time (s)
ax
ay
az
20
0
−20
−40
−60
−80
−100
0
1
2
3
4
5

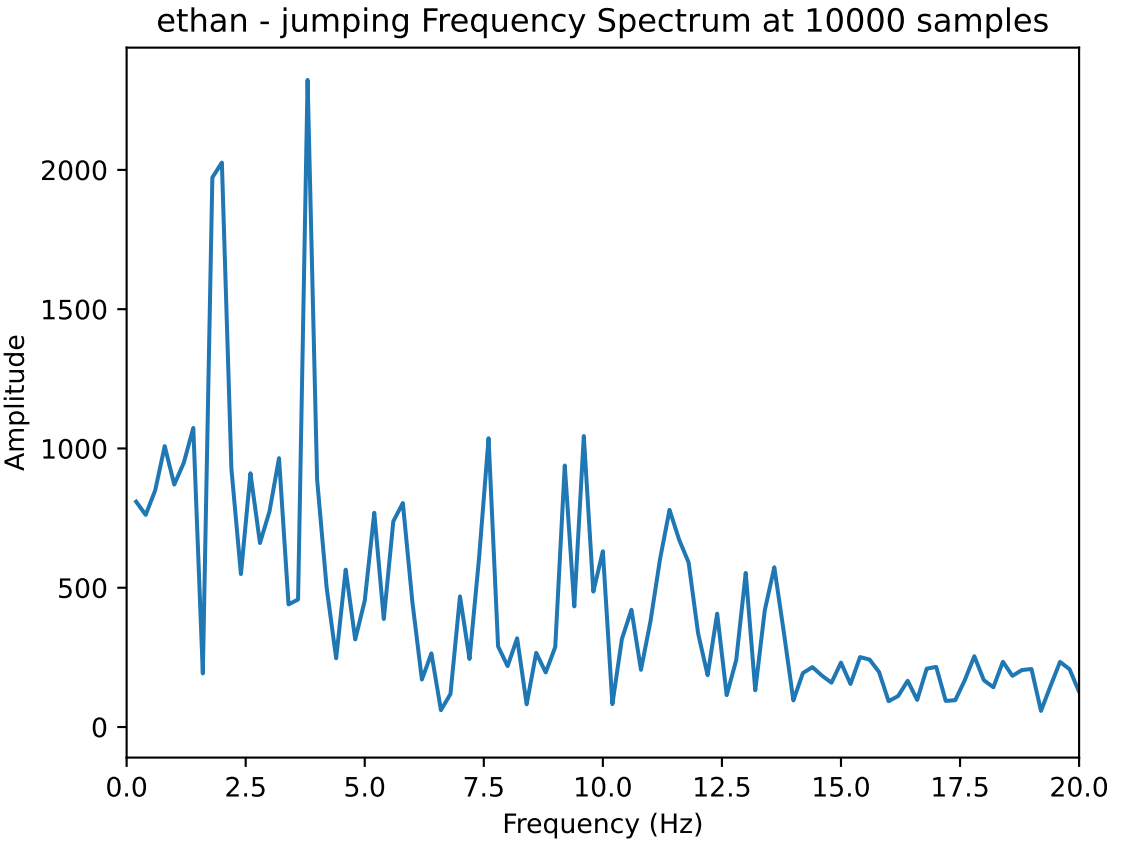

ethan - jumping Acceleration Magnitude at 5500 samples
|a|
Acceleration Magnitude
Time (s)
0
20
40
60
80
100
0
1
2
3
4
5

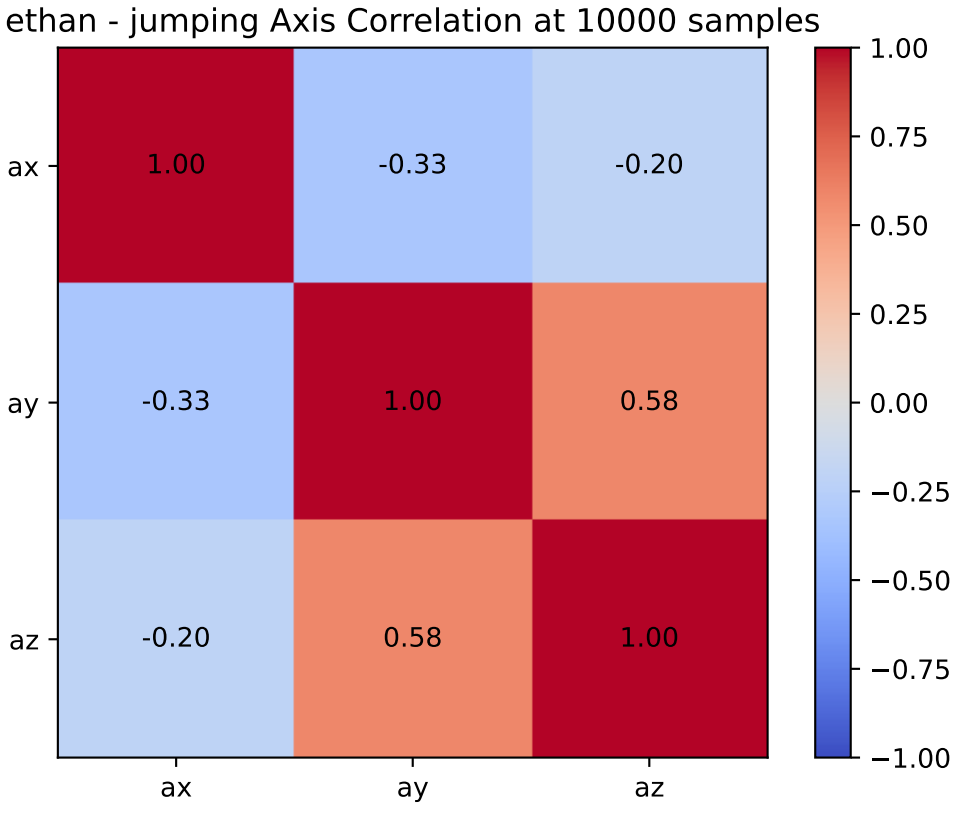

ethan - jumping Frequency Spectrum at 5500 samples
Amplitude
Frequency (Hz)
1750
1500
1250
1000
750
500
250
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


ethan - jumping Axis Correlation at 5500 samples
ax
ay
az
1.00
-0.53
-0.09
-0.53
1.00
0.72
-0.09
0.72
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


ethan - jumping Acceleration at 10000 samples
Acceleration
Time (s)
0
−20
−40
−60
−80
0
1
2
3
4
5
ax
ay
az


ethan - jumping Acceleration Magnitude at 10000 samples
|a|
Acceleration Magnitude
0
20
40
60
80
0
1
2
3
4
5
Time (s)


ethan - jumping Frequency Spectrum at 10000 samples
Amplitude
2000
1500
1000
500
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Frequency (Hz)


ethan - jumping Axis Correlation at 10000 samples
ax
ay
az
1.00
-0.33
-0.20
-0.33
1.00
0.58
-0.20
0.58
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00